

Introduction to Astrophysics and Supernova Phenomenology

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Abstract

Alongside accelerator experiments and cosmology, astrophysics provides a window into new physics beyond the Standard Model. In this JCP HEP School lecture, we follow the evolutionary process of stars, one of the most important objects in astrophysics, and examine the underlying physical laws and observational properties. We also introduce the basic equations and theorems required to describe astrophysical objects. Furthermore, we discuss the explosion mechanism, observational signatures, and phenomenological applications of core-collapse supernovae, which are particularly useful for high-energy particle physics.

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Chapter 1.

Landscape of Stellar Evolution

In this chapter, we follow the evolutionary stages of stars to examine the overall flow of stellar evolution, as well as phenomena that can be applied to research and their key characteristics. The aim of this chapter is not to present detailed stages of stellar evolution or provide precise physical backgrounds, theoretical developments, or experimental discussions, but rather to introduce the key concepts and provide an overview. For a more detailed discussion of the physical concepts in this chapter, see Chapter 2. The structure and content of this chapter is mostly informed by [1–3], and also referred by [4, 5].

The objects we consider are stars. When the following conditions are satisfied, we may assume spherical symmetry for a star.

1. Angular isotropy of physical quantities: All macroscopic physical quantities depend only on the radial coordinate r , and are independent of angular coordinates θ and ϕ .
2. Negligible rotation: The stellar rotation is sufficiently slow such that centrifugal effects can be neglected compared to gravity.
3. Negligible magnetic fields and external forces: Magnetic fields, tidal forces, and other anisotropic external influences are sufficiently weak and do not affect the global structure.
4. Hydrostatic or quasi-static equilibrium: The star evolves on timescales much longer than the dynamical timescale, so that time-dependent deviations from equilibrium can be neglected.
5. Effective isotropy of turbulent motion: Although convection or turbulence may be present, their effects are statistically isotropic and can be treated in an averaged sense.
6. Spherically symmetric gravitational field and chemical composition: The mass distribution and chemical composition is spherically symmetric.

Therefore, in the following discussion, we will assume spherical symmetry when treating stars.

1.1. Measurements of Stellar Objects

We can measure astrophysical objects in various ways, and based on these measurements, we can characterize them. Among the available observables, luminosity, temperature, and magni-

tude are the most widely used quantities, as they both capture the essential physical properties of stars and provide a practical basis for stellar classification.

1.1.1. Luminosity

In the high-temperature, high-density interior of a star, the photon mean free path is extremely small compared to the stellar radius, so photons cannot freely streaming. Consequently, when a star is observed via electromagnetic radiation, the detected photons originate from the photosphere—where photons begin to free-stream—or from regions exterior to it.

Assume that the star radiates as a blackbody, the luminosity L is given by the Stefan-Boltzmann law. If R denotes the radius of the stellar photosphere, then L is expressed as follows.

$$L = 4\pi R^2 \sigma T^4. \quad (1.1)$$

Here, $\sigma = \frac{\pi^2 k_B^4}{60c^2 \hbar^3}$ is the Stefan-Boltzmann constant.

The luminosity and the flux are related according to an inverse square law. If the observer is at a distance d from the star, then the following relation holds.

$$F = \frac{L}{4\pi d^2}. \quad (1.2)$$

Therefore, the luminosity of a star can be determined by measuring the flux.

1.1.2. Temperature

From Eq. (1.1), we may define the effective temperature of a star as follows.

$$T_{\text{eff}} = \left(\frac{L}{4\pi R^2 \sigma} \right)^{1/4}. \quad (1.3)$$

This corresponds to the temperature of a blackbody with the same radius and luminosity as the star.

A blackbody at temperature T emits blackbody radiation according to Planck's law. Fig. 1.1 shows the wavelength-spectral radiance relation of blackbody radiation for different temperatures. The energy density U_ν of radiation with frequency ν emitted by a blackbody at temperature T is given by Planck's law as below.

$$U_\nu(\nu, T) = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/k_B T} - 1}. \quad (1.4)$$

Accordingly, the photospheric temperature of a star can be determined by measuring its spectrum.

Stellar spectra have traditionally been classified using the spectral classes O, B, A, F, G, K, M, L, T, and Y. The temperature increases toward class O and decreases toward class Y, and

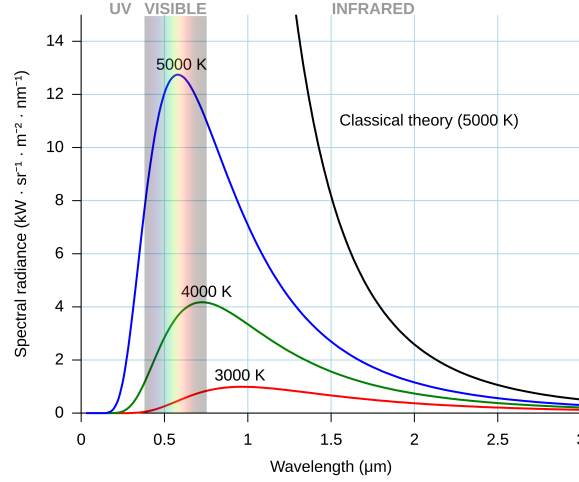


Figure 1.1: Spectral radiance of blackbody radiation as function of the wavelength and the temperature. [6].

each class is further divided into subclasses such as F0-F9. The midpoint of each spectral class is denoted by 5, and the corresponding classification is summarized in Table 1.1. A commonly used mnemonic is “Oh Be A Fine Girl/Guy, Kiss Me, Love To You” to remember the order of spectral classes.

Spectra class	Temperature
O	40 000 K
B	17 000 K
A	9 000 K
F	7 000 K
G	5 500 K
K	4 500 K
M	3 000 K
L	2 000 K
T	1 000 K
Y	350 K

Table 1.1: Median temperatures of each spectral class.

In addition, the absorption and emission lines in a stellar spectrum provide information on the composition of the stellar atmosphere and related properties.

1.1.3. Magnitude

Magnitude is classified into apparent magnitude and absolute magnitude. The apparent magnitude m is defined as follows.

$$m = -2.5 \log \left(\frac{F}{F_0} \right). \quad (1.5)$$

Here, F is the observed flux of the star, and F_0 is the reference flux of the photometric filter used for the measurement. As seen from Eq. (1.2), since $F \propto d^{-2}$, the apparent magnitude m

depends on distance. A difference of 1 magnitude corresponds to a factor of approximately 2.512 in brightness, and objects with apparent magnitude up to about 6 can be observed with the naked eye.

The absolute magnitude is defined by removing the distance dependence of the apparent magnitude. It is defined as the magnitude a star would have if it were located at a distance of 10, pc from the observer, and is given by

$$M = m + 5 - 5 \log d - A, \quad (1.6)$$

where d is the distance to the star in pc, and A accounts for interstellar extinction due to intervening material. From this relation, if m and M are known, the distance d can be determined. The difference between the two magnitudes is defined as follows and is called the distance modulus.

$$m - M \equiv \mu = 5 \log d - 5 + A. \quad (1.7)$$

The magnitude depends on the wavelength band used in measuring the flux. These bands are classified as U, B, V, R, I, J, H and K, as summarized in Table 1.2. The band used in determining the magnitude is indicated by a subscript, e.g., m_V .

Waveband	Central wavelength	Width of band
U	365 nm	70 nm
B	440 nm	100 nm
V	550 nm	90 nm
R	700 nm	220 nm
I	900 nm	240 nm
J	1.25 μm	0.24 μm
H	1.65 μm	0.4 μm
K	2.2 μm	0.6 μm

Table 1.2: The standard photometric wavebands.

1.1.4. The Hertzsprung-Russell Diagram

By measuring physical quantities for many stars, one can classify them on the basis of these observables. A representative example is the Hertzsprung-Russell diagram (H-R diagram), in which stars are plotted on the $T_{\text{eff}}-L$ plane, as shown in Fig. 1.2. The H-R diagram is widely used in the study of stellar types, stellar evolution, and stellar populations such as clusters. Stars on the H-R diagram can be classified according to their luminosity, as summarized in Table 1.3.

Observationally, stars are often organized using their color indices and magnitudes. Accordingly, the color-magnitude diagram (CMD), in which the horizontal axis represents color and the vertical axis represents magnitude, is also commonly used. The color is defined, for example, as $M_B - M_V$, and in practice the absolute magnitude is typically employed. A CMD constructed with absolute magnitude can be put into one-to-one correspondence with the H-R

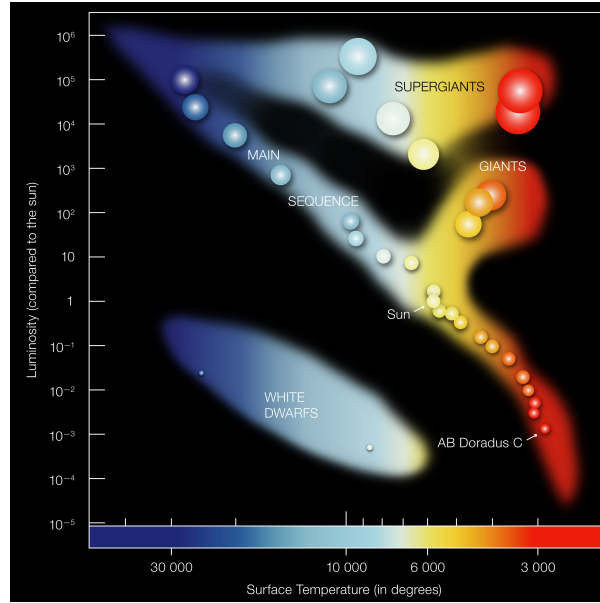


Figure 1.2: The Hertzsprung-Russell diagram [7]. Stars are classified by their temperature (or color) and luminosity (or absolute magnitude).

Luminosity class	Description
Ia	bright supergiants
Ib	supergiants
II	bright giants
III	giants
IV	subgiants
V	main sequences and dwarfs
VI	subdwarfs
VII	white dwarfs

Table 1.3: Luminosity classification.

diagram.

From a research perspective, such classifications are truly useful. As an analogy, consider how a mayfly might attempt to understand the human life cycle. One approach would be to observe people in a crowded place, such as a public square. Although the observation time is negligible compared to a human lifetime, the mayfly can infer the stages of human life by observing many individuals at different stages simultaneously. We can adopt an essentially identical strategy for astrophysical systems. Throughout our following discussions, we will encounter various examples of this approach.

1.2. Pre-Main-Sequence

What is a star? There exists a cliched Korean explanation of the definition of a star: the word “star” is phonetically rendered as “스타,” and the definition is described as “something that burns (타는 것; taneun geot) by itself (스스로; seuseuro).” In a physical sense, a star is an

astronomical object bound by its own gravity that releases energy through nuclear fusion.

A star spends most of its lifetime undergoing hydrogen fusion in its core; this stage is referred to as the main sequence (MS). The entire process leading up to the onset of stable hydrogen fusion at the stellar center, i.e., until the star reaches the zero-age main sequence (ZAMS), is called the pre-main-sequence (PMS), and corresponds to the formation stage of the star.

1.2.1. Gravitational Collapse of Interstellar Clouds

Stars form through the gravitational contraction of interstellar clouds composed of gas and dust. If an interstellar cloud is in hydrostatic equilibrium, it can remain stable without collapsing under gravity. However, if a small pressure perturbation does not dissipate but instead grows, hydrostatic equilibrium can no longer be maintained, and the cloud undergoes gravitational collapse. This condition is known as the Jeans criterion: if the mass of the cloud exceeds the Jeans mass M_J given below, the cloud collapses under its own gravity (for more details, see Section 2.5).

$$M_J = \left(\frac{375 k_B^3 T^3}{4\pi \rho G^3 \mu^3 m_p^3} \right)^{1/2}. \quad (1.8)$$

Here, μ is the mean molecular weight. (Note: In astrophysics, $\mu \equiv \frac{\rho}{nm_u}$ is dimensionless. See Section 2.3 for details.)

Once collapse begins, the interstellar cloud enters the PMS evolutionary phase. This process is quasi-static, because the dynamical timescale (τ_{ff}) is much shorter than the thermal timescale (τ_{KH}). These timescales are approximately given as follows.

$$\begin{aligned} \tau_{\text{ff}} &\simeq \left(\frac{3\pi}{32G\rho} \right)^{1/2}, \\ \tau_{\text{KH}} &\simeq \frac{GM^2}{2RL}. \end{aligned} \quad (1.9)$$

Thus, the evolution proceeds as a slowly evolving hydrostatic sequence, with the internal structure of the cloud continuously changing. During this stage, thermal equilibrium is maintained, but energy equilibrium is not.

As the cloud contracts under gravity, its internal temperature increases according to the virial theorem. For an object in hydrostatic equilibrium, the virial theorem gives the following relation between the gravitational potential energy E_g and the total internal energy E_i (for more details, see Section 2.4).

$$E_g + 3(\gamma - 1)E_i = 0. \quad (1.10)$$

Here, $\gamma \equiv \frac{c_p}{c_v}$ is the heat capacity ratio, and for a non-relativistic monatomic ideal gas, $\gamma \sim \frac{5}{3}$. Therefore, a decrease in E_g leads to an increase in E_i , ultimately raising the internal temperature.

1.2.2. Hayashi and Henyey Tracks

As an interstellar cloud contracts, its core temperature (T_c) gradually increases, and the internal structure of the protostar evolves accordingly. The evolutionary track on the H-R diagram is determined by the changes in T_{eff} and L . Since gravitational contraction reduces the stellar radius, the luminosity change is closely related to how T_{eff} evolves through Eq. (1.1). Therefore, the PMS evolution is controlled by the photospheric boundary condition and by whether the envelope below the photosphere is convective or radiative, which determines T_{eff} .

The local condition for convective motion to occur is determined by comparing the temperature gradient required for radiative transport with the adiabatic temperature gradient. In the absence of a composition gradient, the condition for convective motion to occur in the stellar interior is given by

$$\nabla_{\text{rad}} > \nabla_{\text{ad}}, \quad (1.11)$$

where $\nabla_{\text{ad}} \equiv \left(\frac{d \ln T}{d \ln P}\right)_s = 1 - \frac{1}{\gamma}$ is the adiabatic temperature gradient, and $\nabla_{\text{rad}} \equiv \left(\frac{d \ln T}{d \ln P}\right)_{\text{rad}} \propto \frac{\kappa L P}{M T^4}$ is the radiative temperature gradient. Here, κ is the photon opacity. A large opacity or a large local luminosity tends to increase ∇_{rad} , making convection more likely. For more details on radiative energy transport and convection, see Sections 2.7 to 2.9.

The photospheric boundary condition determines the outer boundary of the stellar structure. The photosphere is approximately defined with the layer where the optical depth $\tau \sim 2/3$. Combining this with hydrostatic equilibrium gives

$$P = \frac{2}{3} \frac{g}{\kappa}, \quad (1.12)$$

at the photosphere. This relation shows that the opacity near the surface affects the allowed value of T_{eff} . In particular, when the envelope is deeply convective, this condition is connected to an approximately adiabatic structure, so the allowed range of T_{eff} becomes narrow.

For low-mass protostars, the temperature is relatively low and the opacity in the envelope is large because the gas is not fully ionized. Therefore, convection tends to be efficient in the envelope, and low-mass protostars can be nearly fully convective. In a deep convective region, the temperature gradient is close to adiabatic, $\nabla \simeq \nabla_{\text{ad}}$, and the structure approximately follows an adiabat. Together with the photospheric boundary condition in Eq. (1.12), this makes T_{eff} change only weakly during contraction. As a result, the temperature remains approximately constant while the luminosity decreases, and the star moves vertically downward on the H-R diagram. This evolutionary path is known as the Hayashi track. In Fig. 1.3, the vertical blue trajectories correspond to the Hayashi track.

For massive protostars, T_c rises more rapidly, and the gas in the interior becomes more highly ionized. The opacity in the interior then decreases, so radiative energy transport becomes more efficient. The envelope is therefore generally more radiative, and the protostar is no longer close to a fully convective structure. In this case, T_{eff} increases during contraction, while the luminosity remains approximately constant or changes only slowly. Thus, the star

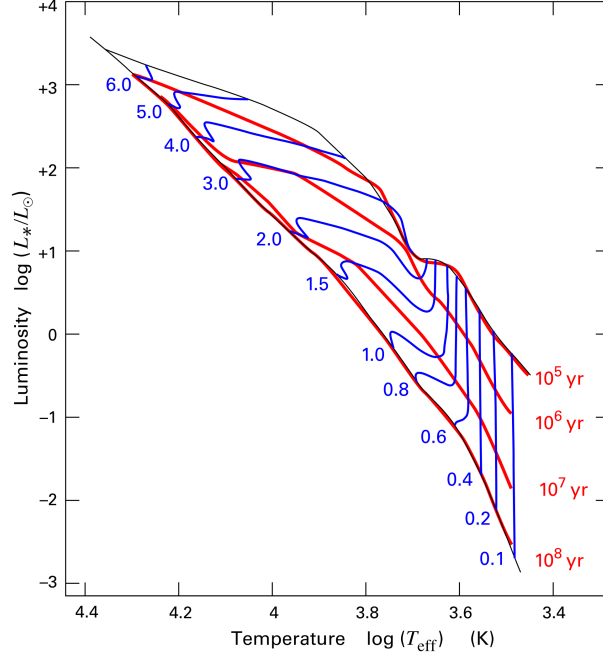


Figure 1.3: Evolutionary tracks on the H-R diagram during the pre-main-sequence phase for protostars of different masses [8]. The blue solid curves represent the evolution of protostars with the indicated masses (in units of $M_{\odot}$) on the H-R diagram, while the red solid curves indicate loci of constant age. The vertical segments of the blue curves correspond to the Hayashi track, and the horizontal segments correspond to the Henyey track.

moves horizontally toward the left on the H-R diagram. This path is called the Henyey track. In Fig. 1.3, the horizontal blue trajectories correspond to the Henyey track.

The evolutionary track followed during the PMS phase mainly depends on the mass of the protostar. Roughly, low-mass protostars reach the main sequence mainly through the Hayashi track, whereas massive protostars evolve mainly along the Henyey track after a very short Hayashi phase. For intermediate masses, the radiative core develops earlier and the convective envelope becomes less dominant as the mass increases. Therefore, the more massive the protostar, the shorter the time it spends on the Hayashi track.

Near the ZAMS, the internal structure also depends on the dominant nuclear reaction. Low-mass stars are mainly powered by the pp chain, whose temperature dependence is relatively weak, so they typically have radiative cores and convective envelopes. In contrast, massive stars are mainly powered by the CNO cycle, whose strong temperature dependence concentrates nuclear energy generation at the center, favoring the formation of a convective core and a radiative envelope.

1.2.3. Zero-Age Main Sequence

As the core temperature of the protostar reaches $T_c \sim 10^7$ K, and the energy production by nuclear reactions in the core balances the energy loss due to radiation and contraction, the protostar finally reaches the ZAMS and becomes a star.

During this process, the main energy source of the star shifts from gravitational contraction to nuclear burning, and nuclear reactions become the primary energy source that sustains the star. The protostar thus becomes a star and enters the main sequence. In general, when referring to the mass of a star, one uses the mass at ZAMS as the reference.

1.2.4. Failure of Hydrogen Ignition

For protostars with masses below approximately $0.08M_{\odot}$, electrons in the central region become degenerate before the core temperature reaches $T_c \sim 10^7$ K. Once electron degeneracy sets in, the increase in internal energy due to gravitational contraction no longer manifests as a rise in temperature, but instead as an increase in the Fermi energy of the electrons.

As a result, gravitational contraction can no longer raise the internal temperature of the protostar, preventing hydrogen ignition. In this case, the object does not enter the main sequence and instead becomes a brown dwarf.

1.3. Main Sequence

A star that successfully initiates hydrogen fusion in its core enters the main sequence. The MS is the evolutionary stage during which hydrogen fusion occurs in the core, and it is the phase in which a star spends the longest fraction of its lifetime.

Hydrogen fusion in the cores of MS stars proceeds primarily through two channels: the proton-proton (pp) chain and the CNO cycle. Which process dominates depends strongly on the stellar mass, as well as on the hydrogen mass fraction X , helium mass fraction Y , and metallicity $Z \equiv 1 - X - Y$. The dominant reaction channel determines the internal structure of the star.

1.3.1. Proton-Proton Chain

For stars with masses below approximately $1.3M_{\odot}$, the pp chain is the dominant fusion channel. The branches of the pp chain are summarized in Fig. 1.4.

The energy production rate per unit mass per unit time due to the pp chain, ϵ_{pp} , has the following approximate temperature dependence around $T \sim 1.5 \times 10^7$ K.

$$\epsilon_{pp} \propto \rho X^2 T_6^4, \quad (1.13)$$

where $T_6 \equiv T/10^6$ K.

1.3.2. CNO Cycle

For stars with masses above approximately $1.3M_{\odot}$, the CNO cycle dominates hydrogen fusion. Around $T \sim 1.7 \times 10^7$ K, the CNO cycle dominates at higher temperatures, while the pp chain

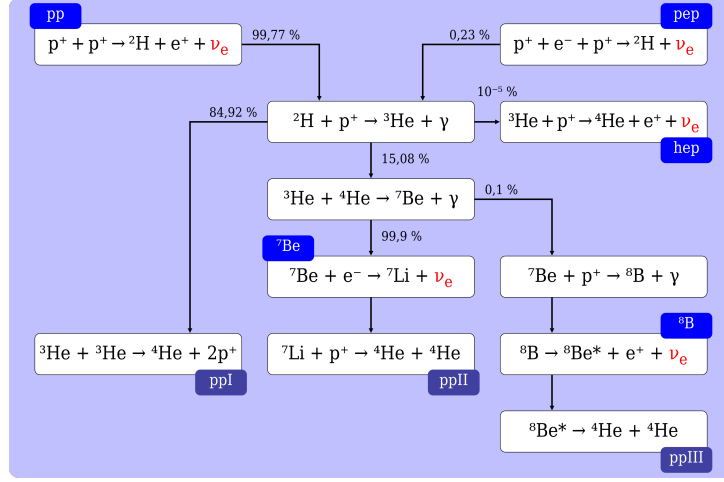


Figure 1.4: Branches of the proton-proton chain [9]. The indicated percentages represent the branching ratios in the Sun.

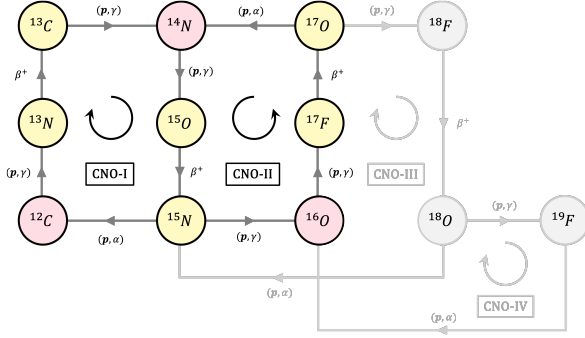


Figure 1.5: Cold CNO cycles [10]. CNO III and IV occur in massive stars.

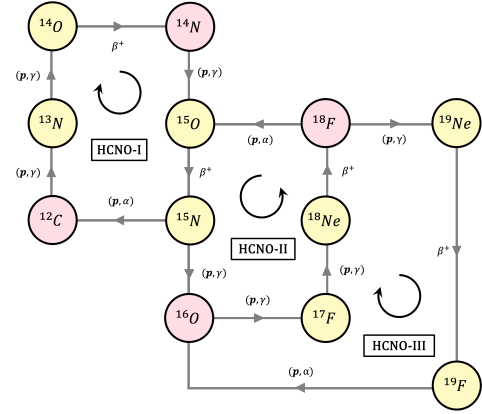


Figure 1.6: Hot CNO cycles [10]. These cycles do not occur in the main sequence stars but arise in extreme environments.

dominates at lower temperatures.

The CNO cycle is broadly divided into cold CNO cycles and hot CNO cycles; in general, “CNO cycle” refers to the cold CNO cycle. The hot CNO cycle occurs in high-temperature, high-density environments such as novae and X-ray bursts. The cold and hot CNO cycles are summarized in Fig. 1.5 and Fig. 1.6, respectively.

The energy production rate of the CNO cycle, ϵ_{CNO} , exhibits the following approximate temperature dependence around $T \sim 1.5 \times 10^7$ K.

$$\epsilon_{\text{CNO}} \propto \rho X X_{\text{CNO}} T_6^{19.9}. \quad (1.14)$$

Here, X_{CNO} is the total mass fraction of C, N, and O.

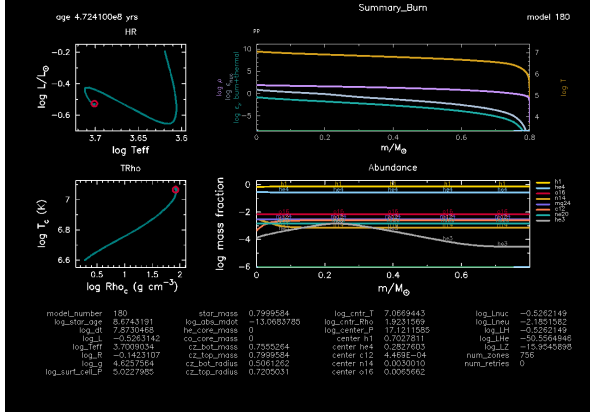


Figure 1.7: Stellar profiles near the ZAMS for $0.8M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. The pp chain dominates hydrogen burning, and the star has a radiative core and a convective envelope.

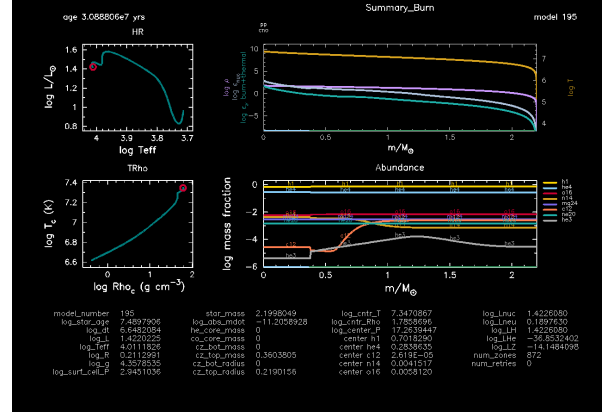


Figure 1.8: Stellar profiles near the ZAMS for $2.2M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. The CNO cycle dominates hydrogen burning, and the star has a convective core and a radiative envelope.

1.3.3. Inner Structure of Main Sequence Stars

From Eqs. (1.13) and (1.14), we find that the energy production rates per unit mass and per unit time for the pp chain and the CNO cycle exhibit different temperature dependences. As a result, in low-mass MS stars where the pp chain dominates, energy production from hydrogen fusion occurs over a relatively extended region, and the increase in luminosity toward the center is gradual. In contrast, in high-mass MS stars where the CNO cycle dominates, hydrogen burning is concentrated in the core region, leading to a steep increase in luminosity toward the center. Consequently, in the pp-chain-dominated case, ∇_{rad} remains small in the core, resulting in the formation of a radiative core, while in the outer layers, a convective region forms due to the still relatively large opacity κ . In the CNO-cycle-dominated case, ∇_{rad} increases sharply in the core, leading to the formation of a convective core, while the stellar envelope becomes radiative.

Figs. 1.7 and 1.8 show results from stellar evolution simulations using the Modules for Experiments in Stellar Astrophysics (MESA) code based on the test suite 1M_pre_ms_to_wd, presenting profiles near the ZAMS for stars with initial masses of $0.8M_{\odot}$ and $2.2M_{\odot}$ ¹. In both simulations, the composition $X = 0.718$, $Y = 0.268$, and $Z = 0.014$ was adopted. From the gray solid curve representing ϵ_{nuc} in the Summary_Burn plot (top right) in each figure, one can see that in the $2.2M_{\odot}$ star, energy production due to hydrogen burning is more strongly concentrated in the core region compared to the $0.8M_{\odot}$ star. The H-R diagram (top left) in Fig. 1.7 shows the clearly distinguishable Hayashi track and Henyey track, while in Fig. 1.8, the star evolves along the Henyey track after the extremely short Hayashi track.

¹If you would like to learn more about MESA and stellar evolution simulations using MESA, visit the official website: <https://docs.mesastar.org/>. In addition, refer to [12–17] for further details.

1.4. Evolution Toward Helium Ignition

As hydrogen is gradually depleted through nuclear fusion, hydrogen burning eventually ceases in the core. At this point, the main sequence phase of the star ends, so called terminal-age main sequence (TAMS). Subsequent stellar evolution proceeds along different paths depending on the stellar mass. In the context of stellar evolution, stars can be broadly classified according to their mass as follows.

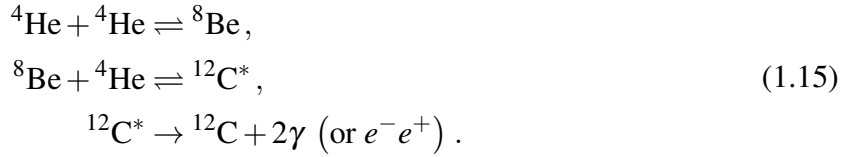
Category	Extremely low	Low	Intermediate	Massive
Mass range	0.08-0.5 $M_{\odot}$	0.5-2.0 $M_{\odot}$	2.0-8.0 $M_{\odot}$	8.0 $M_{\odot}$ -

Table 1.4: Stellar mass classification by evolutionary behavior.

The evolutionary path leading to helium ignition in the stellar core depends on the stellar mass, with different intermediate stages for different mass ranges.

1.4.1. Triple Alpha Process

Helium fusion proceeds via the triple-alpha process (as is well known, the alpha particle corresponds to ${}^4\text{He}$), producing ${}^{12}\text{C}$. The triple-alpha process can be divided into the following three stages.



The intermediate nucleus ${}^8\text{Be}$ produced in the first stage has a very short half-life, 8.19×10^{-17} s, and rapidly decays into two ${}^4\text{He}$ nuclei. As a result, the probability that a third ${}^4\text{He}$ nucleus interacts with ${}^8\text{Be}$ is small, making this step a bottleneck in the overall reaction. Consequently, the process can be approximated as a three-body interaction, rendering helium fusion intrinsically difficult to initiate.

However, the presence of two resonance states allows the triple-alpha process to proceed efficiently in stellar interiors. The first resonance occurs between two ${}^4\text{He}$ nuclei and ${}^8\text{Be}$. The ${}^8\text{Be}$ nucleus lies approximately 92 keV above the energy of two ${}^4\text{He}$ nuclei, and a narrow resonance exists between these states. As a result, the reaction ${}^4\text{He} + {}^4\text{He} \rightleftharpoons {}^8\text{Be}$ proceeds rapidly, and if ${}^4\text{He}$ is in thermal equilibrium with sufficiently high number density and temperature, ${}^8\text{Be}$ can exist in quasi-statistical equilibrium (QSE) with two ${}^4\text{He}$ nuclei. This enables the intermediate state of the triple-alpha process to exist with non-negligible probability, although it is not sufficient by itself to produce significant amounts of ${}^{12}\text{C}$.

The second resonance occurs between ${}^8\text{Be} + {}^4\text{He}$ and the excited state ${}^{12}\text{C}^*$. The Hoyle state, ${}^{12}\text{C}^*$, is an excited state of ${}^{12}\text{C}$, specifically the 0_2^+ state, located about 7.654 MeV above the ground state 0_1^+ . This state lies approximately 0.285 MeV above the ${}^8\text{Be} + {}^4\text{He}$ threshold, allowing ${}^{12}\text{C}^*$ to be produced efficiently via resonance. Thus, the triple-alpha process effec-

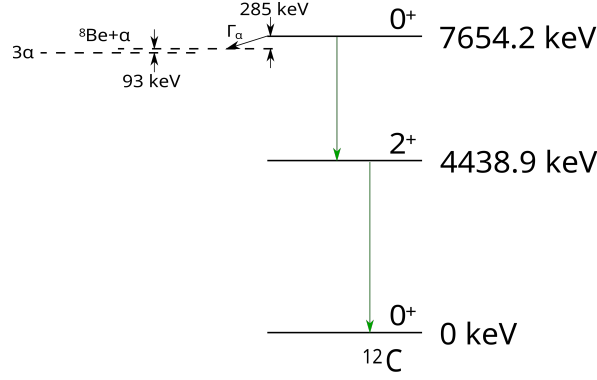


Figure 1.9: Hoyle state and the possible transition states [19]. The two resonance states shown on the left side of the figure play a crucial role in the triple-alpha process.

tively proceeds through successive two-body reactions mediated by resonances, rather than a direct three-body interaction.

The produced $^{12}\text{C}^*$ is unstable, with a half-life of about 2.4×10^{-16} s. Most $^{12}\text{C}^*$ nuclei decay back into three ^4He nuclei via $^8\text{Be} + ^4\text{He}$. However, approximately 0.04% decay to the ground state of ^{12}C . In this process, about 98% proceed via two-photon emission, while about 2% proceed via electron-positron pair emission [18]. The resonance structure associated with the Hoyle state and the energy levels of ^{12}C are summarized in Fig. 1.9.

Examining the individual steps of the triple-alpha process given in Eq. (1.15), Fig. 1.9 shows that the first two reactions are endothermic. This highlights the difficulty of initiating helium fusion.

The few portions of produced ^{12}C undergo the following reaction to form ^{16}O .



The triple-alpha process occurs at temperatures $T \gtrsim 10^8$ K in helium-rich environments. The energy production rate per unit mass per unit time, $\epsilon_{3\alpha}$, exhibits the following approximate temperature dependence near $T \sim 10^8$ K.

$$\epsilon_{3\alpha} \propto \rho^2 Y^3 T_8^{41}, \quad (1.17)$$

where $T_8 \equiv T/10^8$ K. This reflects a much stronger and more sensitive temperature dependence compared to the pp chain or the CNO cycle.

1.4.2. Extremely Low Mass Stars; Failure to Ignite Helium

Stars with masses below $0.5 M_\odot$ remain on the main sequence for times longer than the current age of the Universe; therefore, no stars with masses below $0.5 M_\odot$ that have evolved off the main sequence have been observed to date. Based on modern physics, it is expected that stars in this mass range cannot reach the temperatures required for helium ignition.

As a result, these stars are predicted to enter a post main sequence phase characterized by

hydrogen burning in the outer layers, evolving into a hypothetical stage known as a blue dwarf. After exhausting their hydrogen, they are expected to become helium white dwarfs.

1.4.3. Low Mass Stars; Helium Flash

Once hydrogen fusion in the core ceases, the He-rich core gradually contracts, leading to an increase in temperature and density. As the internal temperature rises, hydrogen fusion begins in a surrounding H-rich shell. The phase up to this point is referred to as the subgiant stage, and once hydrogen fusion starts in the H-rich shell, the star is said to have entered the red giant (RG) phase.

Hydrogen fusion in the H-rich shell causes the stellar envelope to expand. As a result, the surface temperature decreases while the luminosity increases. Due to these characteristics, RG stars move along a path on the H-R diagram known as the red giant branch (RGB). The outer envelope of RG stars develops a convective outer shell due to increased opacity. When convection becomes efficient, nuclear burning products (heavier elements) from the interior are transported outward. This alters the chemical composition of the star, leading to a temporary decrease in the nuclear reaction rate and a corresponding temporary decrease in luminosity. This process is called the first dredge-up (FDU), and as a result, a feature known as the RGB bump appears, where stars accumulate at a characteristic luminosity along the RGB.

While the star remains on the RGB, the core continues to contract, and both temperature and density increase. In low-mass stars, electron degeneracy sets in before the temperature required for helium fusion is reached. Electron degeneracy has two important consequences. First, it increases the conductivity, making energy transport by radiation (In the astrophysical context, both radiation and conduction can be described as diffusive processes; accordingly, radiative energy transport generally includes conduction. See Section 2.7 for details.) more efficient. As a result, the core develops an approximately isothermal temperature profile. Second, it enhances neutrino emission, leading to efficient cooling, particularly in the central regions. Due to these two effects, the He-rich core of an RG becomes nearly isothermal, with the maximum temperature occurring off-center.

As seen from Eq. (1.17), the triple-alpha process has a strong temperature dependence. Consequently, helium ignition in low-mass stars occurs off-center, leading to a thermal runaway known as the helium flash.

The helium flash is an explosive event that occurs in the final stage of the red giant phase of low-mass stars, where a degenerate core undergoes off-center thermal runaway helium fusion. During the flash, the outer core releases, over a few seconds, an amount of energy comparable to the total luminosity of all stars in a galaxy, causing a rapid increase in temperature in that region. However, this energy is not directly observable, as it is primarily used to lift electron degeneracy within the star.

Figs. 1.10 and 1.11 show results from stellar evolution simulations using MESA code based on the test suite 1M_pre_ms_to_wd, presenting profiles before the He flash and at the first He

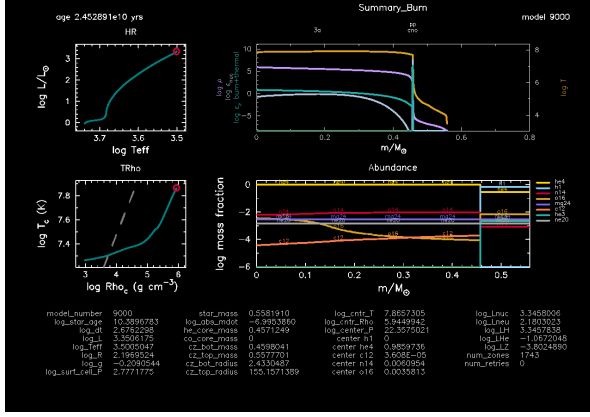


Figure 1.10: Stellar profiles before the first helium flash for $0.8 M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. In the outer regions of the star, H-shell burning is ongoing. The maximum temperature is located off-center.

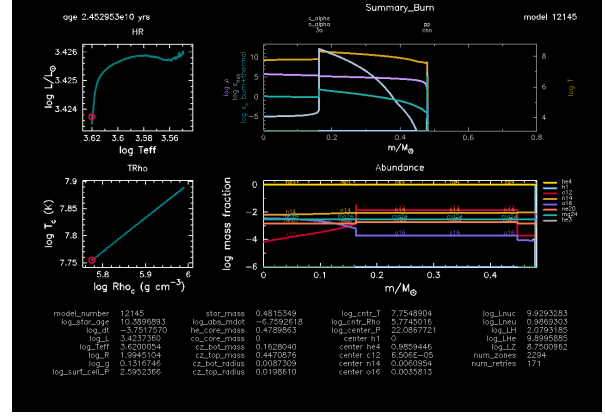


Figure 1.11: Stellar profiles at the first helium flash for $0.8 M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. A rapid onset of helium fusion in the outer core leads to the release of an enormous amount of energy in a short time, resulting in a sharp increase in temperature.

flash for stars with initial masses of $0.8 M_{\odot}$. From the yellow solid curve representing T in the Summary_Burn plot (top right) in the left panel, one can see that prior to the He flash, the core is nearly isothermal due to electron degeneracy, while neutrino cooling shifts the location of the maximum temperature to an off-center region. From the gray solid curve representing ϵ_{nuc} in the Summary_Burn plot (top right) in the right panel, it is evident that the He flash triggers a substantial release of energy through helium fusion. As a result, the temperature in the outer core increases, as shown by the yellow solid curve.

After several helium flashes, the internal structure of the star gradually stabilizes. During this process, the star contracts, its temperature increases, and its luminosity decreases, producing a distinct jump on the H-R diagram. The point of maximum luminosity along the RGB is called the tip of the red giant branch (TRGB), and since this luminosity is nearly constant, it is used as a standard candle for distance measurements in astronomy.

Following the helium flash, stable helium fusion begins at the center, forming a convective core. This stage is referred to as the zero-age horizontal branch (ZAHB). The star then resides on the horizontal branch (HB) of the H-R diagram, characterized by nearly constant luminosity but a wide spread in temperature. During this phase, as time progresses, the luminosity increases slightly while the temperature decreases, causing the star to move roughly horizontally to the right, with a slight upward trend on the H-R diagram.

When core helium is depleted and helium fusion ceases at the center, the star evolves into the asymptotic giant branch (AGB) phase. Fig. 1.12 shows a CMD of stars in the globular cluster Messier 5, classified into RGB, HB, and AGB populations. It provides a representative example of the evolutionary behavior of low-mass stars on the H-R diagram.

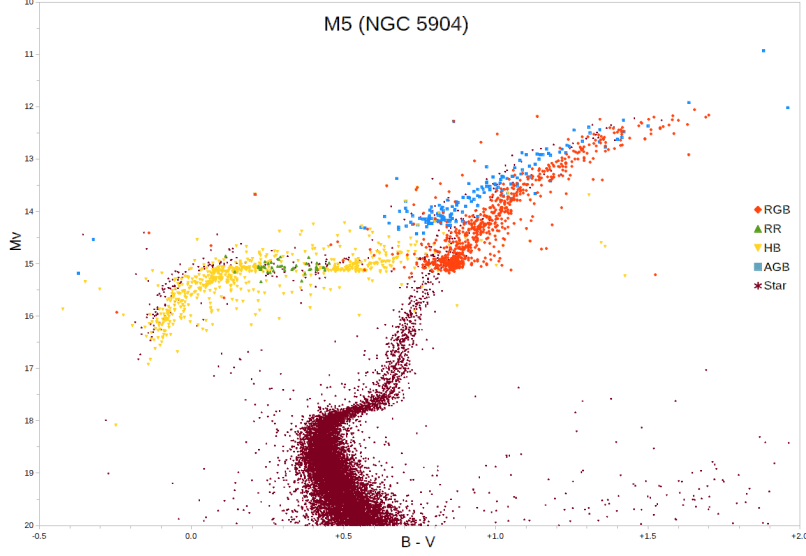


Figure 1.12: Color-magnitude diagram of the globular cluster Messier 5 [20]. The red points correspond to the late main sequence, subgiant branch, and lower RGB; the orange points correspond to the upper RGB; the yellow points correspond to the HB; the green points correspond to RR Lyrae variables; and the light-blue points correspond to the AGB.

1.4.4. Intermediate Mass Stars; Core Ignition

Intermediate-mass stars, like low-mass stars, enter the RGB after the TAMS. However, in intermediate-mass stars the outer convective zone formed during the RG phase is relatively shallow, so the first dredge-up (FDU) is weaker than in low-mass stars.

The most important difference is that, in intermediate-mass stars, helium fusion ignites in the core before electron degeneracy develops. Consequently, no helium flash occurs, and core He ignition takes place before reaching the TRGB. As a result, the pronounced TRGB-ZAHB gap seen for low-mass stars on the H-R diagram does not appear for intermediate-mass stars.

After He ignition, the H-R diagram behavior of intermediate-mass stars depends on mass. Stars with masses of $2.0\text{--}2.5M_{\odot}$ enter the red clump (RC), characterized by small dispersions in both temperature and luminosity; during the core He-burning phase, they maintain nearly constant T and L . Fig. 1.13 shows an H-R diagram based on GALAH data, illustrating how the red clump (RC) appears on the RGB.

Stars with masses of $2.5\text{--}8.0M_{\odot}$ exhibit a characteristic blue loop. Immediately after core He ignition, H-shell burning remains strong, and the star is classified as a cool red giant. As He burning proceeds, the He core becomes more compact and the luminosity of the H-burning shell decreases; the envelope contracts and the temperature increases. The star then moves leftward on the H-R diagram and becomes a blue giant. As contraction raises the temperature of the H-rich shell, H burning strengthens again, the envelope re-expands, the temperature decreases, and the luminosity increases slightly, returning the star to the red giant region.

After depleting core He and ceasing central He fusion, both RC stars and those that have undergone blue-loop evolution proceed to the AGB phase.

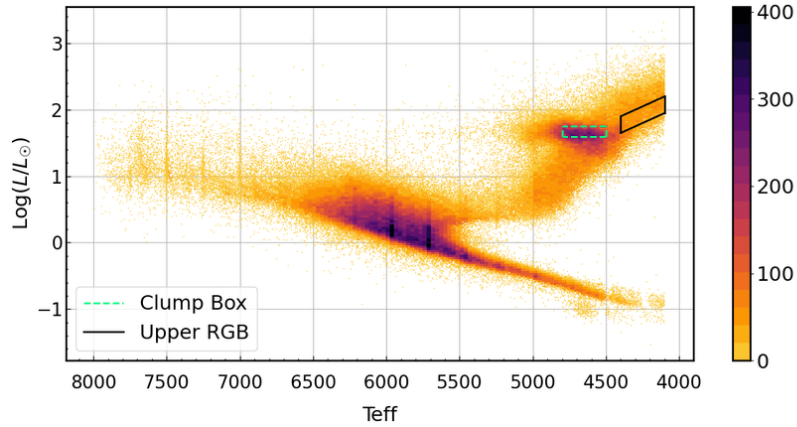


Figure 1.13: H-R diagram based on GALAH data [21]. Darker colors indicate regions populated by a larger number of stars, while lighter (yellow) colors correspond to regions with fewer stars. The cyan dashed box marks the approximate location of the red clump, where a higher concentration of stars is evident compared to other regions along the RGB.

1.4.5. Massive Stars; Form of the Red Supergiant

Massive stars with masses above about $8.0 M_{\odot}$ begin H-shell burning almost immediately after core H burning ends, causing the envelope to expand rapidly and the star to become a red supergiant (RSG). On the H-R diagram, the behavior of RSGs is similar to that of the RGB, but with higher luminosities and lower temperatures; that is, they have larger radii.

Upon entering the RSG phase, the increasing core temperature leads to helium ignition in a non-degenerate He core. Unlike low-mass stars, this does not produce a pronounced change on the H-R diagram. This is primarily because He burning begins under non-degenerate conditions, as in intermediate-mass stars, and additionally because of the influence of the extended, massive H envelope.

As with intermediate-mass stars, the post-He-ignition evolution of massive stars depends on mass. During the core He-burning phase, their luminosities remain approximately constant but tend to increase gradually. Stars with masses below about $20 M_{\odot}$ may exhibit temperature variations similar to the blue loops of intermediate-mass stars; however, their evolution is more restricted due to their massive envelopes.

For stars with masses above about $20 M_{\odot}$, mass loss through stellar winds becomes significant, leading to a continuous increase in temperature. As a result of this irreversible blueward evolution, the star may become a blue supergiant (BSG). In very massive cases, extensive mass loss strips away most of the outer layers, exposing the core and producing a Wolf-Rayet (WR) star.

After core He is depleted and He fusion ceases, stars with masses below approximately $10 M_{\odot}$ enter the super-AGB phase. For stars with masses above about $10 M_{\odot}$, carbon ignition begins in the core.

1.5. Final Destiny of Low-Mass and Intermediate-Mass Stars

So far, we have classified stellar evolution using three main criteria:

1. Whether the next stage of nuclear fusion is possible.
2. Where the next stage of fusion ignites.
3. The evolutionary behavior on the H-R diagram.

For post-MS evolution, if the first is not satisfied, the star belongs to the class of extremely low-mass stars; if the first is satisfied and the second is off-center, the star is a low-mass star; if the first is satisfied and the second occurs in the core, the star is an intermediate-mass or massive star. According to the third criterion, these are further classified into RGB, RSG, and related evolutionary phases.

A similar classification can be applied after the cessation of core He fusion. AGB, which is reached by low mass and intermediate mass stars, corresponds to a stage in which the next fusion process, carbon burning, cannot occur.

1.5.1. Early Asymptotic Giant Branch

The situation after core He burning is analogous to that prior to the RGB phase. The C-O core formed by helium fusion is initially too cool to ignite carbon fusion and therefore contracts, leading to an increase in temperature. As a result, a layered structure forms outside the core, consisting (from the inside outward) of a He-burning shell, an inter-shell region, an H-burning shell, and a convective envelope. On the H-R diagram, the star moves toward lower temperatures and higher luminosities, similar to the RGB, i.e., toward the upper-right. At this stage, the star is said to have entered the AGB phase.

At the early AGB stage, for stars with masses above approximately $3-4 M_{\odot}$, the high opacity of the greatly expanded envelope allows convection to extend deep into the stellar interior. As a result, material from the inter-shell region between the He-burning and H-burning shells is dredged up into the envelope, mixing the chemical composition and reducing the core mass. This process is known as the second dredge-up (SDU).

Fig. 1.14 show results from stellar evolution simulation using MESA code based on the test suite 1M_pre_ms_to_wd, presenting profiles around the early-AGB for a star with initial mass of $1.0 M_{\odot}$. In this simulation, the composition $X = 0.718$, $Y = 0.268$, and $Z = 0.014$ was adopted. From the gray solid curve representing ϵ_{nucl} in the Summary_Burn plot (top right) in the figure, helium fusion is taking place in the inner shell, hydrogen fusion in the outer shell, and an inter-shell region is present in between.

1.5.2. Thermal Pulsing Asymptotic Giant Branch

After entering the AGB phase, the contracting C-O core becomes electron-degenerate before reaching the temperature required for carbon ignition, and begins to cool via neutrino emission.

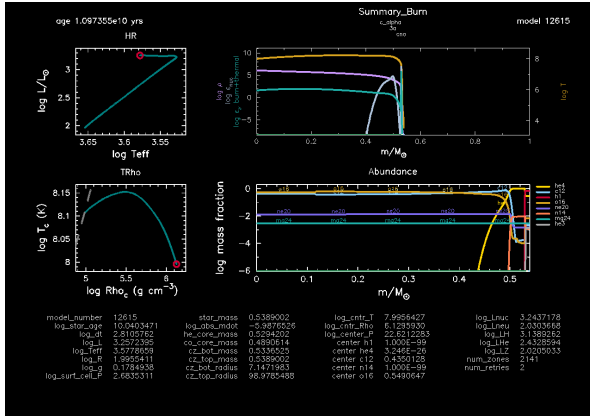


Figure 1.15: Stellar profiles around the post-AGB for $1.0M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. Due to mass loss from the envelope, the total stellar mass decreases, and this is reflected as a leftward shift on the H-R diagram during the post-AGB evolution.

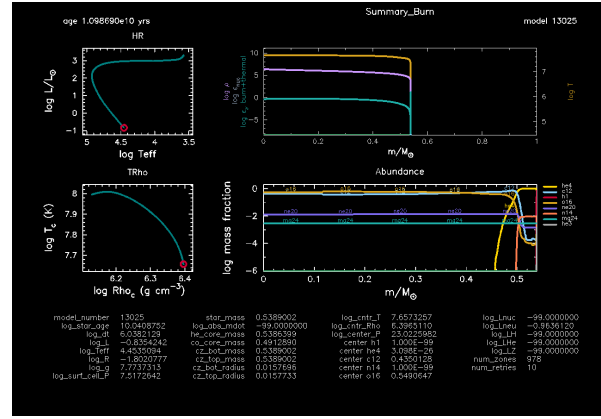


Figure 1.16: Stellar profiles of the C-O white dwarf for $1.0M_{\odot}$ star based on MESA test suite 1M_pre_ms_to_wd [11]. The C-O white dwarf, no longer capable of nuclear fusion, continues to cool via neutrino emission, moving toward the lower-right on the H-R diagram as its temperature decreases and its density increases.

1.5.3. White Dwarf and Planetary Nebula

After losing most of its envelope during the TP-AGB phase, the star is left with a non-burning C-O core surrounded by a thin shell. Upon entering the post-AGB phase, the stellar radius decreases rapidly, while the luminosity is temporarily maintained by the core luminosity and residual shell burning. During this stage, the luminosity remains nearly constant while the temperature increases sharply, causing the star to move leftward on the H-R diagram.

At the same time, the high temperature of the exposed C-O core heats the previously ejected envelope, forming dust and ionizing the surrounding material. As a result, a planetary nebula is formed around the star.

Eventually, shell burning also ceases, and the star can no longer sustain nuclear fusion. The remaining C-O core, having a mass below the Chandrasekhar limit, is stabilized by electron degeneracy pressure. This remnant becomes a C-O white dwarf (WD), which cools over time, primarily through neutrino emission. As it cools, both luminosity and temperature decrease, and the star moves toward the lower-right region of the H-R diagram.

Figs. 1.15 and 1.16 show results from stellar evolution simulations using MESA code based on the test suite 1M_pre_ms_to_wd, presenting profiles around the post-AGB and the C-O WD stage for stars with initial masses of $1.0M_{\odot}$. In Fig. 1.15, the mass loss during the TP-AGB phase due to pulsations can be identified, showing the reduction in stellar mass. On the H-R diagram, this corresponds to a horizontal leftward motion, while the core temperature decreases due to neutrino cooling and the density continues to increase. In Fig. 1.16, a cooling C-O white dwarf can be seen, steadily losing thermal energy via neutrino emission. Its trajectory on the H-R diagram proceeds toward the lower-right, with decreasing core temperature and increasing

density.

1.6. Final Destiny of Massive Stars

In massive stars, carbon fusion does occur; however, the subsequent evolutionary pathways differ significantly depending on the stellar mass. The post-core He-burning evolutionary stages for massive stars of different masses are summarized in Table 1.5.

Mass range	8.0-10 $M_{\odot}$	10-70 $M_{\odot}$	70 $M_{\odot}$ -
Evolution stage	super-AGB	Advanced fusion	Pair instability

Table 1.5: Post-core He-burning evolutionary stages categorized by stellar mass.

1.6.1. Super Asymptotic Giant Branch

Stars with masses in the range 8.0-10 $M_{\odot}$ enter the super-AGB phase after core helium burning ceases. This phase is broadly similar to the AGB phase experienced by low-mass and intermediate-mass stars, but with two important differences:

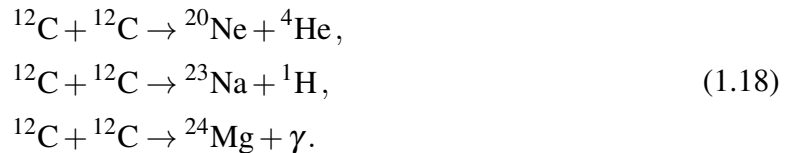
1. Carbon fusion can ignite off-center.
2. Depending on the evolution of the core mass during the super-AGB phase, the star may end its life either as an O-Ne-Mg WD or as an electron-capture supernova (ECSN).

Recalling the AGB evolution discussed in Section 1.5, we focus on the aspects in which the super-AGB phase differs from the standard AGB evolution.

The first key difference is the occurrence of off-center carbon ignition, which arises due to the formation of an electron-degenerate C-O core. Once off-center carbon fusion ignites, the star can be regarded as having entered the super-AGB phase. The energy released by this localized carbon burning propagates inward, leading to successive carbon ignition within the core. This propagating burning front is referred to as the carbon flame.

An important point is that carbon burning does not occur simultaneously throughout the entire core. Instead, a localized carbon-burning region forms off-center and gradually propagates toward the center. This is illustrated in Fig. 1.17, which shows the internal structure of a super-AGB star undergoing carbon burning. Thus, carbon fusion proceeds in the form of a moving burning front rather than as a global core process. The effects of carbon burning, similar to the case of the helium flash, do not produce a prominent signature on the H-R diagram. During this stage, a super-AGB star exhibits behavior similar to that of the early-AGB phase.

The dominant reactions for carbon fusion are given below.



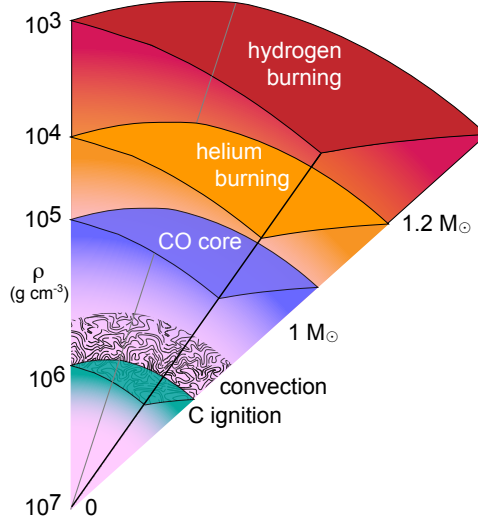


Figure 1.17: Illustration of the structure of a super-AGB star during carbon burning [22]. The carbon flame formed off-center gradually propagates toward the center, forming an O-Ne-Mg core.

As a result of the propagation of the carbon flame and following carbon burning, the composition of the C-O core in a super-AGB star is progressively transformed, ultimately leading to the formation of an O-Ne-Mg core. After carbon burning ceases, the internal structure of a super-AGB star becomes similar to that of an AGB star, with the key difference being that the core is composed of O-Ne-Mg rather than C-O.

The second important difference lies in the final destiny of the star, which depends on the evolution of the core mass. The processes that vary the core mass during the super-AGB phase are as follows.

1. SDU: Convection transports heavier elements outward and lighter elements inward, leading to a decrease in the core mass. In general, SDU is stronger in super-AGB stars than in AGB stars.
2. H/He-shell burning: Heavy elements produced by H and He-shell burning accumulate onto the core, increasing the core mass.
3. Thermal pulsations: During thermal pulses, the envelope loses mass continuously, eventually halting the growth of the core mass due to shell burning. In super-AGB stars, thermal pulsations are characterized by higher luminosities, shorter interpulse periods, and a larger number of pulses compared to AGB stars.

The fate of a super-AGB star depends on whether the core mass reaches the Chandrasekhar mass before the envelope is mostly lost. The Chandrasekhar mass is given by

$$M_{\text{Ch}} = 5.81 Y_e^2 M_{\odot}. \quad (1.19)$$

The point is, the Chandrasekhar mass depends on the electron fraction Y_e . Thus, whether the core mass reaches the Chandrasekhar mass is not determined solely by the increase of the core

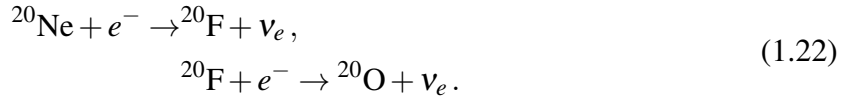
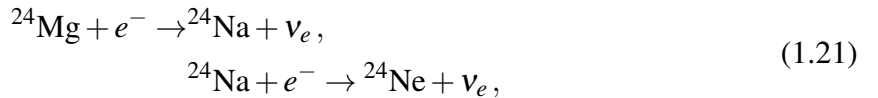
mass itself, but also depends on the evolution of Y_e in the core. In general, the reduction of the Chandrasekhar mass plays a more dominant role than the growth of the core mass. For a C-O core, the Chandrasekhar mass is the well-known value $M_{\text{Ch}} \simeq 1.44 M_{\odot}$, whereas for an O-Ne-Mg core it is approximately $M_{\text{Ch}} \simeq 1.38 M_{\odot}$.

If the envelope is removed before the core mass reaches the Chandrasekhar mass, the super-AGB star evolves similarly to an AGB star, forming a white dwarf and a planetary nebula. On the H-R diagram, it follows a similar path: first moving horizontally to the left, and then toward the lower-right as it cools via neutrino emission. The key difference from the AGB case is that the remnant is an O-Ne-Mg white dwarf.

Conversely, if the core mass reaches the Chandrasekhar mass before the envelope is lost, electron degeneracy pressure can no longer support hydrostatic equilibrium, and the core undergoes collapse. This process is driven by electron capture. After carbon burning ceases, the core continues to contract, increasing the Fermi energy of degenerate electrons. As a result, electron capture onto nuclei becomes energetically allowed.

$$e^{-} + (Z, A) \rightarrow (Z - 1, A) + \nu_e. \quad (1.20)$$

This process reduces the electron fraction Y_e , thereby decreasing the Chandrasekhar mass given in Eq. (1.19). In O-Ne-Mg cores, electron capture proceeds through the Mg chain and the Ne chain [23],



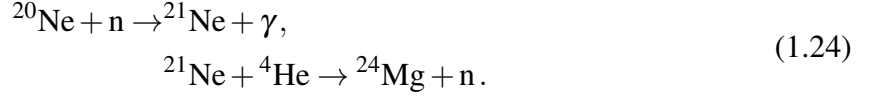
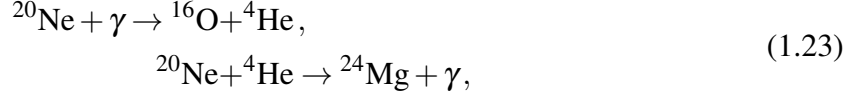
Through these reactions, Y_e decreases continuously, reducing the Chandrasekhar mass, while the emitted ν_e cool the core. Thus, electron capture accelerates the collapse. Although local heating from the de-excitation of daughter nuclei may ignite oxygen fusion, the collapse driven by electron capture is generally not halted, and oxygen burning cannot support the core.

As the collapse proceeds, the core density increases and approaches the nuclear saturation density, $\rho_{\text{sat}} \sim 10^{14} \text{ g/cm}^3$. At this point, the core undergoes bounce, forming a shock wave that leads to a supernova explosion. Such an explosion, triggered and dominated by electron capture, is referred to as an ECSN, leaving behind a neutron star (NS) at the center. The explosion energy of an ECSN is typically $E_{\text{ECSN}} \sim \mathcal{O}(10^{50})$ erg, which is weaker than that of a typical iron-core supernova.

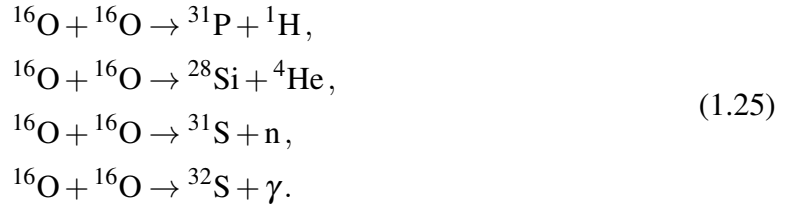
1.6.2. Advanced Fusion

Stars with masses of approximately $10\text{-}70 M_{\odot}$ ignite carbon fusion in cores where electrons are non-degenerate or only partially degenerate. These stars then evolve through C, Ne, O, and Si burning in sequence, eventually forming an Fe core.

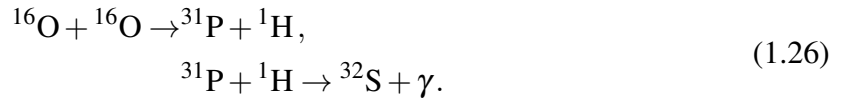
The main C-burning phase, dominated by the processes shown in Eq. (1.18), lasts for about $\mathcal{O}(10^3)$ yr. The main processes in the Ne-burning phase, which lasts for about $\mathcal{O}(1)$ yr, are as follows.



The first process involves photodisintegration, while both processes involve α capture. Thus, Ne burning is not fusion of Ne, but rather a process of photodisintegration and nuclear rearrangement. After Ne burning, O burning begins, with the main processes given as follows.

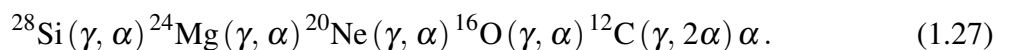


The daughter particles produced by these reactions are further transformed into other nuclei through p , n , and α captures. As a result, about 90% of the products become ${}^{28}\text{Si}$ and ${}^{32}\text{S}$, with additional production of Cl, Ar, K, and Ca [24]. For example, the process by which ${}^{31}\text{P}$ produced in the first reaction of Eq. (1.25) ultimately forms ${}^{32}\text{S}$ is given by



This process can also be written as ${}^{16}\text{O}({}^{16}\text{O}, p){}^{31}\text{P}(p, \gamma){}^{32}\text{S}$, and this notation has also been used in Figs. 1.5 and 1.6. The O-burning phase lasts for about $\mathcal{O}(1)$ month.

Si burning, which follows O burning, is not a set of specific reactions in the same sense as H, He, or C fusion. Rather it is a process in which nuclear matter in quasi-statistical equilibrium (QSE) evolves rapidly toward nuclear statistical equilibrium (NSE). In other words, detailed balance, which previously held only within individual clusters of nuclei, gradually comes to hold among all nuclei composing the core. Just before Si burning, nuclei with mass numbers $A = 24$ -46 and heavier nuclei form separate clusters, each internally in QSE. During Si burning, these clusters merge into a single cluster. Representative photodisintegration processes during Si burning produce p , n , and α particles as follows.



These processes are in equilibrium with their inverse reactions. The produced p , n , and α particles are captured by heavier nuclei, leading to the formation of nuclei such as ${}^{56}\text{Fe}$ and ${}^{56}\text{Ni}$. Si burning lasts for about $\mathcal{O}(1)$ day.

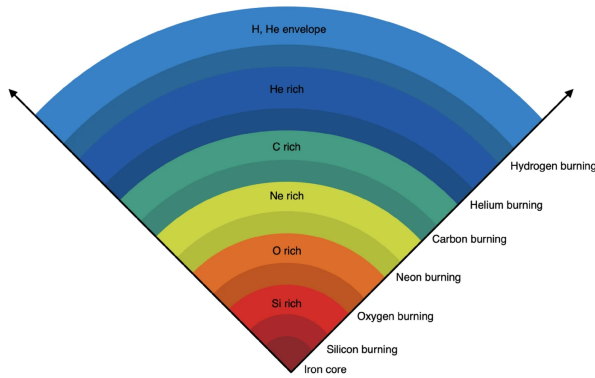


Figure 1.18: Internal structure of a star at the stage when an iron core has formed through advanced fusion [25]. Regions where fusion of each nuclear species occurs alternate with regions enriched in the corresponding nuclear species. This layered structure is called the onion structure.

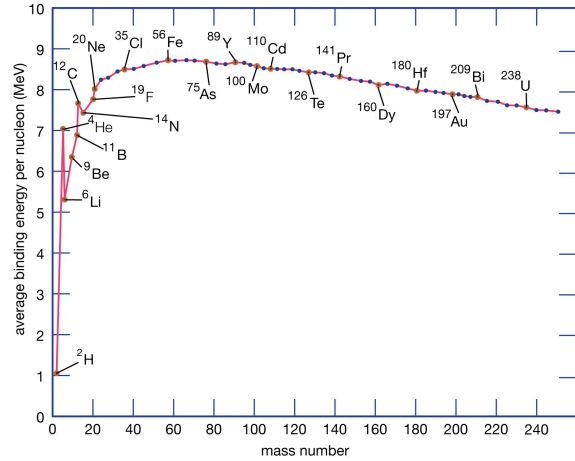


Figure 1.19: Nuclear binding energy per nucleon as a function of atomic mass number [26]. Nuclear reactions proceeding toward higher binding energy per nucleon are exothermic, while those proceeding toward lower binding energy per nucleon are endothermic. Once ^{56}Fe , which has the highest binding energy per nucleon, is produced, further fusion into heavier elements can no longer support the stellar structure.

From C fusion until the end of Si burning, the star barely moves on the H-R diagram. This is because the energy generated during advanced burning stages is lost through neutrinos, and the thermal timescale is longer than the burning timescale, so changes in the stellar interior are not directly reflected at the stellar surface.

A star that has formed an Fe core through advanced fusion develops an onion structure, as shown in Fig. 1.18. As can be seen from the binding energy per nucleon plot in Fig. 1.19, ^{56}Fe , or more precisely ^{62}Ni , is among the most stable nuclei; fusion into heavier nuclei is endothermic rather than exothermic. Therefore, nuclear fusion can no longer support the structure inside the Fe core. At this time, the Fe core is supported by electron degeneracy pressure and thermal pressure. Since the core is composed of heavy isotopes and has a low electron fraction, the Chandrasekhar mass is approximately $M_{\text{Ch}} \simeq 1.25 M_{\odot}$ according to Eq. (1.19).

Once the Fe core forms and enters the initial stage of collapse, mass accretion from the Si-burning shell and electron capture reduce the electron fraction, causing the core to contract gradually and its temperature to increase. As the core temperature rises, photodisintegration of iron occurs as follows.



This reaction is strongly endothermic, absorbing about 124.4 MeV per reaction, and accelerates the collapse together with electron capture. This is one of the important differences between core-collapse supernova (CCSN) and ECSN, which are triggered and dominated by electron capture.

During collapse, when the density reaches about 10^{12} g/cm^3 , neutrinos become trapped. When the density reaches the nuclear saturation density, about 10^{14} g/cm^3 , the EOS stiffens, and nuclear matter becomes less compressible. Infalling matter from the outer layers accumulates at this surface, and the increased pressure of nuclear matter causes the core to bounce. A shock forms and begins to propagate outward, converting the collapse process into an explosion process.

This shock dissociates material along its propagation path into free protons and neutrons, losing energy in the process and eventually stalling. In this stage, due to the strong gravitational potential of the star, matter outside the shock falls inward at $\mathcal{O}(1-10)\%$ of the speed of light. Neutrino from the proto-neutron star (PNS), together with effects such as strong magnetic fields and stellar rotation, can reheat the stalled shock. After this point, the evolution can branch into three main possibilities.

The first possibility is that the shock is successfully revived and propagates outward through the star, producing an explosion. In this case, the central PNS successfully forms a neutron star, and a type II supernova occurs. Unlike type I supernovae, type II supernovae show hydrogen spectral features. In this case, the explosion energy is about 10^{52} erg , and the time from iron-core formation to shock revival is about $\mathcal{O}(0.1) \text{ s}$.

The second possibility is that the shock is revived, but some of the material falls back. In this case, a supernova explosion still occurs, but the fallback material accretes onto the PNS, causing it to exceed the critical mass and form a black hole (BH). A BH formed in this way is called a fallback BH.

The third possibility is that shock revival fails. In this case, the shock fails to eject the stellar envelope, and the supernova explosion fails. The shock that fails to revive falls back, forming a BH at the center; this case corresponds to a failed SN.

These branches are determined by whether the shock is revived and ejects the stellar envelope, as well as by the amount of material that falls back toward the center. Thus, lower-mass stars tend to follow the first branch, while higher-mass stars tend to follow the third branch, although this trend is not absolute.

1.6.3. Pair Instability

Stars with masses heavier than approximately $70 M_{\odot}$ can undergo distinctive phenomena due to pair instability. The O-rich core formed after He burning is in an environment dominated by radiation pressure. When radiation pressure dominates, the radiation energy density satisfies $u \propto T^4$ by the Stefan-Boltzmann law, and the isotropic distribution of photons gives $p = u/3$. Therefore, from the first law of thermodynamics given by Eq. (2.28), since there is no heat exchange between the system and its surroundings during an adiabatic process, the following relation holds.

$$0 = dU + pdV \propto d(VT^4) + \frac{T^4}{3}dV = VdT^4 + \frac{4}{3}T^4dV. \quad (1.29)$$

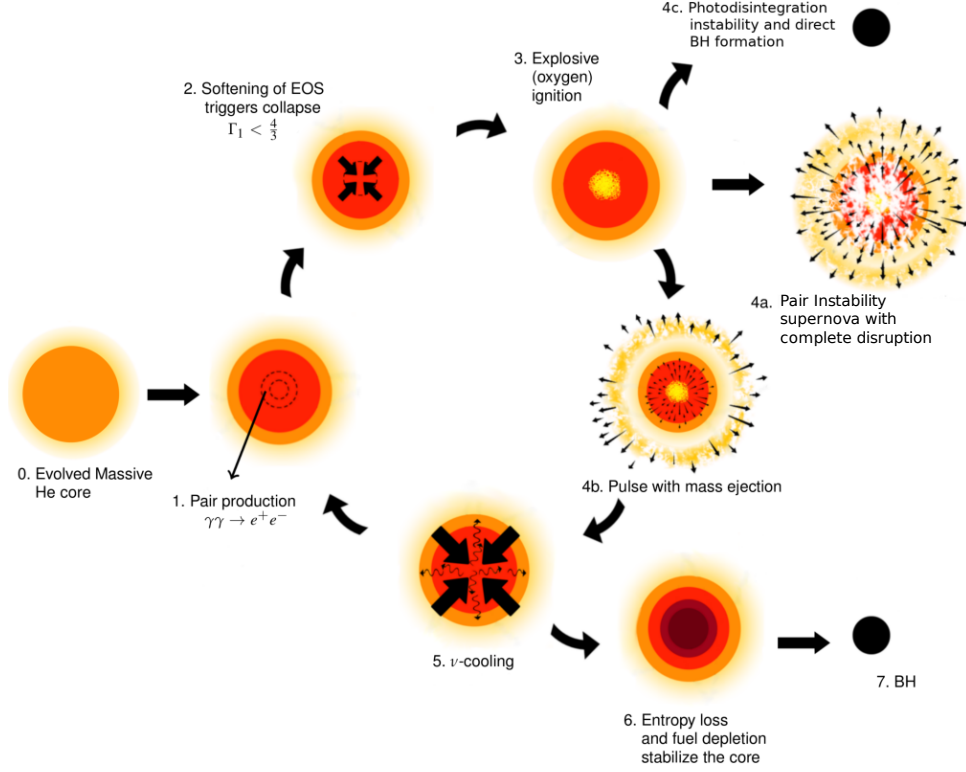


Figure 1.20: Evolution of a star undergoing pair instability evolution [27]. After the collapse triggered by the pair instability, pulsational pair instability, pair instability supernova, and black hole formation during collapse are possible.

Thus, in a radiation-pressure-dominated environment, the (first) adiabatic exponent defined in Section 2.3.6 is approximately $\Gamma_1 \simeq \frac{4}{3}$ because $\rho \propto V^{-1}$. If electron-positron pair production by photons occurs in such an environment,

$$\gamma + \gamma \rightarrow e^- + e^+, \quad (1.30)$$

the energy gained through processes such as stellar contraction is partially consumed in producing electron-positron pairs. This reduces the radiation pressure and creates a condition with adiabatic exponent $\Gamma_1 < \frac{4}{3}$, making the stellar interior dynamically unstable (see Section 2.13 for why a star becomes dynamically unstable under this condition). This is called pair instability, and a star that becomes dynamically unstable in this way begins to collapse. The subsequent evolutionary outcomes after the onset of pair instability are broadly classified into pulsational pair instability (PPI), pair instability supernova (PISN), and BH formation during collapse induced by pair instability, as summarized in Fig. 1.20.

PPI is a supernova imposter event and occurs in stars with masses of approximately $70\text{--}140M_\odot$. During collapse driven by pair instability, the temperature inside the star increases and triggers explosive thermonuclear O burning (3. in Fig. 1.20). The energy released by O burning generates a hydrodynamic pulse in the interior and ejects part of the stellar envelope (4b. in Fig. 1.20). If pair instability no longer occurs in the remaining core after these processes,

the star proceeds through ordinary Si burning and Fe core collapse.

PISN occur in stars with masses of approximately $140\text{-}250M_{\odot}$. In this case, explosive O burning completely disrupts the star, leaving no compact remnant such as an NS or BH (4a. in Fig. 1.20). This characteristic of PISN is expected to produce a mass gap in the mass distribution of stellar-mass BHs.

For stars with masses above approximately $250M_{\odot}$, the energy released by explosive O burning is absorbed through photodisintegration, a strongly endothermic reaction. As a result, the star fails to explode and instead collapses, forming a BH at the center (4c. in Fig. 1.20).

Chapter 2.

Basic Equations and Theorems

In this chapter, we focus on the physics required to describe the basic concepts of astrophysics. The structure and content of this chapter is mostly informed by [1], and also referred by [5, 28].

Still, the objects we consider are stars, and we assume spherical symmetry when discussing them. Based on this, in astrophysics, there are five expressions, the so-called basic equations, which are

$$\begin{aligned}\frac{\partial r}{\partial m} &= \frac{1}{4\pi r^2 \rho}, & (\text{mass conservation}) \\ \frac{\partial P}{\partial m} &= -\frac{Gm}{4\pi r^4}, & (\text{hydrostatic equilibrium}) \\ \frac{\partial L}{\partial m} &= \varepsilon - \varepsilon_v + \varepsilon_g, & (\text{energy conservation}) \\ \frac{\partial T}{\partial m} &= -\frac{Gm}{4\pi r^4} \frac{T}{P} \nabla, & (\text{energy transport}) \\ \frac{\partial X_i}{\partial t} &= \left(\frac{\partial X_i}{\partial t} \right)_{\text{nucl.}} + \left(\frac{\partial X_i}{\partial t} \right)_{\text{diff.}} + \left(\frac{\partial X_i}{\partial t} \right)_{\text{mix}}. & (\text{variance of chemical composition})\end{aligned}\tag{2.1}$$

L is the luminosity, and ε is the energy per unit mass per unit time. The nabla function ∇ in the energy transport equation depends on the specific energy transport mechanism. X_i is the mass fraction of nucleus i . These five equations constitute the fundamental equations for describing astrophysical objects.

In addition, there are various theorems required to describe such objects, including the collapse condition of interstellar clouds, changes in stellar interior conditions due to atomic ionization and electron degeneracy, and the conditions under which hydrostatic equilibrium can be maintained by degeneracy pressure.

We will explore these equations, the physics required to derive them, and the basic theorems needed to explain astrophysical and astronomical phenomena.

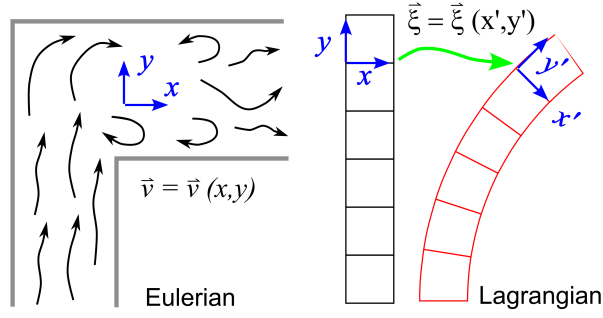


Figure 2.1: Eulerian and Lagrangian description [29].

2.1. Mass Conservation

In the Lagrangian description, mass conservation is expressed by the following relation.

$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho}. \quad (2.2)$$

In the Eulerian description, it is written as follows:

$$\frac{\partial m}{\partial r} = 4\pi r^2 \rho. \quad (2.3)$$

The Lagrangian and Eulerian descriptions provide two complementary frameworks for describing a fluid. In the Lagrangian description, physical quantities are defined by following individual fluid elements as they move through spacetime. In contrast, in the Eulerian description, physical quantities are specified at fixed locations in space, and the fluid is described as it flows through these locations. The two descriptions are illustrated in Fig. 2.1.

In the Lagrangian formulation, each fluid element is labeled (for example, by its initial position or enclosed mass), and its evolution is tracked over time. This approach naturally captures the history of each fluid element and eliminates explicit advection terms in the evolution equations. In the Eulerian formulation, space is discretized into fixed grid cells, and physical quantities are evolved at each spatial location. In this case, changes in the fluid arise both from local time variation and from the transport of fluid across spatial gradients.

In practice, both descriptions are widely used, and the choice between them depends on the problem under consideration. The Eulerian description is often adopted in computational fluid dynamics, particularly when dealing with shocks, turbulence, or multidimensional flows, where fixed grids provide numerical stability and flexibility. The Lagrangian description is commonly used in contexts where tracking individual fluid elements or mass shells is advantageous, such as in stellar structure and evolution. We will adopt the Lagrangian description in equations throughout this material.

Returning to the discussion of mass conservation, referring to Fig. 2.2, the mass enclosed

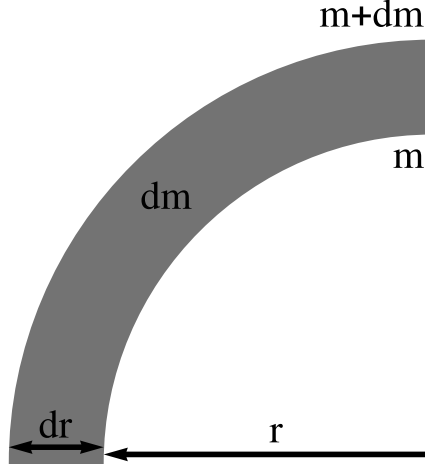


Figure 2.2: Mass dm contained in a spherical shell.

within a spherical shell of a spherically symmetric star can be expressed as follows:

$$dm = 4\pi r^2 \rho (dr - v dt). \quad (2.4)$$

The first term on the right-hand side corresponds to the mass of the spherical shell. The partial derivative of Eq. (2.4) with respect to r at constant t , yields Eq. (2.3). Applying the chain rule, we then obtain Eq. (2.2). This represents mass conservation for a spherical shell in the absence of any mass inflow or outflow. The second term on the right-hand side of Eq. (2.4) represents the mass flux due to the radial velocity. Taking the partial derivative on Eq. (2.4) with respect to t at constant r gives the following equation,

$$\frac{\partial m}{\partial t} = -4\pi r^2 \rho v. \quad (2.5)$$

Additionally, from Eq. (2.3) we have

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial m}{\partial r} &= 4\pi r^2 \frac{\partial \rho}{\partial t} \\ &= -4\pi \frac{\partial(\rho r^2 v)}{\partial r}. \end{aligned} \quad (2.6)$$

and thus we obtain

$$\frac{\partial \rho}{\partial t} = -\frac{1}{r^2} \frac{\partial(\rho r^2 v)}{\partial r}. \quad (2.7)$$

This is the continuity equation of hydrodynamics in the special case of spherical symmetry.

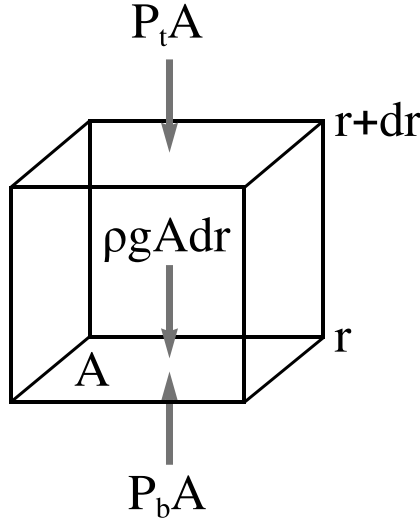


Figure 2.3: Pressure on the upper and lower side of a mass cube.

2.2. Hydrostatic Equilibrium

In classical mechanics, hydrostatic equilibrium is expressed by the following relation,

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4}. \quad (2.8)$$

When we consider general relativity, the hydrostatic equilibrium is described by the Tolman-Oppenheimer-Volkoff(TOV) equation,

$$\frac{\partial P}{\partial r} = -\frac{Gm}{r^2} \rho \left(1 + \frac{P}{\rho c^2}\right) \left(1 + \frac{4\pi r^3 P}{mc^2}\right) \left(1 - \frac{2Gm}{rc^2}\right)^{-1}. \quad (2.9)$$

Eq. (2.9) reduces to Eq. (2.8) in the limit $c \rightarrow \infty$. Retaining only terms of Eq. (2.9) up to the order $1/c^2$ gives the so-called post-Newtonian approximation. Let us examine each of these cases in detail.

2.2.1. Classical Limit

As depicted in Fig. 2.3, the hydrostatic equilibrium condition is given by

$$P_b - P_t = -\frac{\partial P}{\partial r} dr = \rho g dr. \quad (2.10)$$

Using $g = Gm/r^2$, we obtain

$$\frac{\partial P}{\partial r} = -\rho g = -\frac{Gm}{r^2} \rho. \quad (2.11)$$

Applying the mass conservation relation in Eq. (2.3), we find

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4}. \quad (2.12)$$

Based on this result, assuming $P_0 = 0$ at the stellar surface allows us to determine the central pressure P_c , and by further employing the ideal gas equation of state, we can estimate the central temperature T_c

2.2.2. General Relativity (Tolman-Oppenheimer-Volkoff Equation)

To obtain the hydrostatic equilibrium equation in the framework of general relativity, we must solve the Einstein field equations given below

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}. \quad (2.13)$$

Let us consider a spherically symmetric spacetime in Schwarzschild coordinates, which are defined as follows:

$$ds^2 = e^{v(r)}c^2dt^2 - e^{\lambda(r)}dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (2.14)$$

The corresponding metric tensor is then given by

$$g_{\mu\nu} = \text{diag}\left(e^v c^2, -e^\lambda, -r^2, -r^2 \sin^2\theta\right). \quad (2.15)$$

First, in order to evaluate the left-hand side of Eq. (2.13), we need to compute the Ricci tensor, $R_{\mu\nu}$, and the Ricci scalar, R . Each of these can be expressed in terms of the metric ($g_{\mu\nu}$), the Christoffel symbol ($\Gamma^\mu_{\nu\rho}$), and the Riemann tensor ($R^\mu_{\nu\rho\sigma}$), which are given as follows:

$$\begin{aligned} \Gamma^\mu_{\nu\rho} &= \frac{1}{2}g^{\mu\alpha}(\partial_\nu g_{\rho\alpha} + \partial_\rho g_{\nu\alpha} - \partial_\alpha g_{\nu\rho}), \\ R^\mu_{\nu\rho\sigma} &= \partial_\sigma \Gamma^\mu_{\nu\rho} - \partial_\rho \Gamma^\mu_{\nu\sigma} + \Gamma^\mu_{\sigma\alpha} \Gamma^\alpha_{\nu\rho} - \Gamma^\mu_{\rho\alpha} \Gamma^\alpha_{\nu\sigma}, \\ R_{\mu\nu} &= R^\alpha_{\mu\alpha\nu}, \\ R &= g^{\mu\nu} R_{\mu\nu}. \end{aligned} \quad (2.16)$$

Next, to evaluate the right-hand side, we must determine the energy-stress tensor. For a perfect fluid, $T^{\mu\nu}$ is given by

$$T^{\mu\nu} = (\rho c^2 + P)u^\mu u^\nu - P g^{\mu\nu}. \quad (2.17)$$

Since both $g_{\mu\nu}$ and $T_{\mu\nu}$ are diagonal tensors, Eq. (2.13) implies that $R_{\mu\nu}$ is also diagonal. The results of the computation are presented as follows:

$$\begin{aligned}
R_{tt} &= \frac{c^2}{4r} e^\nu e^\lambda (r v'^2 + 4v' - r v' \lambda' + 2r v'') , \\
R_{rr} &= \frac{1}{4r} (-r v'^2 + r v' \lambda' + 4\lambda' - 2r v'') , \\
R_{\theta\theta} &= \frac{1}{2} e^{-\lambda} (-r v' + r \lambda' + 2e^\lambda - 2) , \\
R_{\phi\phi} &= \frac{1}{2} e^{-\lambda} \sin^2 \theta (-r v' + r \lambda' + 2e^\lambda - 2) , \\
R &= \frac{1}{2r} e^{-\lambda} (r^2 v'^2 + 4r v' - r^2 v' \lambda' - 4r \lambda' + 2r^2 v'' - 4e^\lambda + 4) .
\end{aligned} \tag{2.18}$$

Here, primes denote derivatives with respect to r . Substituting the results obtained so far into Eq. (2.13), we obtain the following results,

$$\frac{8\pi G}{c^2} \rho = e^{-\lambda} \left(\frac{\lambda'}{r} - \frac{1}{r^2} \right) + \frac{1}{r^2} , \tag{2.19}$$

$$\frac{8\pi G}{c^4} P = e^{-\lambda} \left(\frac{v'}{r} + \frac{1}{r^2} \right) - \frac{1}{r^2} , \tag{2.20}$$

$$\frac{8\pi G}{c^4} P = \frac{1}{2} e^{-\lambda} \left(\frac{v' - \lambda'}{r} - \frac{v' \lambda'}{2} + \frac{v'^2}{2} + v'' \right) . \tag{2.21}$$

Multiplying the first equation Eq. (2.19) by $4\pi r^2$ and integrating with respect to r , we find

$$\begin{aligned}
\frac{8\pi G}{c^2} m &= 4\pi r (1 - e^{-\lambda}) , \\
\therefore e^{-\lambda} &= 1 - \frac{2Gm}{rc^2} ,
\end{aligned} \tag{2.22}$$

where $m = \int_0^r dr 4\pi r^2 \rho$.

Now we make use of the energy-momentum conservation,

$$0 = \nabla_\mu T^{\mu\nu} = \partial_\mu T^{\mu\nu} + \Gamma^\nu_{\mu\alpha} T^{\mu\alpha} + \Gamma^\mu_{\mu\alpha} T^{\nu\alpha} . \tag{2.23}$$

From this, the condition for hydrostatic equilibrium is given by

$$\begin{aligned}
\nabla_\mu T^{\mu r} &= \partial_\mu T^{\mu r} + \Gamma^r_{\mu\alpha} T^{\mu\alpha} + \Gamma^\mu_{\mu\alpha} T^{r\alpha} \\
&= e^{-\lambda} \left(P' + \frac{\rho c^2 + P}{2} v' \right) , \\
\therefore P' &= -\frac{\rho c^2 + P}{2} v' .
\end{aligned} \tag{2.24}$$

For the calculation, we use the results

$$\begin{aligned}
\partial_\mu T^{\mu r} &= \partial_r T^{rr} = \partial \left(P e^{-\lambda} \right) = P' e^{-\lambda} - \lambda' P e^{-\lambda}, \\
\Gamma^r_{\mu\alpha} T^{\mu\alpha} &= \Gamma^r_{tt} T^{tt} + \Gamma^r_{rr} T^{rr} + \Gamma^r_{\theta\theta} T^{\theta\theta} + \Gamma^r_{\phi\phi} T^{\phi\phi} \\
&= \frac{c^2 v' e^{v-\lambda}}{2} \rho e^{-v} + \frac{\lambda'}{2} P e^{-\lambda} - \frac{2}{r} P e^{-\lambda}, \\
\Gamma^\mu_{\mu\alpha} T^{r\alpha} &= \left(\Gamma^t_{tr} + \Gamma^r_{rr} + \Gamma^\theta_{\theta r} + \Gamma^\phi_{\phi r} \right) T^{rr} \\
&= \left(\frac{v'}{2} + \frac{\lambda'}{2} + \frac{2}{r} \right) P e^{-\lambda}.
\end{aligned} \tag{2.25}$$

Eq. (2.24) can also be obtained by differentiating Eq. (2.20) with respect to r and substituting Eqs. (2.19) to (2.22) into the result. By substituting Eqs. (2.20) and (2.22) into Eq. (2.24), we obtain

$$\begin{aligned}
P' &= -\frac{\rho c^2 + P}{2} \frac{2G}{r^2 c^2} \frac{m + \frac{4\pi r^3 P}{c^2}}{1 - \frac{2Gm}{rc^2}} \\
&= -\frac{Gm}{r^2} \rho \left(1 + \frac{P}{\rho c^2} \right) \left(1 + \frac{4\pi r^3 P}{mc^2} \right) \left(1 - \frac{2Gm}{rc^2} \right)^{-1}.
\end{aligned} \tag{2.26}$$

which is the TOV equation.

For a weak gravitational field, the TOV equation may be approximated by retaining terms only up to order $1/c^2$. This is referred to as the post-Newtonian approximation. In this limit, the TOV equation can be written as follows:

$$P' = -\frac{Gm}{r^2} \rho \left(1 + \frac{P}{\rho c^2} + \frac{4\pi r^3 P}{mc^2} + \frac{2Gm}{rc^2} \right). \tag{2.27}$$

2.3. Brief Review on Thermodynamics

Before proceeding to the next basic equation, we briefly review the thermodynamics needed for the subsequent discussion. Here, we assume a system with fixed composition and a fixed number of particles.

2.3.1. First Law of Thermodynamics

The relation between heat $dq = Tds$ and internal energy u follows from the first law of thermodynamics and is given by

$$dq = du + Pdv. \tag{2.28}$$

Here, both u and $v = 1/\rho$ are defined per unit mass. We wish to rewrite Eq. (2.28) in terms of the specific heats

$$\begin{aligned} c_P &= \left(\frac{dq}{dT} \right)_P = \left(\frac{\partial u}{\partial T} \right)_P + P \left(\frac{\partial v}{\partial T} \right)_P, \\ c_v &= \left(\frac{dq}{dT} \right)_v = \left(\frac{\partial u}{\partial T} \right)_v. \end{aligned} \quad (2.29)$$

Since we assume a fixed composition, we adopt $\rho = \rho(P, T)$ and $u = u(\rho, T)$ for the general equations of state. Therefore,

$$du = \left(\frac{\partial u}{\partial v} \right)_T dv + \left(\frac{\partial u}{\partial T} \right)_v dT. \quad (2.30)$$

The equation of state can be written as

$$\frac{d\rho}{\rho} = \alpha \frac{dP}{P} - \delta \frac{dT}{T}, \quad (2.31)$$

where α and δ are given by

$$\begin{aligned} \alpha &= \left(\frac{\partial \ln \rho}{\partial \ln P} \right)_T, \\ \delta &= - \left(\frac{\partial \ln \rho}{\partial \ln T} \right)_P. \end{aligned} \quad (2.32)$$

On the other hand, from Eqs. (2.28) and (2.30),

$$ds = \frac{1}{T} \left[\left(\frac{\partial u}{\partial T} \right)_v dT + \left(P + \left(\frac{\partial u}{\partial v} \right)_T \right) dv \right]. \quad (2.33)$$

If we use $\partial^2 s / \partial T \partial v = \partial^2 s / \partial v \partial T$, we obtain

$$\frac{\partial}{\partial T} \left[\frac{1}{T} \left(P + \left(\frac{\partial u}{\partial v} \right)_T \right) \right] = \frac{1}{T} \frac{\partial^2 u}{\partial T \partial v}. \quad (2.34)$$

Therefore,

$$\left(\frac{\partial u}{\partial v} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_v - P. \quad (2.35)$$

Consequently, from Eqs. (2.30) and (2.35) we have

$$\left(\frac{\partial u}{\partial T} \right)_P = \left(\frac{\partial u}{\partial T} \right)_v + \left(\frac{\partial v}{\partial T} \right)_P \left[T \left(\frac{\partial P}{\partial T} \right)_v - P \right]. \quad (2.36)$$

By the Eqs. (2.29) and (2.36), we obtain

$$c_P - c_v = \left(\frac{\partial v}{\partial T} \right)_P \left(\frac{\partial P}{\partial T} \right)_v T. \quad (2.37)$$

In addition, from Eq. (2.32) together with $d \ln x = dx/x$,

$$\begin{aligned} \left(\frac{\partial P}{\partial T} \right)_v &= \frac{P}{T} \frac{\delta}{\alpha}, \\ T \left(\frac{\partial v}{\partial T} \right)_P &= v \delta \end{aligned} \quad (2.38)$$

and therefore arrive at

$$c_P - c_v = \frac{P \delta^2}{\rho T \alpha}. \quad (2.39)$$

Finally, Eq. (2.28) can be written as

$$\begin{aligned} dq &= du + P dv = \left(\frac{\partial u}{\partial T} \right)_v dT + \left[\left(\frac{\partial u}{\partial v} \right)_T + P \right] dv \\ &= c_v dT - \frac{P}{\rho} \frac{\delta}{\alpha} \frac{d\rho}{\rho} \\ &= c_v dT - \frac{P \delta}{\rho \alpha} \left(\alpha \frac{dP}{P} - \delta \frac{dT}{T} \right) \\ &= \left(c_v + \frac{P \delta^2}{\rho T \alpha} \right) dT - \frac{\delta}{\rho} dP \\ &= c_P dT - \frac{\delta}{\rho} dP. \end{aligned} \quad (2.40)$$

2.3.2. Adiabatic Temperature Gradient

Since s is constant for an adiabatic process (i.e. $ds = dq/T = 0$), the adiabatic temperature gradient is defined by

$$\nabla_{\text{ad}} \equiv \left(\frac{d \ln T}{d \ln P} \right)_s. \quad (2.41)$$

This quantity is often used in astrophysics. From Eq. (2.40), it can be expressed as

$$\nabla_{\text{ad}} = \left(\frac{P}{T} \frac{\partial T}{\partial P} \right)_s = \frac{P \delta}{\rho T c_P}. \quad (2.42)$$

This describes the thermal variation of a single mass element during adiabatic compression.

2.3.3. Molecular Weight

The equation of state for an ideal gas is

$$P = nk_B T = \frac{\mathcal{R}}{\mu} \rho T, \quad (2.43)$$

where μ denotes the mean molecular weight and $\mathcal{R} \equiv k_B/m_u$ is the gas constant. Neglecting the electron mass, we define the physical quantities of the nuclei as follows:

- X_i : weight fraction of element i
- Z_i : charge number of element i
- μ_i : molecular weight of element i

Using these definitions, we obtain

$$n_i = \frac{\rho_i}{\mu_i m_u} = \frac{\rho}{m_u} \frac{X_i}{\mu_i}, \quad (2.44)$$

$$\begin{aligned} P &= P_e + \sum_i P_i = \left(n_e + \sum_i n_i \right) k_B T \\ &= \sum_i (1 + Z_i) n_i k_B T \\ &= \sum_i \frac{X_i (1 + Z_i)}{\mu_i} \rho \mathcal{R} T, \end{aligned} \quad (2.45)$$

where a charge neutral condition and a fully ionized medium are assumed. Based on these results, define the mean molecular weight as

$$\mu \equiv \left(\sum_i \frac{X_i (1 + Z_i)}{\mu_i} \right)^{-1}. \quad (2.46)$$

With this definition, a mixture of perfect gases can be treated as a uniform monatomic ideal gas. For a neutral (non-ionized) gas, the mean molecular weight is

$$\mu_0 = \left(\sum_i \frac{X_i}{\mu_i} \right)^{-1}, \quad (2.47)$$

while for a fully ionized gas, the mean molecular weight per free electron is

$$\mu_e = \left(\sum_i \frac{X_i Z_i}{\mu_i} \right)^{-1} = \frac{1}{Y_e}. \quad (2.48)$$

Here, Y_e is the number of electrons per nucleon. Approximately, for elements heavier than ^4He , we typically have $\mu_i/Z_i \approx 2$; therefore,

$$\mu_e = \left(X + \frac{1}{2}Y + \frac{1}{2}(1 - X - Y) \right) = \frac{2}{1 + X}. \quad (2.49)$$

Here, $X = X_{\text{H}}$ and $Y = X_{\text{He}}$. Note that this should not be confused with Y_i , which means the number of i per nucleon.

2.3.4. Heat Capacity Ratio

The ratio of specific heats, γ , is defined as

$$\gamma \equiv \frac{c_P}{c_V}. \quad (2.50)$$

By the equipartition theorem, the internal energy per unit mass of a non-relativistic monatomic ideal gas is given by

$$u = \frac{f}{2} \frac{n}{\rho} k_B T = \frac{f}{2} \frac{1}{\rho} P, \quad (2.51)$$

where f denotes the number of degrees of freedom, assume that it does not depend on T . Accordingly, the heat capacities are

$$\begin{aligned} c_V &= \left(\frac{\partial u}{\partial T} \right)_V = \frac{f}{2} \frac{n}{\rho} k_B, \\ c_P &= \left(\frac{\partial u}{\partial T} \right)_P + P \left(\frac{\partial v}{\partial T} \right)_P = \frac{f+2}{2} \frac{n}{\rho} k_B. \end{aligned} \quad (2.52)$$

Here, Eq. (2.39) has been used to calculate c_P . For the ideal gas, note from Eqs. (2.32) and (2.43) that $\alpha = \delta = 1$. Accordingly,

$$\gamma = 1 + \frac{2}{f}. \quad (2.53)$$

This result then leads to the following relation between P and u ,

$$P = (\gamma - 1) \rho u = (\gamma - 1) u_V, \quad (2.54)$$

where u_V is the internal energy per unit volume. From the above results, the adiabatic temperature gradient of the ideal gas can be rewritten as

$$\nabla_{\text{ad}} = \frac{P\delta}{\rho T c_P} = \frac{c_P - c_V}{c_P} = 1 - \frac{1}{\gamma}. \quad (2.55)$$

2.3.5. Adiabatic Sound Speed

For hydrodynamical perturbations, the relevant response of the pressure to a density perturbation is determined by the adiabatic sound speed, which is defined by

$$c_s^2 \equiv \left(\frac{\partial P}{\partial \rho} \right)_s. \quad (2.56)$$

This definition is appropriate because sound waves correspond to small and rapid compressions of a fluid element, for which heat exchange with the surroundings can be neglected. For a small

adiabatic perturbation, the pressure perturbation is given by

$$\delta P = \left(\frac{\partial P}{\partial \rho} \right)_s \delta \rho = c_s^2 \delta \rho. \quad (2.57)$$

The adiabatic sound speed can be expressed in terms of γ , ρ , and P for an ideal gas. Using Eq. (2.31) along an adiabatic variation, together with Eq. (2.41), we obtain

$$d \ln \rho = (\alpha - \delta \nabla_{\text{ad}}) d \ln P. \quad (2.58)$$

Therefore,

$$\left(\frac{\partial \ln P}{\partial \ln \rho} \right)_s = \frac{1}{\alpha - \delta \nabla_{\text{ad}}}. \quad (2.59)$$

It is useful to define the first adiabatic exponent as

$$\Gamma_1 \equiv \left(\frac{\partial \ln P}{\partial \ln \rho} \right)_s. \quad (2.60)$$

Then, from Eqs. (2.56) and (2.60), the sound speed can be written as

$$c_s^2 = \Gamma_1 \frac{P}{\rho}. \quad (2.61)$$

For an ideal gas, Eq. (2.43) gives $\alpha = \delta = 1$. In addition, we obtained the relation between ∇_{ad} and γ in Eq. (2.55). Hence,

$$\Gamma_1 = \frac{1}{1 - \nabla_{\text{ad}}} = \gamma. \quad (2.62)$$

Consequently, the adiabatic sound speed of an ideal gas becomes

$$c_s^2 = \gamma \frac{P}{\rho} = \gamma \frac{\mathcal{R}}{\mu} T. \quad (2.63)$$

2.3.6. Adiabatic Exponents

Describing the thermodynamic response of stellar matter using a single heat capacity ratio γ is generally insufficient, and therefore adiabatic exponents are introduced. These were first formulated by Eddington and later employed by Chandrasekhar in the development of stellar stability theory.

There are three adiabatic exponents, Γ_1 , Γ_2 , and Γ_3 , defined as follows.

$$\Gamma_1 \equiv \left(\frac{\partial \ln P}{\partial \ln \rho} \right)_s, \quad (2.64)$$

$$\frac{\Gamma_2}{\Gamma_2 - 1} \equiv \left(\frac{\partial \ln P}{\partial \ln T} \right)_s = \frac{1}{\nabla_{\text{ad}}}, \quad (2.65)$$

$$\Gamma_3 \equiv \left(\frac{\partial \ln T}{\partial \ln \rho} \right)_s + 1. \quad (2.66)$$

The following relation holds among the adiabatic exponents.

$$\frac{\Gamma_1}{\Gamma_3 - 1} = \frac{\Gamma_2}{\Gamma_2 - 1}. \quad (2.67)$$

2.4. Virial Theorem

The virial theorem is a relation that holds when a system is in hydrostatic equilibrium. Multiplying both sides of the hydrostatic equilibrium equation, Eq. (2.12), by $4\pi r^3$ and integrating over m yields the following result,

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = 4\pi r^3 P \Big|_0^M - \int_0^M 12\pi r^2 P \frac{\partial r}{\partial m} dm. \quad (2.68)$$

The first term on the right-hand side vanishes if we assume $r = 0$ at the stellar center and $P = 0$ at the stellar surface. By rearranging the second term on the right-hand side using Eq. (2.2), we define the gravitational potential energy as

$$E_g \equiv - \int_0^M \frac{Gm}{r} dm = - \int_0^M 3 \frac{P}{\rho} dm. \quad (2.69)$$

Applying Eq. (2.54) and introducing the definition of total internal energy $E_i \equiv \int_0^M \frac{u}{\rho} dm$, we obtain

$$E_g + 3(\gamma - 1)E_i = 0. \quad (2.70)$$

Consequently, the total energy W can be written as follows:

$$W = E_g + E_i = \left(1 - \frac{1}{3(\gamma - 1)} \right) E_g = \frac{3\gamma - 4}{3\gamma - 3} E_g. \quad (2.71)$$

A stellar structure is bounded when $W < 0$, and from Eq. (2.71), this condition implies $\gamma > \frac{4}{3}$. For a non-relativistic monatomic ideal gas, $\gamma = \frac{5}{3}$, whereas for a relativistic monatomic ideal gas, $\gamma \sim \frac{4}{3}$. Thus, stars dominated by relativistic gas are unstable. In such a case, the star releases or removes energy through explosion or collapse. If a nonzero P is present on the stellar surface, the first term on the right-hand side of Eq. (2.68) must be retained.

The luminosity of a star is $L = -\frac{dW}{dt}$, and therefore we can define the Kelvin-Helmholtz timescale,

$$\begin{aligned} \tau_{\text{KH}} &\equiv \frac{E_i}{L} \approx \frac{|E_g|}{L} \\ &= \frac{GM^2}{2RL}. \end{aligned} \quad (2.72)$$

This timescale represents the period required for a star to radiate away its total internal energy, providing a rough estimate of a star's lifetime. In most cases, however, the actual stellar lifetime

exceeds this timescale. The reason is that when nuclear burning of one species ceases in the stellar interior, another species typically begins to fuse, so that when one component stops contributing to the internal energy, another begins to contribute.

2.5. Jeans Criterion

The Jeans criterion, or Jeans instability, specifies the condition under which the gas pressure in an interstellar cloud can no longer support it against gravity, leading to gravitational collapse. When this condition is satisfied, the cloud enters the PMS stage. The critical mass at which gravitational collapse sets in is called the Jeans mass, M_J , and is given as follows.

$$M_J = \frac{\pi^{5/2}}{6} \frac{c_s^3}{\rho^{1/2} G^{3/2}} \simeq \left(\frac{375 k_\beta^3 T^3}{4 \pi \rho G^3 \mu^3 m_p^3} \right)^{1/2}, \quad (2.73)$$

where μ is the mean molecular mass. The Jeans mass is typically derived in two ways. The first approach considers a finite spherical gas cloud and derives the condition using the virial theorem or energy balance. The second approach considers an infinite, homogeneous medium and derives the condition using linear perturbation. While both approaches yield the same physical scaling, they do not define exactly the same Jeans mass in a strict mathematical sense. In what follows, we examine both derivations and compare their results.

2.5.1. Derivation Based on the Energy Balance (Virial Theorem)

Consider a spherical gas cloud of finite radius R , total mass M , and uniform density ρ . For the cloud to collapse under gravity, the gravitational binding energy must exceed the internal energy of the gas. For a non-relativistic monatomic ideal gas, Eq. (2.70) gives $E_g = -2E_i$, so the condition for collapse can be written as

$$|E_g| > 2E_i. \quad (2.74)$$

The gravitational energy E_g is given from Eq. (2.69) as

$$E_g = - \int_0^M \frac{Gm}{r} dm = G \left(\frac{4}{3} \pi \rho \right)^{1/3} \int_0^M m^{2/3} dm = \frac{3}{5} \frac{GM^2}{R}. \quad (2.75)$$

In this calculation, we have used $r = \left(\frac{m}{\frac{4}{3} \pi \rho} \right)^{1/3}$. Also, assuming that the internal energy E_i is entirely thermal, the ideal gas equation of state yields

$$E_i = \frac{3}{2} N k_\beta T = \frac{3}{2} \frac{M}{\mu m_p} k_\beta T. \quad (2.76)$$

From Eqs. (2.74) to (2.76), we obtain the Jeans mass M_J as

$$M_J = \frac{5k_\beta T}{G\mu m_p} R. \quad (2.77)$$

Substituting $M = \frac{4}{3}\pi R^3 \rho$ into the above expression, we obtain the Jeans radius R_J .

$$R_J = \left(\frac{15k_\beta T}{4\pi\rho G\mu m_p} \right)^{1/2}. \quad (2.78)$$

Finally, substituting this result into Eq. (2.77), the Jeans mass derived from the virial theorem can be written as follows.

$$M_J = \left(\frac{375k_\beta^3 T^3}{4\pi\rho G^3 \mu^3 m_p^3} \right)^{1/2}. \quad (2.79)$$

2.5.2. Derivation Based on the Linear Perturbation

Consider an ideal gas with uniform density ρ under Newtonian gravity. This fluid is described by the continuity equation, Euler equation, and Poisson equation given below.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0, \quad (2.80)$$

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{1}{\rho} \nabla P - \nabla \Phi, \quad (2.81)$$

$$\nabla^2 \Phi = 4\pi G \rho. \quad (2.82)$$

Assuming that the medium is at rest, we introduce first-order linear perturbations, i.e.,

$$\rho = \rho_0 + \delta\rho, \quad P = P_0 + \delta P, \quad \vec{v} = \vec{0} + \delta\vec{v}, \quad \Phi = \Phi_0 + \delta\Phi. \quad (2.83)$$

Then, neglecting terms of second order and higher, the equations describing the fluid reduce to the following form.

$$\frac{\partial \delta\rho}{\partial t} + \rho_0 \nabla \cdot \delta\vec{v} = 0, \quad (2.84)$$

$$\frac{\partial \delta\vec{v}}{\partial t} = -\frac{1}{\rho_0} \nabla \delta P - \nabla \delta\Phi, \quad (2.85)$$

$$\nabla^2 \delta\Phi = 4\pi G \delta\rho. \quad (2.86)$$

Differentiating the continuity equation with respect to time, substituting the Euler and Poisson equations, and using Eq. (2.57), we obtain

$$\frac{\partial^2 \delta\rho}{\partial t^2} = c_s^2 \nabla^2 \delta\rho + 4\pi G \rho_0 \delta\rho. \quad (2.87)$$

We now assume a plane-wave solution for $\delta\rho$ in this result, i.e.,

$$\delta\rho \propto e^{i(\vec{k}\cdot\vec{r}-\omega t)}, \quad \nabla^2\delta\rho \propto -k^2\delta\rho, \quad \frac{\partial^2\delta\rho}{\partial t^2} \propto -\omega^2\delta\rho. \quad (2.88)$$

Therefore, the dispersion relation implied by Eq. (2.87) is

$$\omega^2 = c_s^2 k^2 - 4\pi G\rho_0. \quad (2.89)$$

The first term on the right-hand side represents the restoring force due to gas pressure, while the second term represents the gravitational instability. From this result, when

$$4\pi G\rho_0 > c_s^2 k^2, \quad (2.90)$$

we have $\omega^2 < 0$, so ω becomes purely imaginary and the perturbation grows exponentially. The Jeans length λ_J is therefore given by

$$\lambda_J = \frac{2\pi}{k_J} = \sqrt{\frac{\pi}{G\rho_0}} c_s. \quad (2.91)$$

If the Jeans mass is defined as the mass of a sphere with radius $R_J = \frac{1}{2}\lambda_J$, then M_J is given by

$$M_J = \frac{\pi^{5/2}}{6} \frac{c_s^3}{\rho_0^{1/2} G^{3/2}}. \quad (2.92)$$

From Eq. (2.63), for a non-relativistic monatomic ideal gas,

$$c_s^2 = \gamma \frac{P}{\rho} = \frac{5}{3} \frac{k_B T}{\mu m_p}, \quad (2.93)$$

and hence,

$$M_J = \frac{\pi^{5/2}}{6} \frac{1}{\rho_0^{1/2} G^{3/2}} \left(\frac{125 k_B^3 T^3}{27 \mu^3 m_p^3} \right)^{1/2} = \left(\frac{\pi}{3} \right)^3 \left(\frac{375 k_B^3 T^3}{4\pi \rho G^3 \mu^3 m_p^3} \right)^{1/2}. \quad (2.94)$$

Comparing this result with Eq. (2.79), we find that there is a numerical difference by a factor of $\left(\frac{\pi}{3}\right)^3 \simeq 1.15$. The two approaches describe the same physical phenomenon, but this numerical difference arises from the different assumptions associated with the perspectives of energy balance and dynamical stability.

2.6. Energy Conservation

Energy conservation is given by

$$\frac{\partial L_r}{\partial m} = \varepsilon - \varepsilon_v + \varepsilon_g. \quad (2.95)$$

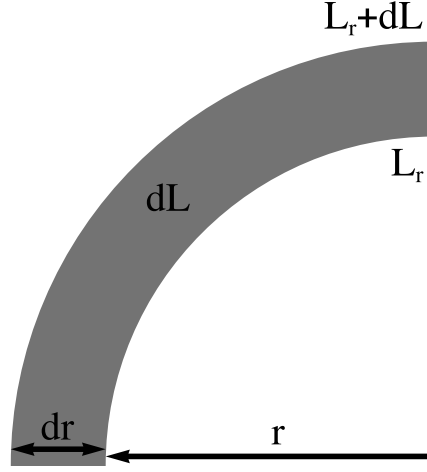


Figure 2.4: Luminosity through a spherical shell.

Here, L_r denotes the luminosity at radius r , and ϵ is the energy per unit mass per unit time. In general, ϵ depends on the temperature, density, and composition of the reacting nuclear species. L_r is determined by the mechanisms of energy transport-radiation, conduction, and convection-and although neutrinos also contribute, their effect can usually be neglected during ordinary stellar evolution.

From Fig. 2.4, assuming a stationary situation and a spherically symmetric system leads to

$$dL_r = 4\pi r^2 \rho \epsilon dr = \epsilon dm. \quad (2.96)$$

Therefore,

$$\frac{dL_r}{dm} = \epsilon. \quad (2.97)$$

In a non-stationary situation, the internal energy changes due to the mechanical work done by the spherical shell, and thus we have

$$dq = \left(\epsilon - \frac{\partial L_r}{\partial m} \right) dt. \quad (2.98)$$

From the above relation, it follows that

$$\frac{\partial L_r}{\partial m} = \epsilon - \frac{dq}{dt} = \epsilon - \frac{\partial u}{\partial t} + \frac{P}{\rho^2} \frac{\partial P}{\partial t} = \epsilon - c_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t} = \epsilon + \epsilon_g, \quad (2.99)$$

where ϵ_g is

$$\epsilon_g \equiv -T \frac{\partial s}{\partial t} = -c_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t} \left(= -c_P T \left(\frac{1}{T} \frac{\partial T}{\partial t} - \frac{\nabla_{\text{ad}}}{P} \frac{\partial P}{\partial t} \right) \right). \quad (2.100)$$

Eq. (2.40) was used for the above calculation.

The quantity ϵ_ν describes the energy loss due to neutrinos produced in nuclear reactions.

Since $\varepsilon_\nu > 0$ by its definition, we can explicitly put the neutrino contribution as

$$\frac{\partial L_r}{\partial m} = \varepsilon - \varepsilon_\nu + \varepsilon_g. \quad (2.101)$$

2.7. Energy Transport by Diffusive Processes

Energy transport by diffusion includes radiation and conduction, and is given as follows;

$$\frac{\partial T}{\partial m} = -\frac{T}{P} \frac{GM}{4\pi r^4} \nabla_{\text{rad}}, \quad (2.102)$$

where ∇_{rad} is given by

$$\nabla_{\text{rad}} = \left(\frac{d \ln T}{d \ln P} \right)_{\text{rad}} = \frac{3}{16\pi a c G} \frac{\kappa l P}{m T^4}. \quad (2.103)$$

Here, $a = \frac{4}{c} \sigma$ is the radiation density constant, and l denotes the local luminosity. The quantity κ is the mean opacity, defined as the frequency-averaged cross section per unit mass. Both radiative and conductive processes contribute to κ .

2.7.1. Radiative Diffusion

The mean free path of photons is defined as follows.

$$l_\gamma = \frac{1}{\kappa \rho}. \quad (2.104)$$

Here, κ is the mean absorption coefficient, obtained by averaging the radiative cross section per unit mass over frequency.

As an example of a main sequence star, let us estimate the photon mean free path and internal conditions in the Sun. For ionized hydrogen in stellar interiors, a lower bound on the opacity is $\kappa \simeq 0.4 \text{ cm}^2/\text{g}$. Since the mean density of the Sun is $\bar{\rho}_\odot \simeq 1.4 \text{ g/cm}^3$, the mean free path of the photon inside the Sun is approximately

$$l_{\gamma, \odot} \sim 1.8 \text{ cm}. \quad (2.105)$$

The temperature gradient between the stellar core and the surface can be estimated approximately as follows;

$$\frac{\Delta T}{\Delta r} = \frac{T_c - T_o}{R}. \quad (2.106)$$

For the Sun, taking $T_c \sim 10^7 \text{ K}$, $T_o \sim 10^4 \text{ K}$, and $R_\odot \sim 7.0 \times 10^{10} \text{ km}$, the characteristic temperature gradient is

$$\left(\frac{\Delta T}{\Delta r} \right)_\odot \sim 1.4 \times 10^{-4} \text{ K} \cdot \text{cm}^{-1}. \quad (2.107)$$

Since the photon mean free path $l_{\gamma, \odot}$ is extremely small compared to $R_\odot$, the temperature

variation over a local mean free path scale is approximately

$$\Delta T_{\text{loc}} = l_{\gamma, \odot} \left(\frac{dT}{dr} \right) \simeq 2.5 \times 10^{-4} \text{ K}. \quad (2.108)$$

According to the Stefan-Boltzmann law, the radiation energy density scales as $U \propto T^4$. Therefore,

$$\frac{\Delta U}{U} \simeq 4 \frac{\Delta T}{T}. \quad (2.109)$$

In the solar core,

$$\frac{\Delta U}{U} \simeq 4 \frac{2.5 \times 10^{-4}}{10^7} \sim 10^{-10}. \quad (2.110)$$

From this result, since $l_{\gamma} \ll R$ inside the star, we conclude that the stellar interior is in approximate thermal equilibrium.

Since the photon mean free path inside a star is extremely short compared to the stellar radius, radiative transport by photons can be described as a diffusion process. The Fick's law for particle diffusion is given by

$$\vec{J} = -D \nabla n, \quad (2.111)$$

where $\vec{J}$ is the diffusive flux and n is the number density. The diffusion coefficient (diffusivity) D for an isotropic random walk is

$$D = \frac{1}{3} v l_p. \quad (2.112)$$

Replacing Eq. (2.111) with the diffusive flux of the radiative energy $\vec{F}$, n with the radiation energy density U , and D with $D_{\text{rad}} = \frac{1}{3} c l_{\gamma}$, we obtain

$$\vec{F}_{\text{rad}} = -D_{\text{rad}} \nabla U. \quad (2.113)$$

The Stefan-Boltzmann law says that the total energy density of radiation is

$$U = a T^4, \quad (2.114)$$

where $a = \frac{4}{c} \sigma$ is the radiation density constant and σ is the Stefan-Boltzmann constant. From Eqs. (2.105), (2.113) and (2.114), and assuming spherical symmetry in the stellar interior, we have $|\vec{F}| = F$ and $\nabla U = \frac{\partial U}{\partial r}$. The diffusive flux of radiative energy is therefore

$$\vec{F}_{\text{rad}} = -\frac{4ac}{3} \frac{T^3}{\kappa \rho} \nabla T \equiv -k_{\text{rad}} \nabla T. \quad (2.115)$$

Because we assume spherical symmetry, the local luminosity is given as $l = 4\pi r^2 F$. Hence,

$$\frac{\partial T}{\partial r} = -\frac{3}{16\pi ac} \frac{\kappa \rho l}{r^2 T^3}, \quad (2.116)$$

or, equivalently,

$$\frac{\partial T}{\partial m} = -\frac{3}{64\pi^2 ac} \frac{\kappa l}{r^4 T^3}. \quad (2.117)$$

2.7.2. Rosseland Mean Opacity

In Eq. (2.105), we used a frequency-averaged value of κ , and in Eq. (2.114) we used the total energy density of radiation. Therefore, $\vec{F}$ and l obtained above are results integrated over all frequencies. Consequently, κ must be a well-defined mean value that adequately reflects the appropriate frequency weighting.

Rewriting Eq. (2.113) in a frequency-dependent form, we obtain the following form.

$$\vec{F}_\nu = -D_\nu \nabla U_\nu. \quad (2.118)$$

The subscript ν indicates that the corresponding physical quantity depends on the frequency ν . Accordingly, $D_\nu = \frac{1}{3} c l_\nu = \frac{c}{3\kappa_\nu \rho}$ and U_ν is given by the Planck's law as follows:

$$U_\nu = \frac{4\pi}{c} B(\nu, T) = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/k_B T} - 1}. \quad (2.119)$$

From this, the gradient of the energy density of radiation is given by the following expression.

$$\nabla U_\nu = \frac{4\pi}{c} \frac{\partial B}{\partial T} \nabla T. \quad (2.120)$$

Therefore, from Eq. (2.118), we obtain the following result.

$$\vec{F}_\nu = -\frac{4\pi}{3\rho} \left(\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu \right) \nabla T. \quad (2.121)$$

Comparing this result with Eq. (2.113), we define the Rosseland mean opacity κ (also frequently denoted by κ_γ) as follows.

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu = \frac{\int_0^\infty \kappa_\nu^{-1} x d\nu}{\int_0^\infty x d\nu}. \quad (2.122)$$

In the last equality, $x \equiv \frac{\partial B}{\partial T}$ and $\frac{\partial}{\partial T} \int_0^\infty B d\nu = \frac{\partial}{\partial T} \left(\frac{\sigma}{\pi} T^4 \right) = \frac{acT^3}{\pi}$ have been used.

2.7.3. Conductive Diffusion

Energy transport by conduction arises from collisions during the random motion of particles. In typical stellar environments, the mean free path of particles $l_p \ll l_\gamma$, and the particle velocity $v \ll c$. Therefore, energy transport by conduction is subdominant compared to radiative transport.

However, when particles are strongly degenerate, both the mean free path and the particle

velocity increase significantly, so conductive energy transport can make a sizable contribution. Representative examples include stellar cores with degenerate electrons and compact objects such as neutron stars.

Since conduction can also be described as particle diffusion, it follows from Fick's law that

$$\vec{F}_{\text{cd}} = -\frac{4ac}{3} \frac{T^3}{\kappa_{\text{cd}}\rho} \nabla T \equiv -k_{\text{cd}} \nabla T. \quad (2.123)$$

2.7.4. Radiative Temperature Gradient

From Eqs. (2.115) and (2.123), we may define the total energy flux due to diffusive processes as follows.

$$\vec{F} = \vec{F}_{\text{rad}} + \vec{F}_{\text{cd}} = -\frac{4acT^3}{3\rho} (\kappa_{\text{rad}}^{-1} + \kappa_{\text{cd}}^{-1}) \nabla T. \quad (2.124)$$

Defining $\kappa^{-1} \equiv \kappa_{\text{rad}}^{-1} + \kappa_{\text{cd}}^{-1}$ and using $l = 4\pi r^2 F$, we obtain the following result.

$$\frac{\partial T}{\partial m} = -\frac{3}{64\pi^2 ac} \frac{\kappa l}{r^4 T^3}. \quad (2.125)$$

Assuming hydrostatic equilibrium and dividing the above expression by Eq. (2.8), we obtain

$$\frac{\partial T}{\partial P} = \frac{3}{16\pi ac G} \frac{\kappa l}{m T^3}. \quad (2.126)$$

Based on this, the radiative temperature gradient ∇_{rad} is defined as follows:

$$\nabla_{\text{rad}} \equiv \left(\frac{d \ln T}{d \ln P} \right)_{\text{rad}} = \frac{3}{16\pi ac G} \frac{\kappa l P}{m T^4}. \quad (2.127)$$

It includes contributions not only from radiation but also from conduction, and relates P and T of two neighboring mass shells. Note that this expression has a form similar to Eq. (2.41), which appeared earlier in the discussion of thermodynamics. This structure repeatedly emerges in the analysis of energy transport. From this form, the energy transport equation in Eq. (2.1) can be written as a function of ∇ , which depends on the specific energy transport mechanism.

2.8. Instability

Unlike radiative and conductive energy transport, which can be described as diffusion processes, energy transport by convection arises from motion driven by instability. Therefore, before discussing convective energy transport, we first examine the nature of instability.

Perturbations generated within a star can induce local fluid motions. Since we are considering a spherically symmetric star, if such local motions do not grow sufficiently, the global average effect of the perturbation can be neglected. However, if the perturbation grows large enough, it can produce non-spherical motions and significantly affect the stellar structure. Al-

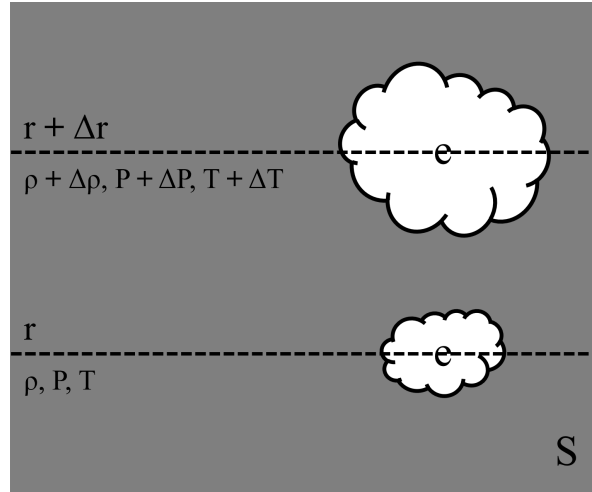


Figure 2.5: Schematic diagram of a mass element (parcel) produced by a fluctuation within the surrounding.

though these motions are locally asymmetric, they are statistically distributed within the star, so the basic equations Eq. (2.1) derived under the assumption of spherical symmetry may still be applied in an averaged sense. Nevertheless, such motions can give rise to convective energy transport.

Accordingly, understanding whether a local perturbation can grow sufficiently—that is, understanding instability—is a necessary prerequisite for discussing convective energy transport. Instabilities are broadly classified into dynamical, vibrational, and secular instabilities. In what follows, we discuss each of these in turn.

2.8.1. Dynamical Instability

The first type of instability we consider is dynamical instability, which arises when the timescale for heat exchange between the mass element and the surrounding medium—i.e., the thermal adjustment time—is sufficiently longer than the timescale for the element’s motion, so that the motion is effectively adiabatic.

As illustrated in Fig. 2.5, we consider a mass element generated by a fluctuation within the surrounding medium. For an arbitrary physical quantity A , we define the difference between the mass element and the surrounding medium as follows.

$$DA \equiv A_e - A_s. \quad (2.128)$$

The subscripts e and s denote the element and the surrounding medium, respectively. Note that this should not be confused with the specific heat s .

First, consider DP . If $DP \neq 0$, the element will expand or contract until it reaches pressure balance with the surrounding medium. This adjustment occurs at the sound speed of the medium. Since this process typically takes place much faster than other motions of the element,

it violates the assumptions made above. Therefore, we impose the following condition.

$$DP = 0. \quad (2.129)$$

Next, consider DT . If $DT > 0$, then for a perfect gas, since $\rho \propto P/T$, it follows that $D\rho < 0$. In this case, the element is lighter than the surrounding medium, so the buoyancy force causes the element to rise. Therefore, $\Delta r > 0$, and the temperature fluctuation induces radial motion of the mass element. As the element rises due to buoyancy, it becomes surrounded by a new environment. The corresponding $D\rho$ can then be expressed as follows.

$$D\rho = \left[\left(\frac{d\rho}{dr} \right)_e - \left(\frac{d\rho}{dr} \right)_s \right] \Delta r. \quad (2.130)$$

The radial component of the buoyancy force per unit volume, $\vec{K}$, arising from $D\rho$, is given as follows.

$$K_r = -gD\rho. \quad (2.131)$$

Recall that we are considering an element that has risen due to buoyancy and is now in a new environment. If $D\rho < 0$, then $K_r > 0$, so $\vec{K}$ acts in the upward direction and the element continues to rise. This corresponds to an unstable situation in which the perturbation keeps growing. Conversely, if $D\rho > 0$, then $K_r < 0$, so $\vec{K}$ acts in the downward direction and the element descends back to its original position. In this case, the perturbation does not grow and instead decays, corresponding to a stable situation. From this, the condition for stability based on Eq. (2.130) is given as follows:

$$\left(\frac{d\rho}{dr} \right)_e - \left(\frac{d\rho}{dr} \right)_s > 0. \quad (2.132)$$

To express the density appearing in this expression in terms of other state variables, we introduce the following equation of state.

$$\frac{d\rho}{\rho} = \alpha \frac{dP}{P} - \delta \frac{dT}{T} + \varphi \frac{d\mu}{\mu}. \quad (2.133)$$

The definitions of α and δ are the same as in Eq. (2.32), and similarly, φ is defined as follows.

$$\varphi \equiv \left(\frac{\partial \ln \rho}{\partial \ln \mu} \right). \quad (2.134)$$

In the partial derivatives defining α , δ , and φ , each quantity is defined under the assumption that the other two variables are held constant. For a perfect gas, since $\rho \sim \frac{P\mu}{T}$, we have $\alpha = \delta = \varphi = 1$.

Eq. (2.133) is similar to Eq. (2.31), but the difference is that here we have included an additional term depending on the molecular weight μ , thereby allowing for changes in the chemical composition. That is, $d\mu$ in Eq. (2.133) represents a change in the chemical composition,

namely changes in quantities such as the concentrations of nuclei or the degree of ionization.

Using Eq. (2.133), the condition for stability given by Eq. (2.132) can be rewritten as follows.

$$\left(\frac{\alpha}{P} \frac{dP}{dr}\right)_e - \left(\frac{\delta}{T} \frac{dT}{dr}\right)_e - \left(\frac{\alpha}{P} \frac{dP}{dr}\right)_s + \left(\frac{\delta}{T} \frac{dT}{dr}\right)_s - \left(\frac{\varphi}{\mu} \frac{d\mu}{dr}\right)_s > 0. \quad (2.135)$$

Here, we have assumed that the element maintains its own chemical composition, i.e., $d\mu = 0$, while for the surrounding medium we assume $d\mu \neq 0$, since the element can pass through layers with different chemical compositions.

In Eq. (2.135), the two terms involving the pressure gradient cancel each other due to the condition $DP = 0$. To rewrite the remaining terms using variables in a form similar to ∇_{ad} and ∇_{rad} defined earlier in Eqs. (2.41) and (2.127), we multiply them by the pressure scale height H_P , defined as follows.

$$H_P \equiv -\frac{dr}{d \ln P} = -P \frac{dr}{dP}. \quad (2.136)$$

Under hydrostatic equilibrium, from Eq. (2.11) we obtain $H_P = \frac{P}{\rho g}$. The quantity H_P is a local characteristic length scale that depends on r , and represents the distance over which P changes by a factor of e . In general, H_P is small in the outer regions of a star and increases toward the center, approaching $H_P \rightarrow \infty$ at $r = 0$.

From Eq. (2.135) and the definition of H_P , we obtain the following expression.

$$\left(\frac{d \ln T}{d \ln P}\right)_s < \left(\frac{d \ln T}{d \ln P}\right)_e + \frac{\varphi}{\delta} \left(\frac{d \ln \mu}{d \ln P}\right)_s. \quad (2.137)$$

This expression can be rewritten by introducing the following gradients.

$$\nabla \equiv \left(\frac{d \ln T}{d \ln P}\right)_s, \nabla_e \equiv \left(\frac{d \ln T}{d \ln P}\right)_e, \nabla_\mu \equiv \left(\frac{d \ln \mu}{d \ln P}\right)_s. \quad (2.138)$$

Therefore, the condition for stability is expressed as follows:

$$\nabla < \nabla_e + \frac{\varphi}{\delta} \nabla_\mu. \quad (2.139)$$

If all the energy is transported by radiation, then $\nabla = \nabla_{\text{rad}}$. If the elements undergo adiabatic motion, then $\nabla_e = \nabla_{\text{ad}}$, and thus,

$$\nabla_{\text{rad}} < \nabla_{\text{ad}} + \frac{\varphi}{\delta} \nabla_\mu \quad (2.140)$$

is satisfied for the layer to be stable. This condition is known as the Ledoux criterion, and in the special case $\nabla_\mu = 0$, it is referred to as the Schwarzschild criterion. In regions where the Ledoux criterion is violated, convection occurs. In regions where the Ledoux criterion is satisfied but the Schwarzschild criterion is not, semiconvection arises, which is neither fully convective nor fully stable. Note that Eq. (2.140) is a local criterion, and convective motion is

influenced by neighboring layers.

When the condition in Eq. (2.140) is satisfied, the layer is dynamically stable and convection does not occur, so $\nabla = \nabla_{\text{rad}}$. Instability arises when ∇_{rad} becomes too large or too small, and in the late stages of stellar evolution, ∇_{μ} can play a significant role.

Let us consider the case where convection occurs in a layer with $\nabla_{\mu} = 0$. In this case, from Eq. (2.140), we have $\nabla > \nabla_e$. In addition, since part of the energy is transported by convection, the required temperature gradient is smaller than in the case where all the energy is transported by radiation. Therefore, $\nabla < \nabla_{\text{rad}}$. Finally, mass elements are not perfectly adiabatic and lose some heat through radiation, so $\nabla_e > \nabla_{\text{ad}}$. Thus, in a convective envelope, the following relation holds.

$$\nabla_{\text{rad}} > \nabla > \nabla_e > \nabla_{\text{ad}}. \quad (2.141)$$

In summary, the first inequality arises from the contribution of convection as an additional channel of energy transport, the second inequality represents the condition required to sustain convection according to the Schwarzschild criterion, and the third inequality originates from the thermodynamic fact that real elements are not perfectly adiabatic.

2.8.2. Vibrational Instability

Let us consider a dynamically stable layer. Mass elements that rise due to buoyancy subsequently descend and return to their original position. During this process, the mass elements acquire momentum, which can induce oscillations. Instability arising from such oscillations is referred to as vibrational instability, or over-instability.

The $D\rho$ of a mass element that has risen by Δr from its initial position can be expressed as follows from Eqs. (2.130), (2.133), (2.136) and (2.138).

$$D\rho = \frac{\rho\delta}{H_p} \left[\nabla_e - \nabla + \frac{\varphi}{\delta} \nabla_{\mu} \right] \Delta r. \quad (2.142)$$

Under the buoyant force per unit volume due to gravity, given by Eq. (2.131), the mass element acquires an acceleration expressed as follows.

$$\frac{\partial^2 \Delta r}{\partial t^2} = -\frac{g\delta}{H_p} \left[\nabla_e - \nabla + \frac{\varphi}{\delta} \nabla_{\mu} \right] \Delta r. \quad (2.143)$$

Let us assume that the mass element has been displaced by an initial amount Δr_0 into a dynamically stable layer. In this case, the mass element is accelerated toward its equilibrium position and undergoes oscillatory motion. From Eq. (2.143), the general solution is obtained immediately as follows.

$$\Delta r = \Delta r_0 e^{i\omega t}. \quad (2.144)$$

In the case where the mass elements move adiabatically, i.e., $\nabla_e = \nabla_{\text{ad}}$, the corresponding

$\omega = \omega_{\text{ad}}$ is called the Brunt-Väisälä frequency, and is given as follows.

$$\omega_{\text{ad}}^2 = \frac{g\delta}{H_P} \left(\nabla_e - \nabla + \frac{\varphi}{\delta} \nabla_\mu \right). \quad (2.145)$$

If a mass element moves into a dynamically unstable layer, i.e., if the Ledoux criterion given by Eq. (2.140) is not satisfied, then $\omega_{\text{ad}}^2 < 0$, and from Eq. (2.144), Δr grows exponentially with time. This result is consistent with our previous discussion of dynamical instability. The Brunt-Väisälä frequency also plays an important role in non-radial oscillations.

In general, for mass elements $DT \neq 0$, and their motion is not adiabatic. If the thermal adjustment time is longer than the oscillation period, the following relation holds.

$$DT = -\frac{T}{H_P} (\nabla_e - \nabla) \Delta r. \quad (2.146)$$

In the dynamically stable case, $\frac{D\rho}{\Delta r} > 0$, and thus the stability condition in Eq. (2.140) is satisfied. If the layer is chemically homogeneous, Eq. (2.140) implies that $\nabla_e - \nabla > 0$, which means that when $\Delta r > 0$, we have $DT < 0$. That is, above its equilibrium position, the element is cooler than the surrounding medium and therefore gains heat via radiation. This reduces both $\nabla_e - \nabla$ and $D\rho$, ultimately leading to a decrease in the oscillation amplitude. Therefore, this type of heat exchange results in a slow damping of the oscillation. This can be represented by a small imaginary part of ω , while the real part of the oscillatory term remains close to the Brunt-Väisälä frequency under our assumptions.

If the layer is chemically inhomogeneous, two possibilities arise. One possibility is that if $\nabla_e - \nabla > 0$, damping occurs due to radiative heat exchange, as in the previous discussion. The other possibility is that ∇_μ is sufficiently large so that Eq. (2.139) is satisfied while $\nabla_e - \nabla < 0$. In this case, conversely, when $\Delta r > 0$, we have $DT > 0$, so the mass element loses energy through radiation. This increases both $\nabla_e - \nabla$ and $D\rho$, ultimately leading to an increase in the amplitude. That is, the oscillation is slowly amplified. This case corresponds to vibrational instability.

Therefore, even if a layer is dynamically stable, it can be vibrationally unstable. Specifically, when the Ledoux criterion is satisfied but the Schwarzschild criterion is not, i.e.,

$$\nabla_{\text{ad}} < \nabla < \nabla_{\text{ad}} + \frac{\varphi}{\delta} \nabla_\mu, \quad (2.147)$$

the layer is dynamically stable but vibrationally unstable. This corresponds to the condition under which semiconvection occurs, as discussed earlier in Eq. (2.140). As in the case of dynamical instability, this is only a local criterion. The response of neighboring layers and the growth of the oscillation amplitude itself can affect convective motion by mixing the chemical composition.

Dynamical instability arises from the purely adiabatic motion of the elements, and from Eq. (2.143) we find that it grows on a timescale of $(H_P/g)^{1/2}$. In contrast, vibrational instability

originates from heat exchange between the elements and the surrounding medium. Therefore, one may expect that the timescale of vibrational instability depends on the thermal adjustment time.

Let us consider the case where $DT > 0$, so that the element loses energy via radiation. In addition to the radial energy flux $\vec{F}$, a local and non-radial flux $\vec{f}$ can transfer energy from the elements to the surrounding medium. As discussed in Section 2.7, from Fick's law and the Stefan–Boltzmann law, the magnitude f of $\vec{f}$ is given as follows.

$$f = \frac{4acT^3}{3\kappa\rho} \left| \frac{\partial T}{\partial n} \right|, \quad (2.148)$$

where $\frac{\partial}{\partial n}$ denotes the derivative along the normal direction to the surface of the element. If we assume that the element is a uniform spherical parcel with diameter d , then $\frac{\partial T}{\partial n} \approx \frac{2DT}{d}$. Therefore, the energy emitted per unit time through radiation across the surface area S of the element, denoted by λ , i.e., the luminosity of the element, is given as follows.

$$\lambda = Sf = \frac{8acT^3}{3\kappa\rho} \frac{S}{D} DT. \quad (2.149)$$

This corresponds to the time variation of the enthalpy of the element. In the situation we consider, the pressure difference between the element and the surrounding medium is always zero, and when $dP = 0$, the change in enthalpy is equal to the heat exchanged. Therefore, from Eq. (2.40),

$$\rho V c_P \frac{\partial T}{\partial t} \simeq \rho V c_P \frac{\partial DT}{\partial t} = -\lambda. \quad (2.150)$$

Here, V is the volume of the element, and ρV appears because Eq. (2.40) is defined per unit mass. The approximation $\frac{\partial T}{\partial t} \simeq \frac{\partial DT}{\partial t}$ is justified because the temperature change of the surrounding medium due to the energy emitted by the element is negligible.

We define the thermal adjustment time τ_{adj} as follows.

$$\frac{DT}{\tau_{\text{adj}}} = -\frac{\partial DT}{\partial t}. \quad (2.151)$$

From Eqs. (2.149) and (2.150),

$$\tau_{\text{adj}} = \frac{\kappa\rho^2 c_P d^2}{16acT^3} = \frac{\rho V c_P DT}{\lambda}. \quad (2.152)$$

This is obtained by dividing the excess thermal energy of the element relative to the surrounding medium by the luminosity, and is identical to the Kelvin-Helmholtz timescale given by Eq. (2.72).

Rewriting the condition assumed at the beginning of the discussion of vibrational instability, we have $\tau_{\text{adj}} \gg 1/\omega_{\text{ad}}$.

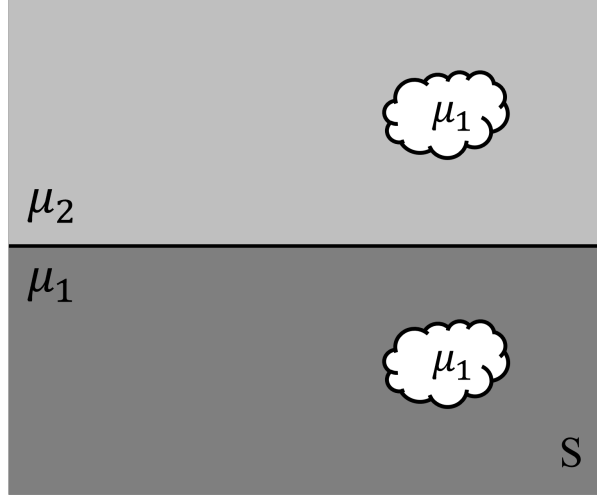


Figure 2.6: Schematic diagram of a mass element moving between layers with different chemical compositions.

2.8.3. Secular Instability

Secular instability, or thermal instability, arises from heat leakage. As in the case of vibrational instability, it can occur even when the system is dynamically stable.

Consider a situation where $D\mu \neq 0$ but $\nabla_\mu = 0$, as illustrated in Fig. 2.6. When the mass element is in mechanical equilibrium, we have $DP = D\rho = 0$. In this case, from the equation of state for ρ given by Eq. (2.133), we obtain the following relation.

$$\delta \frac{DT}{T} = \varphi \frac{D\mu}{\mu}. \quad (2.153)$$

In typical stellar interior conditions, $\delta, \varphi > 0$, so if $D\mu > 0$, the parcel is hotter than the surrounding medium and releases heat to it. The energy loss increases ρ due to the condition $DP = 0$, and the parcel sinks until $D\rho = 0$ is restored. Conversely, if $D\mu < 0$, the parcel rises.

Under the assumed conditions, if there is no heat exchange, then from Eq. (2.153) we have $DT = \text{const.}$, and thus the parcel continues to move with velocity v_μ . Even in the absence of heat exchange, the displacement of the element induces adiabatic expansion or compression, which changes DT . In addition, radiative heat exchange directly modifies DT .

When the time variation of the temperature of the surrounding medium can be neglected, the following result is obtained from Eqs. (2.136), (2.146) and (2.151).

$$\frac{1}{T} \frac{\partial DT}{\partial t} = (\nabla_{\text{ad}} - \nabla) \frac{\partial \ln P}{\partial t} - \frac{DT}{T \tau_{\text{adj}}}. \quad (2.154)$$

In addition, from Eq. (2.136), v_μ can be expressed as follows.

$$\frac{\partial \ln P}{\partial t} = -\frac{v_\mu}{H_P}. \quad (2.155)$$

In the case $\frac{\partial DT}{\partial t} = 0$, from Eqs. (2.153) to (2.155) we obtain

$$v_\mu = - \frac{H_P}{(\nabla_{\text{ad}} - \nabla)} \frac{\varphi}{\tau_{\text{adj}}} \frac{D\mu}{\delta \mu}. \quad (2.156)$$

Thus, in this case, when $D\mu > 0$, the parcel sinks on a timescale of τ_{adj} due to radiative energy loss in a dynamically stable layer, i.e., in a region where $\nabla_{\text{ad}} > \nabla$ and the Schwarzschild criterion is satisfied.

A situation such as $D\mu > 0$ can occur in stellar interiors. For example, stars with masses approximately in the range $0.8 M_\odot < M < 2.0 M_\odot$ form a He core after the main sequence, and due to neutrino cooling, the maximum temperature is located not at the center but off-center. If carbon is produced via He burning in this region, a configuration with $D\mu \neq 0$ and $\nabla_\mu = 0$ can arise, corresponding to the conditions assumed at the beginning of the discussion of secular instability.

2.9. Energy Transport by Instability-Driven Motion

Energy transport by convection arises from instability-driven motion and is expressed as follows;

$$\frac{\partial T}{\partial m} = - \frac{T}{P} \frac{GM}{4\pi r^4} \nabla, \quad (2.157)$$

This expression has exactly the same form as that for energy transport by diffusive processes. The difference is that the function ∇ , which depends on the energy transport mechanism, cannot be determined in a simple manner. The function ∇ can be obtained as a solution of a cubic equation in mixing length theory, and in the deep interior, $\nabla \simeq \nabla_{\text{ad}}$.

2.9.1. Mixing Length Theory

The analogue of mixing length theory is similar to the mean free path in thermodynamics. As illustrated in Fig. 2.7, the mean free path of particles in thermodynamics corresponds to the mixing length of a mass element in fluid dynamics. After the parcel has traveled a distance equal to the mixing length, l_m , it mixes with the medium and loses its identity as a parcel. Based on this approach, our goal is to organize the equations involving ∇ and ultimately derive the cubic equation of mixing length theory, which provides an expression for determining ∇ .

First, if the total energy flux at a given point, $F = \frac{l}{4\pi r^2}$, is transported only by the radiative flux, then from Eqs. (2.102) and (2.124) we obtain the following result.

$$F = \frac{4acG}{3} \frac{mT^4}{\kappa r^2 P} \nabla_{\text{rad}}. \quad (2.158)$$

In fact, convection also contributes to F . That is, $F = F_{\text{rad}} + F_{\text{conv}}$. From the discussion in

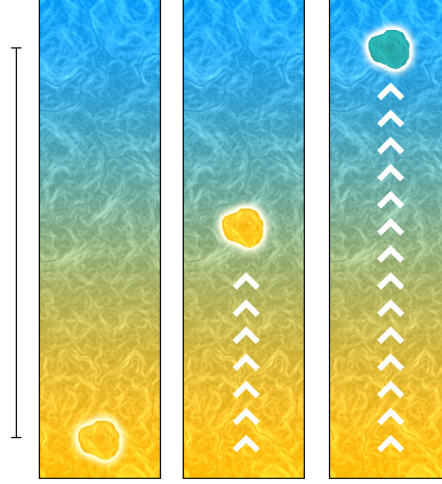


Figure 2.7: Formation, motion, and dissipation of a fluid parcel in mixing length theory [30]. The mass element travels a distance equal to the mixing length l_m (shown as a black bar on the left side of the figure), after which it loses its original characteristics.

Section 2.7, when convection contributes to F , F_{rad} is given by follows;

$$F_{\text{rad}} = \frac{4acG}{3} \frac{mT^4}{\kappa r^2 P} \nabla. \quad (2.159)$$

Note that F_{rad} depends on $\frac{\partial T}{\partial r}$, and when energy transport occurs only through the radiative flux, $\frac{\partial T}{\partial r}$ can be expressed in terms of ∇_{rad} through Eqs. (2.125) to (2.127). As discussed in Eq. (2.141), once convection sets in as a new transport channel, $\nabla_{\text{rad}} > \nabla$.

Under pressure balance, $DP = 0$, the change in enthalpy is equal to the heat exchanged. Therefore, from Eq. (2.40) and the mass flux per unit area per unit time, ρv , the local convective flux can be written as follows.

$$F_{\text{conv}} = \rho v c_p DT. \quad (2.160)$$

Since individual fluid parcels can have different v and DT , the average convective flux can be obtained by using appropriately averaged values of these quantities. The quantity DT can be expressed as follows:

$$\frac{DT}{T} = \frac{1}{T} \frac{\partial (DT)}{\partial r} \frac{l_m}{2} = (\nabla - \nabla_e) \frac{l_m}{2} \frac{1}{H_P}. \quad (2.161)$$

In the first equality, it is assumed that the average distance traveled by mass elements crossing a spherical surface at fixed r is $l_m/2$. The second equality follows from Eq. (2.146). Since the radial buoyant force per unit mass is $k_r = -g \frac{D\rho}{\rho}$, when $D\rho = D\mu = 0$, the work done by buoyancy is obtained from Eqs. (2.133) and (2.161) as follows.

$$\int_0^{l_m/2} k_r dr = \int_0^{l_m/2} g \delta (\nabla - \nabla_e) \frac{r}{H_P} dr = g \delta (\nabla - \nabla_e) \frac{l_m^2}{8H_P}. \quad (2.162)$$

Assuming that all the work done by buoyancy is converted into the kinetic energy of the parcel,

the velocity v of the element is given by the following expression.

$$v^2 = g\delta(\nabla - \nabla_e) \frac{l_m^2}{8H_P}. \quad (2.163)$$

Substituting Eqs. (2.161) and (2.163) into Eq. (2.160), we obtain the following result.

$$F_{\text{conv}} = \frac{1}{4\sqrt{2}} \rho c_P T \sqrt{g\delta} \left(\frac{\nabla - \nabla_e}{H_P} \right)^{3/2} l_m^2. \quad (2.164)$$

Next, let us consider the change in the temperature T_e of a mass element moving with velocity v . As noted in the discussion of secular instability, this change arises from adiabatic expansion or compression and radiative energy exchange with the surrounding medium. From Eq. (2.150), the total temperature change per unit length is given by

$$\left(\frac{dT}{dr} \right)_e = \left(\frac{dT}{dr} \right)_{\text{ad}} - \frac{\lambda}{\rho V c_P v T}. \quad (2.165)$$

Multiplying both sides by H_P/T , we obtain the following result from Eqs. (2.41), (2.136) and (2.138).

$$\nabla_e - \nabla_{\text{ad}} = \frac{\lambda H_P}{\rho V c_P v T}. \quad (2.166)$$

Substituting Eqs. (2.149) and (2.161) into the above expression and rearranging, we obtain

$$\frac{\nabla_e - \nabla_{\text{ad}}}{\nabla - \nabla_e} = \frac{6acT^3}{\kappa \rho^2 c_P l_m v}. \quad (2.167)$$

In the calculation, the heuristic form factor $\frac{l_m S}{Vd} \approx \frac{9}{2l_m}$ has been used. For a sphere with diameter l_m , $\frac{l_m S}{Vd} = \frac{6}{l_m}$.

From the five equations obtained so far, Eqs. (2.158), (2.159), (2.163), (2.164) and (2.167), if the local quantities P , T , ρ , l , m , c_P , ∇_{ad} , and ∇_{rad} are given, then F_{rad} , F_{conv} , v , ∇_e , and ∇ can be determined. Of course, this is possible only once the free parameter l_m has been specified. That is, once the mixing length l_m is given, ∇ can be determined from the results obtained so far. In general, l_m is specified in the following form.

$$l_m = \alpha_{\text{MLT}} H_P, \quad (2.168)$$

where α_{MLT} is the mixing length parameter, a dimensionless constant of $\mathcal{O}(1)$. It is difficult to apply a single universal value of α_{MLT} to all stars. For example, solar calibration shows that $\alpha_{\text{MLT}} \simeq 1.931$ in MESA yields results in good agreement with solar observations [31]. Values of α_{MLT} in the range of about 1.7-2.0 can account for various stars with relatively small errors [32]. In 1D supernova simulations based on GR1D, $\alpha_{\text{MLT}} \simeq 1.5$ gives results most similar to those of three-dimensional supernova simulations [33].

To obtain an expression for determining ∇ , we define the following two dimensionless

quantities.

$$W \equiv \nabla_{\text{rad}} - \nabla_{\text{ad}},$$

$$U \equiv \frac{3acT^3}{\kappa\rho^2 c_P l_m^2} \sqrt{\frac{8H_P}{g\delta}}. \quad (2.169)$$

The physical meaning of W is clear. In the case of U , if we define the remaining factors in Eqs. (2.159) and (2.164) other than the ∇ part as the corresponding conductivities σ_{rad} and σ_{conv} , then from Eqs. (2.11) and (2.136),

$$U = \frac{9}{8} \frac{\sigma_{\text{rad}}}{\sigma_{\text{conv}}} \simeq \frac{\sigma_{\text{rad}}}{\sigma_{\text{conv}}}, \quad (2.170)$$

that is, it corresponds to the ratio of the conductivities. In addition, if we define the time required for a mass element to fall freely over a distance H_P as $\tau_{\text{ff}} \equiv (2H_P/g)^{1/2}$, then from Eq. (2.151),

$$U = \frac{3}{8\sqrt{\delta}} \frac{d^2}{l_m^2} \frac{\tau_{\text{ff}}}{\tau_{\text{adj}}}. \quad (2.171)$$

Here, d is the diameter of the parcel. Assuming that $d \approx l_m$, the factor $3/(8\sqrt{\delta})$ is of $\mathcal{O}(1)$, so this represents the ratio of τ_{ff} to τ_{adj} .

Eliminating v in Eq. (2.167) using Eq. (2.163), we obtain the following relation.

$$\nabla_e - \nabla_{\text{ad}} = 2U \sqrt{\nabla - \nabla_e}. \quad (2.172)$$

Rewriting the left-hand side of this equation as $(\nabla - \nabla_{\text{ad}}) - (\nabla - \nabla_e)$, we see that it is a quadratic equation for $\nabla - \nabla_e$. Therefore, from the quadratic formula, we obtain the solution

$$\sqrt{\nabla - \nabla_e} = -U + \xi. \quad (2.173)$$

Here, ξ is the positive root of the expression given by the quadratic formula, namely,

$$\xi^2 = \nabla - \nabla_{\text{ad}} + U^2. \quad (2.174)$$

Eliminating F_{rad} and F_{conv} using Eqs. (2.158), (2.159) and (2.164), and applying the hydrostatic equilibrium equation Eq. (2.11) and the definition of H_P , Eq. (2.136), to Eq. (2.161), we obtain the following relation.

$$(\nabla - \nabla_e)^{3/2} = \frac{8}{9} U (\nabla_{\text{rad}} - \nabla). \quad (2.175)$$

Substituting Eq. (2.173) into this expression, we obtain the cubic equation of mixing length theory, given by

$$(\xi - U)^3 + \frac{8}{9} U (\xi^2 - U^2 - W) = 0, \quad (2.176)$$

from which ∇ can be determined.

2.9.2. Convective Efficiency

From Eq. (2.172), we define the convective efficiency Γ as follows.

$$\Gamma \equiv \frac{(\nabla - \nabla_e)^{1/2}}{2U} = \frac{\nabla - \nabla_e}{\nabla_e - \nabla_{\text{ad}}}, \quad (2.177)$$

and from Eqs. (2.161) and (2.164),

$$\nabla - \nabla_e = \frac{4H_P}{\rho V c_P \nu T} F_{\text{conv}} A. \quad (2.178)$$

Here, $A = \pi \left(\frac{d}{2}\right)^2 \simeq \frac{1}{4} \pi l_m^2$ is the cross-sectional area of the parcel perpendicular to its direction of motion. Rewriting Γ using Eqs. (2.172) and (2.178), we obtain

$$\Gamma = \frac{4}{3} \frac{F_{\text{conv}} A}{\lambda} \simeq \frac{F_{\text{conv}} A}{\lambda}. \quad (2.179)$$

In the final expression, the numerator corresponds to the energy transported by convection, while λ corresponds to the energy lost of the parcel through radiation. That is,

$$\Gamma \approx \frac{\text{energy transported by convection}}{\text{radiative energy loss of element}}. \quad (2.180)$$

This represents the efficiency of convection. In addition, defining the lifetime of a mass element as $\tau_l = l_m/\nu$, we obtain from Eqs. (2.151), (2.161) and (2.172),

$$\Gamma = \frac{\nabla - \nabla_e}{\nabla_e - \nabla_{\text{ad}}} = 2 \frac{\tau_{\text{adj}}}{\tau_l}. \quad (2.181)$$

That is, Γ approximately corresponds to the ratio of the thermal adjustment time to the lifetime of the parcel.

Let us examine the limiting cases of Γ for Eqs. (2.172), (2.175) and (2.176).

1. $\Gamma \rightarrow \infty$ ($U \rightarrow 0$): From Eq. (2.172), $\nabla_e \rightarrow \nabla_{\text{ad}}$, and from Eq. (2.175), $\nabla \rightarrow \nabla_{\text{ad}}$. In this case, Eq. (2.162) shows that the magnitude of the buoyancy becomes very small. There are two possible situations. The first is when $\nabla \rightarrow \nabla_{\text{ad}}$ even in the absence of convective instability, as in cases where the opacity is very low or conduction is very strong. Such situations occur only in limited cases, for example in dense, highly degenerate white dwarfs and neutron stars. The second possibility is when convection is extremely strong. Weak buoyancy cannot initiate convection or make it stronger, but as can be seen from Eqs. (2.154) and (2.162), if buoyancy can offset thermal dissipation, then it is possible to sustain strong convection. In this case, almost all the energy is transported by convection, and radiation can be almost neglected. During stellar evolution, most cases with $\nabla \rightarrow \nabla_{\text{ad}}$ correspond to strong convection. Both of the above situations occur mainly in the dense central regions of stars. Since $\nabla \rightarrow \nabla_{\text{ad}}$ in this situation, there is no need to calculate ∇ .

2. $\Gamma \rightarrow 0$ ($U \rightarrow \infty$): Since the left-hand side of Eq. (2.175) must remain finite, we have $\nabla \rightarrow \nabla_{\text{rad}}$. In this case, by contrast, most of the energy is transported by radiation. Since $\nabla \rightarrow \nabla_{\text{rad}}$, there is likewise no need to calculate ∇ . Such situations arise mainly near the low-density photosphere.

The region between the two limiting cases above, i.e., the region where $\nabla_{\text{rad}} > \nabla > \nabla_{\text{ad}}$, is called superadiabatic.

Γ and W are quantities defined in the context of mixing length theory. When quantifying convection in stars, it is more common to use dimensionless numbers frequently employed in fluid dynamics, such as the Reynolds number (Re), Péclet number ($\text{Pe} = \frac{\text{advective transport rate}}{\text{diffusive transport rate}}$), and Prandtl number ($\text{Pr} = \frac{\text{momentum diffusivity}}{\text{thermal diffusivity}}$) [34]. For example, the Reynolds number, which represents the ratio of inertial forces to viscous forces arising from relative velocity differences within a fluid, is given by

$$\text{Re} = \frac{\rho v l_m}{\eta}. \quad (2.182)$$

Here, η is the dynamic viscosity. When $\text{Re} \rightarrow 0$, the flow is laminar (sheet-like), while when $\text{Re} \rightarrow \infty$, the flow is turbulent. In general, Re takes large values in stellar interiors and decreases toward the outer regions. This implies that fluid parcels undergoing convective motion in stellar interiors can travel large distances l_m .

In strongly turbulent regimes, i.e., when Re is large, the contribution of nonlinear terms in equations describing fluid motion, such as the Navier-Stokes equation, becomes significant. Including these nonlinear terms makes numerical calculations more difficult and demands substantial computational resources. However, for most of the stellar evolution process, the contribution of nonlinear terms can be neglected, and a linearized approximation remains valid. This is because the velocity of convective motion is much smaller than the sound speed of the stellar medium.

2.10. Chemical Composition

The variation of the chemical composition can be expressed as follows.

$$\left(\frac{\partial X_i}{\partial t} \right) = \left(\frac{\partial X_i}{\partial t} \right)_{\text{nuclear}} + \left(\frac{\partial X_i}{\partial t} \right)_{\text{diffusion}} + \left(\frac{\partial X_i}{\partial t} \right)_{\text{mix}}. \quad (2.183)$$

Each term corresponds to contributions from nuclear reactions, diffusion, and mixing due to convective motion. The mass fraction X_i for nucleus i is defined as follows.

$$X_i \equiv \frac{m_i n_i}{\rho}. \quad (2.184)$$

The following conservation law holds for X_i .

$$\sum_i X_i = 1. \quad (2.185)$$

This expression is directly related to mass conservation. In astrophysics, the mass fractions of isotopes are typically denoted as follows.

$$X \equiv X_{\text{H}}, \quad Y \equiv X_{\text{He}}, \quad Z \equiv 1 - X - Y. \quad (2.186)$$

Note that this should not be confused with the number of i per nucleon, Y_i . The quantity Z is referred to as the metallicity; in the context of astrophysics, all elements other than the isotopes of hydrogen and helium are called metals. The known solar values of X , Y , and Z are given as follows [35].

$$X_{\odot} = 0.7438, \quad Y_{\odot} = 0.2423, \quad Z_{\odot} = 0.0139. \quad (2.187)$$

Before proceeding with the discussion of chemical composition, we briefly mention one advantage of using the Lagrangian description. In fluid dynamics, the material derivative is given by

$$\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + \vec{v} \cdot \nabla Q. \quad (2.188)$$

Here, Q is an arbitrary physical quantity of a fluid element. In the Lagrangian description, $Q = Q(m, t)$, and $\frac{dQ}{dt}$ itself is a fundamental variable. That is, $\frac{dQ}{dt}$ directly describes the change of the physical quantity of the fluid element.

In contrast, in the Eulerian description, $Q = Q(r, t)$, and in addition to the temporal change $\frac{\partial Q}{\partial t}$ of the physical quantity of a fluid parcel, an advection term, $\vec{v} \cdot \nabla Q$, appears, corresponding to the spatial variation of the fluid element. This advantage is particularly evident when describing chemical components, and Eq. (2.183) is expressed in this form because we adopt the Lagrangian formulation. In the Eulerian formulation, the expression becomes more complicated.

2.10.1. Nuclear Reactions in Radiative Regions

In radiative regions, convection is absent, and if diffusion is neglected, there is no mass exchange between different mass shells. Therefore, X_i changes only through nuclear reactions.

Nuclei i can be created or destroyed through nuclear reactions such as fusion or fission. We define the reaction rate r_{ab} as follows.

$$r_{ab} \equiv \frac{\text{Number of reactions that transform nuclei from type } a \text{ into type } b}{(\text{unit time}) \times (\text{unit volume})}. \quad (2.189)$$

From this, the time variation of X_i is given from Eq. (2.184) as follows.

$$\frac{\partial X_i}{\partial t} = \frac{m_i}{\rho} \left[\sum_j r_{ji} - \sum_k r_{ik} \right]. \quad (2.190)$$

This expression corresponds to the case where one nucleus i participates in the reaction. Therefore, if a given reaction channel produces or destroys more than one i , the corresponding term should be appropriately normalized.

In general, a reaction $a \rightarrow b$ either absorbs or releases energy. Most reactions occurring in astrophysical environments are exothermic. In Section 2.6, we introduced ε , the energy change per unit mass per unit time. The quantity ε depends on the reaction rates as follows.

$$\varepsilon = \sum_{a,b} \varepsilon_{ab} = \frac{1}{\rho} \sum_{a,b} e_{ab} r_{ab}. \quad (2.191)$$

Here, e_{ab} is the energy released per reaction $a \rightarrow b$. For the reaction $a \rightarrow b$, we define the energy generated per unit mass of a as follows.

$$q_{ab} = \frac{e_{ab}}{m_a} \quad (2.192)$$

Using this, Eq. (2.190) can be rewritten as follows.

$$\frac{\partial X_i}{\partial t} = \sum_j \frac{\varepsilon_{ji}}{q_{ji}} - \sum_k \frac{\varepsilon_{ik}}{q_{ik}}. \quad (2.193)$$

This corresponds to $\left(\frac{\partial X_i}{\partial t} \right)_{\text{nucl.}}$. When N different nuclei are simultaneously subject to nuclear transformations, these equations form a system of N differential equations known as a nuclear reaction network.

As a simple example, consider the production of helium through hydrogen burning.

$$\frac{\partial X}{\partial t} = -\frac{\varepsilon_H}{q_H}, \quad \frac{\partial Y}{\partial t} = -\frac{\partial X}{\partial t}. \quad (2.194)$$

Here, ε_H and q_H are the corresponding quantities for the conversion of hydrogen into helium.

2.10.2. Diffusion

Microscopic effects such as diffusion can also vary the chemical composition. Depending on the driving force, various types of diffusive effects may arise, but here we consider concentration diffusion, temperature diffusion, and pressure diffusion.

Let us begin with concentration diffusion. Since we are dealing here with phenomena on microscopic scales, we adopt the fluid rest frame based on the Lagrangian formulation. We

define the concentration of particles of species i as $c_i \equiv \frac{n_i}{n_{\text{tot}}}$. According to Fick's law,

$$\vec{j}_{D_i} = -D\nabla c_i. \quad (2.195)$$

Here, D is the diffusion coefficient. Since the flux of concentration is $\vec{j}_D = c\vec{v}_D$, where $\vec{v}_D$ is the diffusion velocity in the fluid rest frame, we obtain

$$\vec{v}_D = -\frac{D}{c}\nabla c. \quad (2.196)$$

From the continuity equation

$$\frac{\partial c}{\partial t} = \nabla \cdot \vec{j}_D, \quad (2.197)$$

and using Eq. (2.195), we obtain the following result when D is constant.

$$\frac{\partial c}{\partial t} = D\nabla^2 c. \quad (2.198)$$

From a dimensional analysis of this equation, the characteristic timescale τ_D can be estimated approximately as follows.

$$\tau_D \approx \frac{L_D^2}{D}. \quad (2.199)$$

Here, L_D is the characteristic length scale associated with variations in c .

Generalizing Eq. (2.196), we can include temperature diffusion and pressure diffusion as below.

$$\vec{v}_D = -\frac{1}{c}D(\nabla c + k_T\nabla \ln T + k_P\nabla \ln P). \quad (2.200)$$

Of course, the coefficients k_T and k_P must be specified appropriately.

To discuss temperature diffusion, let us assume in Cartesian coordinates that ∇T is directed along the z -axis. Consider a flux of particles with mean free path l and speed $\bar{v}$ crossing the x - y plane at $z = 0$. Let θ be the angle between the z -axis and a particle contributing to the flux. Then the flux j^+ crossing in the $+\hat{z}$ direction receives a contribution from particles whose last scattering occurred at $z = -l \cos \theta \equiv -l\mu$, with velocity component $v_z = \bar{v}\mu$.

$$j_\mu^+ = c(-l\mu)\bar{v}(-l\mu)\mu = \left(c(0) - l\mu\frac{\partial c}{\partial z}\right)\left(\bar{v}(0) - l\mu\frac{\partial \bar{v}}{\partial z}\right)\mu. \quad (2.201)$$

The subscript μ indicates the dependence on μ , and in the second equality a Taylor expansion has been performed. Taking the angular average over the lower hemisphere of the $z = 0$ plane, we obtain

$$\begin{aligned} j^+ &= \frac{1}{4\pi} \int d\Omega j_\mu^+ \simeq \frac{1}{2} \int_{-1}^0 d\mu \left[c(0)\bar{v}(0)\mu - l\mu^2 \left(c(0)\frac{\partial \bar{v}}{\partial z} + \bar{v}(0)\frac{\partial c}{\partial z} \right) \right] \\ &= \frac{1}{2} \left[\frac{1}{2}c(0)\bar{v}(0) - \frac{1}{3}l \left(\bar{v}(0)\frac{\partial c}{\partial z} + c(0)\frac{\partial \bar{v}}{\partial z} \right) \right]. \end{aligned} \quad (2.202)$$

In the calculation, the higher-order term $\frac{\partial c}{\partial z} \frac{\partial \bar{v}}{\partial z}$ has been neglected. By the same method, one can obtain j^- and then compute the net flux j as follows.

$$j = j^+ - j^- = -\frac{1}{3}l \left(\bar{v}(0) \frac{\partial c}{\partial z} + c(0) \frac{\partial \bar{v}}{\partial z} \right). \quad (2.203)$$

If $\bar{v}$ is constant, we recover the same expression as in Eq. (2.112).

The relative diffusion velocity between two different species of particles is given as follows.

$$v_{D1} - v_{D2} = \frac{j_1}{c_1} - \frac{j_2}{c_2}. \quad (2.204)$$

The subscripts on the physical quantities indicate that they correspond to the respective particle species. The rms velocity of gas i obeying the Maxwell-Boltzmann distribution is given by

$$\begin{aligned} v_{\text{rms},i} &= \left[\int_0^\infty v^2 f(v) d^3v \right]^{1/2} \\ &= \left[4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4 e^{-mv^2/2k_B T} dv \right] \\ &= \sqrt{\frac{3k_B T}{m}} \\ &= \sqrt{\frac{3\mathcal{R}T}{\mu_i}}. \end{aligned} \quad (2.205)$$

In the calculation, we used $\int_{-\infty}^\infty e^{-Ax^2} dx = \sqrt{\frac{\pi}{A}}$. Using this result together with Eqs. (2.203) and (2.204), the relation $c_1 + c_2 = 1$ for a binary mixture, and $\frac{\partial v_{\text{rms}}}{\partial z} = \frac{\partial v_{\text{rms}}}{\partial T} T \frac{\partial \ln T}{\partial z}$, we obtain the following result.

$$v_{D1} - v_{D2} = -\frac{D}{c_1 c_2} \left(\frac{\partial c_1}{\partial z} + k_T \frac{\partial \ln T}{\partial z} \right). \quad (2.206)$$

D and k_T are given by

$$\begin{aligned} D &= \frac{1}{3} (c_2 l_1 \bar{v}_1 + c_1 l_2 \bar{v}_2) = \sqrt{\frac{\mathcal{R}T}{3}} \left(\frac{c_2 l_1}{\sqrt{\mu_1}} + \frac{c_1 l_2}{\sqrt{\mu_2}} \right), \\ k_T &= \frac{1}{2} \frac{l_1 \sqrt{\mu_2} - l_2 \sqrt{\mu_1}}{l_1 c_1 \sqrt{\mu_2} + l_2 c_2 \sqrt{\mu_1}} c_1 c_2. \end{aligned} \quad (2.207)$$

It is clear that $D > 0$ and that the magnitude of k_T is $\mathcal{O}(1)$ or less, while the sign of k_T depends on the composition inside the star.

For a stellar interior composed of hydrogen and helium, such as the Sun, if we define $v_D = v_H - v_{\text{He}}$, then in a generally ionized H-He mixture have $k_T > 0$, so for temperature diffusion we have $\text{sign}(v_D) = -\text{sign}\left(\frac{\partial \ln T}{\partial z}\right)$. That is, hydrogen diffuses in the direction of decreasing temperature, namely outward in the star.

Next, let us discuss pressure diffusion. First, consider an isothermal layer in hydrostatic

equilibrium in the presence of a gravitational field, i.e., the final hydrostatic equilibrium state. From Eqs. (2.11), (2.43) and (2.136), the pressure scale height H_{P_i} of nucleus i is given by

$$H_{P_i} = \frac{P}{g\rho_i} = \frac{\mathcal{R}T}{g\mu_i}. \quad (2.208)$$

Since H_{P_i} becomes smaller as μ_i increases, heavier nuclei move below lighter nuclei. That is, under the same temperature and the same gravitational field, nuclei with different μ_i have different H_{P_i} , and therefore compositional separation occurs.

However, actual pressure diffusion is not determined simply by the pressure gradient alone. More precisely, it is determined by the equation of motion for each species i , and the factors affecting it are as follows.

1. Partial pressure gradient: diffusion driven by the pressure gradient of each species itself.
2. Gravity: due to differences in the molecular weight μ_i of each species, heavier species settle below lighter ones.
3. Electric field: since electrons and ions respond differently to gravity, an electric field is induced so that charge separation does not occur.
4. Inter-species collisional friction: when one species moves, Coulomb collisions with other species act like a frictional force and can slow down the motion.
5. Thermal coupling: residual heat flow can affect the pressure coefficient itself.

Taking these effects into account, the Burgers momentum equation of nucleus i is given by [36]

$$\frac{dP_i}{dr} + \rho_i g - \rho_{ei} E = \sum_{j \neq i} K_{ij} [(v_j - v_i) + 0.6(x_{ij}r_i - y_{ij}r_j)]. \quad (2.209)$$

Here, the charge density is defined by $\rho_{ei} \equiv n_i q_i$, K_{ij} is the friction coefficient between the two species i and j , and v_i and v_j are the velocities of the respective species in the fluid rest frame. Also, $x_{ij} \equiv \mu_{ij}/m_i$ and $y_{ij} \equiv \mu_{ij}/m_j$, where $\mu_{ij} \equiv \frac{m_i m_j}{m_i + m_j}$ is the reduced mass for species i and j .

A detailed discussion of this lies beyond the scope of the basics, so we conclude our discussion of pressure diffusion here. What should be remembered is that, despite the complicated origin of pressure diffusion, it can still be written in the form of Eq. (2.200).

So far, we have examined three types of diffusion—concentration diffusion, thermal diffusion, and pressure diffusion—as well as the corresponding expressions for their diffusion velocities. We are now in a position to relate them to the time variation of the mass fraction X_i . From Eq. (2.184), the mass density of nucleus i is given by

$$\rho_i = \rho X_i. \quad (2.210)$$

Neglecting the source term due to nuclear reactions, the continuity equation for species i is given by

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i \vec{u}_i) = 0. \quad (2.211)$$

Here, $\vec{u}_i$ is the velocity of nucleus i in the lab frame. If the mean fluid velocity in the lab frame is defined as $\vec{u}$, then the relation between $\vec{u}_i$ and the velocity $\vec{v}_i$ of species i in the fluid rest frame is given by

$$\vec{u}_i = \vec{u} + \vec{v}_i. \quad (2.212)$$

By the definition of the fluid rest frame,

$$\sum_i \rho_i \vec{v}_i = 0. \quad (2.213)$$

Substituting Eqs. (2.210) and (2.212) into Eq. (2.211), we obtain the following result.

$$\frac{\partial \rho X_i}{\partial t} + \nabla \cdot (\rho X_i \vec{u}) + \nabla \cdot (\rho X_i \vec{v}_i) = 0. \quad (2.214)$$

For the first two terms on the left-hand side of the above equation, using the material derivative given by Eq. (2.188), we obtain

$$\begin{aligned} \frac{\partial \rho X_i}{\partial t} + \nabla \cdot (\rho X_i \vec{u}) &= \rho \frac{\partial X_i}{\partial t} + X_i \frac{\partial \rho}{\partial t} + \rho \vec{u} \cdot \nabla X_i + X_i \nabla \cdot (\rho \vec{u}) \\ &= \rho \frac{dX_i}{dt} - \rho \vec{u} \cdot \nabla X_i + X_i \frac{\partial \rho}{\partial t} + \rho \vec{u} \cdot \nabla X_i + X_i \nabla \cdot (\rho \vec{u}) \\ &= \rho \frac{dX_i}{dt} + X_i \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) \right]. \end{aligned} \quad (2.215)$$

At this stage, the terms inside the square brackets are canceled by the continuity equation for the total fluid, given by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0. \quad (2.216)$$

Therefore, substituting Eqs. (2.215) and (2.216) into Eq. (2.214), we obtain

$$\rho \frac{dX_i}{dt} = -\nabla \cdot (\rho X_i \vec{v}_i). \quad (2.217)$$

Assuming spherical symmetry, simplifying the right-hand side of the above equation, and using the fact that $\frac{d}{dt}$ can be written as $\frac{\partial}{\partial t}$ in the local fluid rest frame, we finally obtain

$$\frac{\partial X_i}{\partial t} = -\frac{1}{\rho r^2} \frac{\partial}{\partial r} (r^2 \rho X_i v_i). \quad (2.218)$$

Thus, the time variation of X_i can be expressed in terms of the diffusion velocity in the fluid rest frame.

Finally, the variation of the chemical composition due to concentration, temperature, and pressure diffusion can be written as follows [36].

$$\frac{\partial X_i}{\partial t} = -\frac{1}{\rho r^2} \frac{\partial}{\partial r} \left[r^2 X_i T^{5/2} \left(A_P(i) \frac{\partial \ln P}{\partial r} + A_T(i) \frac{\partial \ln T}{\partial r} + \sum_{j \neq e, {}^4\text{He}} A_j(i) \frac{\partial \ln C_j}{\partial r} \right) \right]. \quad (2.219)$$

Here, $A_P(i)$, $A_T(i)$, and $A_J(i)$ are the diffusion constants for pressure, temperature, and concentration diffusion, respectively, and $C_i \equiv \frac{c_i}{c_e} = \frac{n_i}{n_e}$. The factor $T^{5/2}$ appears because, in a Coulomb plasma, the diffusivity scales approximately as $D \propto T^{5/2}/\rho$.

2.10.3. Mixing in Convective Regions

In convective regions, convection causes mass exchange between different mass shells. In general, the change in chemical composition due to convection is faster than that due to nuclear reactions, so in turbulent convective regions,

$$\frac{\partial X_i}{\partial t} = 0, \quad (2.220)$$

can be assumed. Under the Lagrangian description, let us consider a convective zone extending between m_1 and m_2 . Owing to strong turbulent convective motion, we have $X_i = \bar{X}_i = \text{const.}$ within this region. However, the values of X_i just outside the boundaries at m_1 and m_2 , denoted by X_{i1} and X_{i2} , may differ from $\bar{X}_i$ inside the convective zone and may even be discontinuous. That is, the variation of X_i within the convective layer depends on the evolution of the boundaries m_1 and m_2 . Therefore,

$$\frac{\partial \bar{X}_i}{\partial t} = \frac{1}{m_2 - m_1} \left(\frac{\partial m_2}{\partial t} (X_{i2} - \bar{X}_i) - \frac{\partial m_1}{\partial t} (X_{i1} - \bar{X}_i) + \int_{m_1}^{m_2} \frac{\partial X_i}{\partial t} dm \right). \quad (2.221)$$

The first two terms on the right-hand side represent the change in X_i due to the motion of the boundaries, while the integral term represents the change in X_i within the convective zone due to nuclear reactions.

If nuclear reactions proceed very rapidly, the assumption in Eq. (2.220) is no longer valid. In that case, convective mixing can be approximated as a diffusive process with the diffusion velocity given in Eq. (2.163). Then, the contribution from convective motion can be included by adding the following term to Eq. (2.219).

$$D_c \frac{\partial X_i}{\partial r} = \left(\frac{1}{3} v_c \alpha_{\text{MLT}} H_P \rho r^2 \right) \frac{\partial X_i}{\partial r}. \quad (2.222)$$

In general, D_c is larger by orders of magnitude than the other diffusion constants, so in this case the contributions from other types of diffusion can be neglected.

2.11. Saha Ionization Equation

In Section 2.3, when we briefly reviewed thermodynamics, we defined the mean molecular weight per free electron, μ_e , given by Eq. (2.48), assuming that the gas inside an astrophysical object is fully ionized. This is an appropriate approximation for the hot and dense interiors of stars. However, in low-temperature and low-pressure environments such as stellar envelopes,

hydrogen, helium, and other elements may remain partially or fully neutral.

The Saha ionization equation describes how atoms or ions are distributed among different ionization states in thermal equilibrium. Let us assume that local thermal equilibrium (LTE) holds inside an astrophysical object. As a representative ionization process, consider the reaction in which an atom X^i in the i -th ionization state loses one electron and becomes X^{i+1} , as written below.



When this reaction proceeds sufficiently fast in thermal equilibrium, chemical equilibrium is established. Therefore, the following relation holds among the chemical potentials of the each components.

$$\mu_i = \mu_{i+1} + \mu_e . \quad (2.224)$$

Care should be taken not to confuse this chemical potential with the mean molecular weight μ . This relation implies that the Helmholtz free energy F does not change in an LTE environment, i.e., at fixed T and V . In other words, regardless of the direction in which the reaction in Eq. (2.223) proceeds, the free energy of the system remains unchanged.

Assume that each component can be treated as an ideal gas and that the particles obey the Maxwell-Boltzmann distribution. Then the number density of a given particle species a is given by

$$n_a = \frac{Z_a}{\lambda_a^3} e^{-(E_a - \mu_a)/k_\beta T} . \quad (2.225)$$

Here, Z_a is the partition function associated with the internal degrees of freedom, and λ_a is the thermal de Broglie wavelength. They are given by

$$Z_a = \sum_i g_i e^{-E_i/k_\beta T} , \quad (2.226)$$

$$\lambda_a = \frac{h}{\sqrt{2\pi m_a k_\beta T}} . \quad (2.227)$$

Solving Eq. (2.225) for the chemical potential μ_a , we obtain

$$\mu_a = k_\beta T \ln \left(\frac{n_a \lambda_a^3}{Z_a} \right) + E_a . \quad (2.228)$$

Substituting this result into Eq. (2.224) gives

$$\ln \left(\frac{n_i \lambda_i^3}{Z_i} \right) + \frac{E_i}{k_\beta T} = \ln \left(\frac{n_{i+1} \lambda_{i+1}^3}{Z_{i+1}} \right) + \ln \left(\frac{n_e \lambda_e^3}{Z_e} \right) + \frac{E_{i+1} + E_e}{k_\beta T} . \quad (2.229)$$

The partition functions Z_i and Z_{i+1} depend on the internal energy levels of the corresponding ions, while for a free electron, if only spin degeneracy is considered, we may set $Z_e = 2$. Using

this, the above expression becomes

$$\frac{n_{i+1}n_e}{n_i} = 2 \left(\frac{\lambda_i}{\lambda_{i+1}\lambda_e} \right)^3 \frac{Z_{i+1}}{Z_i} e^{-\chi_i/k_\beta T}. \quad (2.230)$$

Here, $\chi_i = E_{i+1} + E_e - E_i$ is the energy difference between the initial and final states of the reaction. The mass difference between the two ionization states, m_i and m_{i+1} , is approximately of order the electron mass and is therefore negligible compared to the atomic mass. Thus, from Eq. (2.227), $\lambda_i \simeq \lambda_{i+1}$. Writing the electron thermal wavelength explicitly, we finally obtain the Saha ionization equation.

$$\frac{n_{i+1}n_e}{n_i} = 2 \frac{(2\pi m_e k_\beta T)^{3/2}}{h^3} \frac{Z_{i+1}}{Z_i} e^{-\chi_i/k_\beta T}. \quad (2.231)$$

This shows that, for an ideal gas obeying Maxwell-Boltzmann statistics in LTE, the degree of ionization depends on the temperature and number density.

2.12. Degenerate Electron Gas

Whether and to what extent electrons are degenerate inside an astrophysical object has a decisive impact on stellar evolution. Representative examples include whether an interstellar cloud in the PMS phase can successfully reach the ZAMS and become a star, and where He ignition occurs, which in turn determines the subsequent behavior on the H-R diagram. In this section, we examine how the state variables of electrons are determined according to electron degeneracy and what form the EOS takes.

2.12.1. Fully Degenerate Electron Gas

Electrons obey the Fermi-Dirac distribution. For a single electron,

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_\beta T} + 1}. \quad (2.232)$$

In the non-relativistic limit, $E(p) \simeq m_e + \frac{p^2}{2m_e}$. Therefore, in astrophysical contexts, it is often useful to employ the following non-relativistic equivalent form of the chemical potential for a massive particle.

$$\mu_x^{\text{nonrel.}} = \mu_x - m_x. \quad (2.233)$$

Regardless of whether the electron is relativistic or non-relativistic, by defining the kinetic energy of a single electron as $E_{\text{kin}}(p) \equiv E(p) - m_e$, Eq. (2.232) can be written as follows.

$$f(p) = \frac{1}{e^{(E_{\text{kin}}(p)-\mu^{\text{nonrel.}})/k_\beta T} + 1}. \quad (2.234)$$

This is because the Fermi-Dirac distribution is derived from the partition function of the grand canonical ensemble, and the partition function of the grand canonical ensemble

$$Z = \sum_i e^{-(E_i - \mu N_i)/k_B T} \quad (2.235)$$

is invariant under the redefinition $E \rightarrow E - m_e$, since the total energy of electrons occupying state i changes as $E_i \rightarrow E_i - m_e N_i$, provided that the chemical potential is simultaneously redefined as $\mu \rightarrow \mu - m_e$.

The Fermi energy is defined as the energy difference, at absolute zero, between the ground state occupied by the particles and the highest occupied energy state. From Eq. (2.234), in the limit $T \rightarrow 0$,

$$\begin{aligned} f(p) &= 1 \quad \text{for} \quad E_{\text{kin}}(p) \leq \mu^{\text{nonrel.}}, \\ f(p) &= 0 \quad \text{for} \quad E_{\text{kin}}(p) > \mu^{\text{nonrel.}}. \end{aligned} \quad (2.236)$$

The Fermi energy is defined as $E_F \equiv \mu^{\text{nonrel.}}$, and the corresponding momentum is called the Fermi momentum, p_F .

When $E_F \gg k_B T$, the particle can be regarded as completely degenerate. In the limit $T \rightarrow 0$, the electron number density is obtained from Eq. (2.236) as follows.

$$n_e = g_e \int_0^\infty \frac{d^3 p}{(2\pi\hbar)^3} f(p) = \frac{8\pi}{h^3} \int_0^{p_F} dp p^2 = \frac{8\pi}{3h^3} p_F^3. \quad (2.237)$$

In the calculation, we assumed the electron spin degeneracy $g_e = 2$ and an isotropic distribution of electrons in phase space. The internal energy per unit volume of the electrons is given by

$$\begin{aligned} u_e &= g_e \int_0^\infty \frac{d^3 p}{(2\pi\hbar)^3} E_{\text{kin}} f(p) = \frac{8\pi}{h^3} \int_0^{p_F} dp p^2 \left(\sqrt{m_e^2 + p^2} - m_e \right) \\ &= \frac{8\pi}{h^3} m_e^4 \int_0^x d\xi \xi^2 \left(\sqrt{1 + \xi^2} - 1 \right) \\ &= \frac{\pi}{h^3} m_e^4 \left[x(2x^2 + 1) \sqrt{1 + x^2} - \ln \left(x + \sqrt{1 + x^2} \right) - \frac{8}{3} x^3 \right]. \end{aligned} \quad (2.238)$$

where $\xi \equiv p/m_e$ and $x \equiv p_F/m_e$, and the same assumptions as in the calculation of n_e have been used. The ξ integral is evaluated by making the substitution $\xi \equiv \sinh u$ and using $\sinh a \cosh a = \frac{1}{2} \sinh 2a$ and $\sinh^2 \frac{a}{2} = \frac{\cosh a - 1}{2}$. Since the internal energy that can vary in thermodynamics is E_{kin} , we used E_{kin} in the integral. The electron pressure is given by

$$\begin{aligned} P_e &= g_e \int_0^\infty \frac{d^3 p}{(2\pi\hbar)^3} \frac{p^2}{3E} f(p) = \frac{8\pi}{3h^3} \int_0^{p_F} dp \frac{p^4}{\sqrt{m_e^2 + p^2}} \\ &= \frac{8\pi}{3h^3} m_e^4 \int_0^x d\xi \frac{\xi^4}{\sqrt{1 + \xi^2}} \\ &= \frac{\pi}{3h^3} m_e^4 \left[x(2x^2 - 3) \sqrt{1 + x^2} + 3 \ln \left(x + \sqrt{1 + x^2} \right) \right]. \end{aligned} \quad (2.239)$$

where the same assumptions, substitution, and ξ integration as in the calculation of u_e have been used.

By the definition of $x \equiv \frac{p_F}{m_e}$ we introduced above, the case $x \ll 1$ corresponds to the situation in which most electrons are non-relativistic, while $x \gg 1$ corresponds to the situation in which most electrons move relativistically. For $x \ll 1$, the Taylor expansions give $\sqrt{1+x^2} \simeq 1 + \frac{x^2}{2} - \frac{x^4}{8} \dots$ and $\ln(x + \sqrt{1+x^2}) \simeq x - \frac{x^3}{6} + \frac{3x^5}{40} \dots$, and therefore

$$x \rightarrow 0: \quad u_e \rightarrow \frac{4\pi}{5h^3} m_e^4 x^5, \quad P_e \rightarrow \frac{8\pi}{15h^3} m_e^4 x^5. \quad (2.240)$$

From this result, together with Eq. (2.237) and $n_e = \frac{\rho}{\mu_e m_u}$, the EOS of a fully degenerate non-relativistic electron gas is given by

$$P_e = \frac{3^{2/3} h^2}{20\pi^{2/3} m_e m_u^{5/3}} \left(\frac{\rho}{\mu_e} \right)^{5/3} \propto \left(\frac{\rho}{\mu_e} \right)^{5/3}. \quad (2.241)$$

In addition, the following relation holds between P_e and u_e :

$$P_e = \frac{2}{3} u_e. \quad (2.242)$$

Therefore, for a completely ionized gas with fully degenerate, non-relativistic electrons that can be treated as an ideal gas, Eq. (2.54) gives

$$\gamma = \frac{5}{3}. \quad (2.243)$$

For $x \gg 1$, since $\sqrt{1+x^2} \simeq x$,

$$x \rightarrow \infty: \quad u_e \rightarrow \frac{2\pi}{h^3} m_e^4 x^4, \quad P_e \rightarrow \frac{2\pi}{3h^3} m_e^4 x^4. \quad (2.244)$$

Similarly, the EOS of a fully degenerate relativistic electron gas is given by

$$P_e = \frac{3^{1/3} h}{8\pi^{1/3} m_u^{4/3}} \left(\frac{\rho}{\mu_e} \right)^{4/3} \propto \left(\frac{\rho}{\mu_e} \right)^{4/3}. \quad (2.245)$$

In addition, the following relation holds between P_e and u_e :

$$P_e = \frac{1}{3} u_e. \quad (2.246)$$

Therefore, for a completely ionized gas with fully degenerate, relativistic electrons that can be treated as an ideal gas, Eq. (2.54) gives

$$\gamma \sim \frac{4}{3}. \quad (2.247)$$

2.12.2. Partially Degenerate Electron Gas

For a finite temperature, electrons may be non-degenerate or partially degenerate. The electron number density, internal energy per unit volume, and pressure can be written as follows.

$$n_e = \frac{8\pi}{h^3} \int_0^\infty dp \frac{p^2}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}, \quad (2.248)$$

$$u_e = \frac{8\pi}{h^3} \int_0^\infty dp \frac{p^2 E_{\text{kin}}}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}, \quad (2.249)$$

$$P_e = \frac{8\pi}{3h^3} \int_0^\infty dp \frac{1}{E_{\text{kin}} + m_e} \frac{p^4}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}, \quad (2.250)$$

where $\eta \equiv \frac{\mu^{\text{non-rel.}}}{k_\beta T}$ is the degeneracy parameter. The limit $\eta \rightarrow -\infty$ corresponds to the non-degenerate limit, while $\eta \rightarrow \infty$ corresponds to the fully degenerate limit.

When the electrons are non-relativistic, $E_{\text{kin}} \simeq \frac{p^2}{2m_e}$. In this case, n_e , u_e , and P_e are given respectively by

$$n_e = \frac{4\pi}{h^3} (2m_e)^{3/2} \int_0^\infty dE_{\text{kin}} \frac{E_{\text{kin}}^{1/2}}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}, \quad (2.251)$$

$$u_e = \frac{4\pi}{h^3} (2m_e)^{3/2} \int_0^\infty dE_{\text{kin}} \frac{E_{\text{kin}}^{3/2}}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}, \quad (2.252)$$

$$P_e = \frac{8\pi}{3h^3} (2m_e)^{3/2} \int_0^\infty dE_{\text{kin}} \frac{E_{\text{kin}}^{3/2}}{e^{(E_{\text{kin}}/k_\beta T - \eta)} + 1}. \quad (2.253)$$

We define the unnormalized complete Fermi-Dirac integral $F_n(\eta)$ as follows.

$$F_n(\eta) \equiv \int_0^\infty dx \frac{x^n}{e^{x-\eta} + 1}. \quad (2.254)$$

Note that, unlike the conventional complete Fermi-Dirac integral, this expression does not contain the normalization factor $\frac{1}{\Gamma(n+1)}$. Using this expression, n_e , u_e , and P_e can be rewritten as follows.

$$n_e = \frac{8\sqrt{2}\pi}{h^3} m_e^{3/2} (k_\beta T)^{3/2} F_{1/2}(\eta), \quad (2.255)$$

$$u_e = \frac{8\sqrt{2}\pi}{h^3} m_e^{3/2} (k_\beta T)^{5/2} F_{3/2}(\eta), \quad (2.256)$$

$$P_e = \frac{16\sqrt{2}\pi}{3h^3} m_e^{3/2} (k_\beta T)^{5/2} F_{3/2}(\eta) = \frac{2}{3} u_e. \quad (2.257)$$

In the non-degenerate limit, $\eta \rightarrow -\infty$, we have $F_n(\eta) \simeq e^\eta \int_0^\infty dx x^n e^{-x} = e^\eta \Gamma(n+1)$. Thus, $F_{1/2}(\eta) \simeq \frac{\sqrt{\pi}}{2} e^\eta$ and $F_{3/2}(\eta) \simeq \frac{3\sqrt{\pi}}{4} e^\eta$. Substituting these results into Eqs. (2.255) and (2.257) yields the ideal gas EOS for non-relativistic, non-degenerate electrons, $P_e = n_e k_\beta T$.

On the other hand, in the fully degenerate limit $\eta \rightarrow \infty$, the integrand of $F_n(\eta)$ effectively

contains a step function, as discussed in the case of a fully degenerate electron gas, and the same results as before are recovered.

If the electrons are relativistic, $E_{\text{kin}} \simeq p$. In this case, n_e , u_e , and P_e are given respectively by

$$n_e = \frac{8\pi}{h^3} \int_0^\infty dp \frac{p^2}{e^{p/k_\beta T - \eta} + 1} = \frac{8\pi}{h^3} (k_\beta T)^3 F_2(\eta), \quad (2.258)$$

$$u_e = \frac{8\pi}{h^3} \int_0^\infty dp \frac{p^3}{e^{p/k_\beta T - \eta} + 1} = \frac{8\pi}{h^3} (k_\beta T)^4 F_3(\eta), \quad (2.259)$$

$$P_e = \frac{8\pi}{3h^3} \int_0^\infty dp \frac{p^3}{e^{p/k_\beta T - \eta} + 1} = \frac{8\pi}{3h^3} (k_\beta T)^4 F_3(\eta) = \frac{1}{3} u_e. \quad (2.260)$$

For $\eta \rightarrow -\infty$, we have $F_2(\eta) \simeq 2e^\eta$ and $F_3(\eta) \simeq 6e^\eta$. Therefore, the ideal gas EOS for non-relativistic, non-degenerate electrons is also found to be $P_e = n_e k_\beta T$. When $\eta \rightarrow \infty$, the same results as in the previous discussion are again recovered.

For a partially degenerate electron gas, n_e , u_e , and P_e are given as functions of η in both the non-relativistic and relativistic limits. In other words, the state variables of a partially degenerate electron gas can be expressed analytically in terms of the degeneracy parameter. However, no analytic approach is known for the case in which the electron behavior is neither non-relativistic nor relativistic. This is because the relation between E and p cannot be approximated by a simple form. In this case, the state variables of the electron gas should be treated numerically.

2.13. Chandrasekhar Limit

When the primary energy source of a star is nuclear fusion, the energy released by fusion can support the star against gravitational collapse and maintain its structure. If nuclear fusion ceases in the interior, the object begins to collapse under gravity. However, as the density increases and the constituent particles become degenerate, degeneracy pressure can support the object against gravity and stabilize it. The Chandrasekhar limit is the maximum mass of an object that can be supported against gravitational collapse by electron degeneracy pressure.

2.13.1. Lane-Emden Equation

The Lane-Emden equation is the dimensionless Poisson equation for the Newtonian gravitational potential of a spherically symmetric object composed of a polytropic fluid. In thermodynamics, processes satisfying $PV^x = \text{const.}$ are called polytropic processes, and this relation is called the polytropic relation. The thermodynamic processes corresponding to different values of x are as follows.

$$x = 0 : \text{isobaric}, \quad x = 1 : \text{isothermal}, \quad x = \gamma : \text{adiabatic}, \quad x \rightarrow \infty : \text{isochoric}.$$

In astrophysics, the following relation is called the polytropic relation.

$$P = K\rho^\gamma \equiv K\rho^{1+\frac{1}{n}}. \quad (2.261)$$

Here, K is the polytropic constant, γ is the polytropic exponent, and n is the polytropic index defined as follows. All of these quantities are constants.

$$n \equiv \frac{1}{\gamma - 1}. \quad (2.262)$$

From Eqs. (2.241) and (2.245), non-relativistic degenerate electrons correspond to $n = \frac{3}{2}$, while relativistic degenerate electrons correspond to $n = 3$.

In general, as in Eqs. (2.31) and (2.133), the EOS determines ρ as a function of P , T , and μ . If the gas follows a polytropic relation such as Eq. (2.261), however, ρ depends only on P . Therefore, in the classical limit, one can determine P and Φ using the hydrostatic equilibrium equation, Eq. (2.11), together with the Poisson equation, Eq. (2.82). Substituting $g = \frac{\partial\Phi}{\partial r}$ and Eq. (2.261) into Eq. (2.11) gives

$$\frac{\partial\Phi}{\partial r} = -\gamma K\rho^{\gamma-2} \frac{\partial\rho}{\partial r}. \quad (2.263)$$

For $\gamma \neq 1$, integrating both sides of the above equation and imposing the boundary condition $\Phi = 0$ at $\rho = 0$ gives

$$\rho = \left(\frac{-\Phi}{(n+1)K} \right)^n. \quad (2.264)$$

Substituting this result into the Poisson equation under spherical symmetry, we obtain

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\Phi}{\partial r} \right) = 4\pi G \left(\frac{-\Phi}{(n+1)K} \right)^n. \quad (2.265)$$

Now define the following dimensionless parameters.

$$\theta \equiv \frac{\Phi}{\Phi_c} = \left(\frac{\rho}{\rho_c} \right)^{1/n}, \quad \xi \equiv \alpha r, \quad \alpha \equiv \sqrt{\frac{4\pi G}{(n+1)K} \rho_c^{(n-1)/n}}. \quad (2.266)$$

The subscript c denotes the center. Thus, at $r = 0$, we have $\xi = 0$, $\rho = \rho_c$, $\Phi = \Phi_c$, and $\theta = 1$. Using these definitions, Eq. (2.265) can be rewritten, finally yielding the Lane-Emden equation.

$$\frac{1}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial\theta}{\partial \xi} \right) + \theta^n = 0. \quad (2.267)$$

For the case $\gamma = 1$, i.e., the isothermal case, integrating Eq. (2.263) and imposing the boundary condition $\Phi_c = 0$ gives

$$\ln \left(\frac{\rho}{\rho_c} \right) = -\frac{\Phi}{K}. \quad (2.268)$$

We define the dimensionless parameters as follows.

$$\theta \equiv \frac{\Phi}{K}, \quad \xi \equiv \alpha r, \quad \alpha \equiv \sqrt{\frac{4\pi G \rho_c}{K}}. \quad (2.269)$$

Similarly, substituting these parameters and Eq. (2.268) into the spherically symmetric Poisson equation and rearranging, we obtain

$$\frac{1}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial \theta}{\partial \xi} \right) - e^{-\theta} = 0. \quad (2.270)$$

This equation is called the isothermal Lane-Emden equation.

For $\gamma \neq 1$, Taylor expanding θ in the Lane-Emden equation around $\xi = 0$ gives

$$\theta(\xi) = 1 + a_1 \xi + a_2 \xi^2 + a_3 \xi^3 + \dots. \quad (2.271)$$

For Eq. (2.267) to remain finite at $\xi = 0$, we must have $\frac{\partial \theta}{\partial \xi} = 0$, and therefore $a_1 = 0$. Substituting this result into Eq. (2.267) gives

$$6a_2 + 12a_3 \xi + 20a_4 \xi^2 + \dots = - \left(1 + a_2 \xi^2 + a_3 \xi^3 + \dots \right)^n. \quad (2.272)$$

From the condition $\theta = 1$ at $\xi = 0$, we obtain $a_2 = -\frac{1}{6}$. In addition, comparing the first-order terms in ξ on the left- and right-hand sides, the left-hand side depends on a_3 , while the right-hand side contains no first-order term in ξ . Therefore, $a_3 = 0$, and by induction all odd-order coefficients vanish. Thus,

$$\theta(\xi) = \sum_{k=0}^{\infty} a_k \xi^{2k}. \quad (2.273)$$

Substituting this expression into the Lane-Emden equation gives

$$\sum_{k=1}^{\infty} 2k(2k+1) a_k \xi^{2k-2} = - \left(\sum_{k=0}^{\infty} a_k \xi^{2k} \right)^n. \quad (2.274)$$

Defining $k \equiv m+1$ and rewriting the above equation, we obtain

$$\sum_{m=-0}^{\infty} 2(m+1)(2m+3) a_{m+1} \xi^{2m} = - \left(\sum_{m=0}^{\infty} b_m \xi^{2m} \right)^n, \quad (2.275)$$

where b_m is given by the n -fold Cauchy product as follows.

$$b_m \equiv \sum_{j_1+j_2+\dots+j_n=m} a_{j_1} a_{j_2} \dots a_{j_n} = \sum_{\substack{q_0+q_1+\dots+q_m=n \\ q_1+2q_2+\dots+m q_m=m}} \frac{n!}{q_0! q_1! \dots q_m!} a_0^{q_0} a_1^{q_1} \dots a_m^{q_m}. \quad (2.276)$$

That is, as n increases, it takes the following form.

$$b_m^{(n=2)} = \sum_{j=0}^m a_j a_{m-j}, \quad b_m^{(n=3)} = \sum_{j=0}^m b_j^{(n=2)} a_{m-j}, \quad \dots, \quad b_m^{(n)} = \sum_{j=0}^m b_j^{(n-1)} a_{m-j}. \quad (2.277)$$

Therefore, for arbitrary m , the following relation holds between the coefficients a and b .

$$a_{m+1} = -\frac{b_m}{2(m+1)(2m+3)}. \quad (2.278)$$

Using this relation, Eq. (2.271) can be written up to fourth order as follows.

$$\theta(\xi) = 1 - \frac{1}{6}\xi^2 + \frac{n}{120}\xi^4 + \dots. \quad (2.279)$$

For specific values of n , this result has the following analytic expressions.

$$\begin{aligned} n=0: \quad \theta(\xi) &= 1 - \frac{1}{6}\xi^2, \\ n=1: \quad \theta(\xi) &= \frac{\sin \xi}{\xi}, \\ n=5: \quad \theta(\xi) &= \frac{1}{\sqrt{1+\xi^2/3}}. \end{aligned} \quad (2.280)$$

From the definition of $\theta(\xi)$, the condition $\rho = 0$ corresponds to $\theta(\xi) = 0$. The value $\xi = \xi_n$ satisfying this condition is called the surface of the polytrope. For $n < 5$, ξ_n is finite, whereas for $n \geq 5$, it diverges to positive infinity.

2.13.2. Chandrasekhar Mass

Substituting the solution of the Lane-Emden equation into the mass conservation equation, Eq. (2.3) gives

$$m(r) = 4\pi \int_0^r dr r^2 \rho = 4\pi \rho_c \int_0^r dr r^2 \theta^n = 4\pi \rho_c \frac{1}{\alpha^3} \int_0^\xi d\xi \xi^2 \theta^n = -4\pi \rho_c \frac{\xi^2}{\alpha^3} \frac{\partial \theta}{\partial \xi}. \quad (2.281)$$

At $\xi = \xi_n$, this result becomes

$$M(R) = -\frac{1}{\sqrt{4\pi}} \left(\frac{(n+1)K}{G} \right)^{3/2} \rho_c^{(3-n)/2n} \left[\xi^2 \frac{\partial \theta}{\partial \xi} \right]_{\xi=\xi_n} \propto \rho_c^{(3-n)/2n}. \quad (2.282)$$

This result, together with the definition of ξ ,

$$r \propto \rho_c^{(1-n)/2n}, \quad (2.283)$$

gives a family of solutions for M and R that can maintain hydrostatic equilibrium for arbitrary ρ_c for a fixed polytropic EOS. Eq. (2.282) shows that, for all values of n other than $n = 3$,

equilibrium solutions exist over a continuous range of masses. For $n = 3$, however, the mass that can maintain hydrostatic equilibrium is uniquely fixed to a particular constant value, $M = \text{const.}$ In this case, the object may have arbitrary ρ_c and a corresponding arbitrary radius R .

Now suppose that hydrostatic equilibrium is broken and the star collapses under gravity. From Eqs. (2.261) and (2.281), the ratio between the pressure and gravitational scales,

$$Q \equiv \frac{P_c}{GM^2/R^4} = \frac{K R^4}{G M^2} \rho_c^{(n+1)/n} = \frac{K}{G} \left(\frac{\xi_n}{-4\pi \frac{\partial \theta}{\partial \xi} |_{\xi=\xi_n}} \right)^{(n+1)/n} M^{(1-n)/n} R^{(n-3)/n}, \quad (2.284)$$

is expected to be smaller than unity for a given fixed mass M . As the object collapses under gravity and R decreases, the following possibilities arise depending on n .

- $n < 3$: Since $\frac{n-3}{n} < 0$, Q increases as R decreases. This means that, as the radius decreases, pressure strengthens more rapidly than gravity, allowing hydrostatic equilibrium to be restored.
- $n = 3$: Q is independent of the radius R and depends on the mass as $M^{-2/3}$. Thus, whether hydrostatic equilibrium can be maintained is determined solely by the mass of the object. If the object exceeds the critical mass for hydrostatic equilibrium, no hydrostatic equilibrium is possible for any choice of ρ_c or R .
- $n > 3$: Although hydrostatic equilibrium solutions exist, $\frac{n-3}{n} > 0$, so Q decreases as R decreases. This corresponds to a dynamically unstable configuration, in which even a small perturbation can trigger gravitational collapse.

A representative example of a polytropic EOS with $n = 3$ is the EOS of degenerate electrons moving relativistically, as discussed above. From Eq. (2.245),

$$K = \frac{3^{1/3} h}{8\pi^{1/3} m_u^{4/3}} \mu_e^{-4/3}, \quad (2.285)$$

and numerical calculation gives $\left[\xi^2 \frac{\partial \theta}{\partial \xi} \right]_{\xi=\xi_3} = -2.01824$. Therefore, from Eqs. (2.48) and (2.282), the limiting mass for which an object dominated by relativistic degenerate electrons can maintain hydrostatic equilibrium is

$$M_{\text{Ch}} = 5.81 Y_e^2 M_{\odot}. \quad (2.286)$$

This is the Chandrasekhar mass. If the mass of a stellar core with degenerate electrons exceeds this mass, the star collapses.

2.13.3. Tolman-Oppenheimer-Volkoff Limit

During the collapse triggered by the exceeding of the Chandrasekhar mass, the core density increases and the constituent particles become more and more degenerate. If the core density reaches the nuclear saturation density, the repulsive interaction between nucleons and neutron degeneracy pressure can halt the collapse and maintain the structure. Even in this case,

however, there exists an upper mass limit for stable configurations; this is called the Tolman-Oppenheimer-Volkoff (TOV) limit.

Unlike the Chandrasekhar limit, for which a universal analytic result exists, the TOV limit does not have a universal closed-form solution. This is because, at the nuclear saturation density $\rho_{\text{sat}} \sim \mathcal{O}(10^{14}) \text{ g/cm}^3$, general relativistic effects make significant contributions to hydrostatic equilibrium, and the result also depends strongly on the EOS of nucleons, which is not yet fully determined. For a given EOS, the mass limit can be calculated by solving the TOV equation.

If only neutron degeneracy pressure is considered, this upper mass is approximately $0.7M_{\odot}$; however, several observed neutron stars have masses exceeding $2.0M_{\odot}$ [37]. This indicates that, in addition to neutron degeneracy pressure, repulsive interactions play an important role. Current theoretical estimates place the TOV limit at approximately $2.3M_{\odot}$. If the mass of a neutron star exceeds the TOV limit, it collapses into a black hole.

Chapter 3.

Core-Collapse Supernova

Supernovae can be broadly classified by their explosion mechanisms into thermonuclear supernova, core-collapse supernova (CCSN), and pair-instability supernova. Among these, a CCSN is a supernova that occurs when the core of a massive star collapses under gravity, leaving behind a neutron star or a black hole at the center.

The collapse process of a CCSN and its subsequent remnants provide extreme environments of high temperature and high pressure, making them useful high-energy systems for BSM studies. In this section, we focus on CCSN, especially for iron-core supernova.

3.1. Categorization of Supernovae

Supernovae can be classified broadly according to their explosion mechanisms and observed spectra. The explosion mechanism is classified according to the energy source that drives the supernova explosion. These include thermonuclear supernovae, triggered by nuclear energy; CCSNe in the broad sense, triggered by gravitational energy; and pair-instability supernovae, in which the collapse is triggered by photon-induced electron-positron pair production, while the explosion itself is driven by thermal runaway.

The classification based on observed spectra is determined by the presence or absence of spectral features associated with specific elements in the supernova spectrum. If hydrogen features are absent, the event is classified as Type I; if hydrogen features are present, it is classified as Type II. Type I supernovae are further divided into Types Ia, Ib, and Ic according to the presence or absence of Si and He features: Type Ia shows Si features, Type Ib shows He features but no Si features, and Type Ic shows neither Si nor He features. Type II supernovae are further classified into Types II-P and II-L according to the shape of their light curves, where P denotes plateau and L denotes linear.

Additional suffixes are often used to describe more specific observational properties. The suffix n denotes narrow emission lines in the spectrum, which usually indicate interaction between the supernova ejecta and dense circumstellar material. Representative examples include Type IIn, characterized by narrow hydrogen emission lines, and Type Ibn, characterized by narrow helium emission lines. The suffix pec denotes peculiar and is used when the observed spectral or photometric properties deviate significantly from those of ordinary members of the

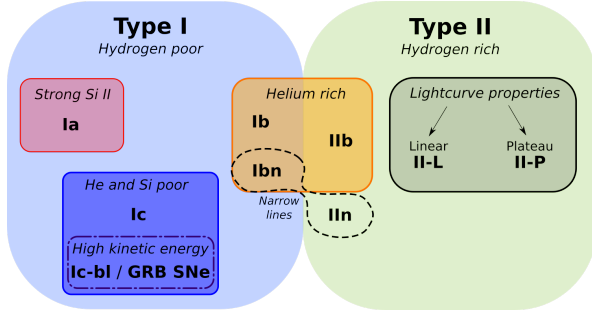


Figure 3.1: Classification of supernovae according to spectra and light curves [38]. Supernovae can be classified based on the spectral features of individual elements appearing in the observed spectra, as well as on the properties of their light curves.

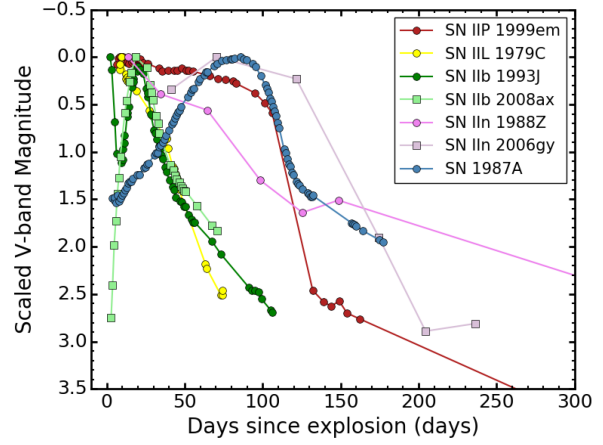


Figure 3.2: V-band light curves of typical individuals of each SN II subtype [39]. Magnitudes are normalized to peak SN magnitudes for comparison. The light curves of SN IIP 1999em (red), SN IIL 1979C (yellow), and SN 1987A (blue) clearly illustrate the differences among the corresponding subtypes.

corresponding class. For example, SN 1987A is commonly classified as Type II-pec, or as an 87A-like supernova, because its light curve differs from those of standard Type II-P and Type II-L events.

Observationally, thermonuclear supernovae are generally associated with Type Ia, whereas CCSNe are mainly associated with hydrogen-rich Type II supernovae or stripped-envelope Type Ib/Ic supernovae, excluding Type Ia. This observation-based classification is well illustrated in Figs. 3.1 to 3.3. Fig. 3.1 shows the classification of supernovae according to their observed spectra and light curves. Fig. 3.2 presents the light curves of the subtypes of Type II SNe, illustrating how the light-curve evolution differs among the subtypes. Fig. 3.3 shows spectral plots of supernovae for each classification type, showing the presence or absence of spectral features associated with specific elements.

3.2. Mechanisms Driving Core Collapse

Mechanisms that drive core collapse include electron capture and photodisintegration. Core collapse generally occurs when electron capture reduces the electron degeneracy pressure, thereby decreasing M_{Ch} given in Eq. (2.286). Once core collapse begins, electron capture and photodisintegration further accelerate the collapse.

3.2.1. Electron Capture

When electrons become degenerate, the increase in internal energy due to core contraction does not raise the internal temperature, but instead appears as an increase in the Fermi energy of the

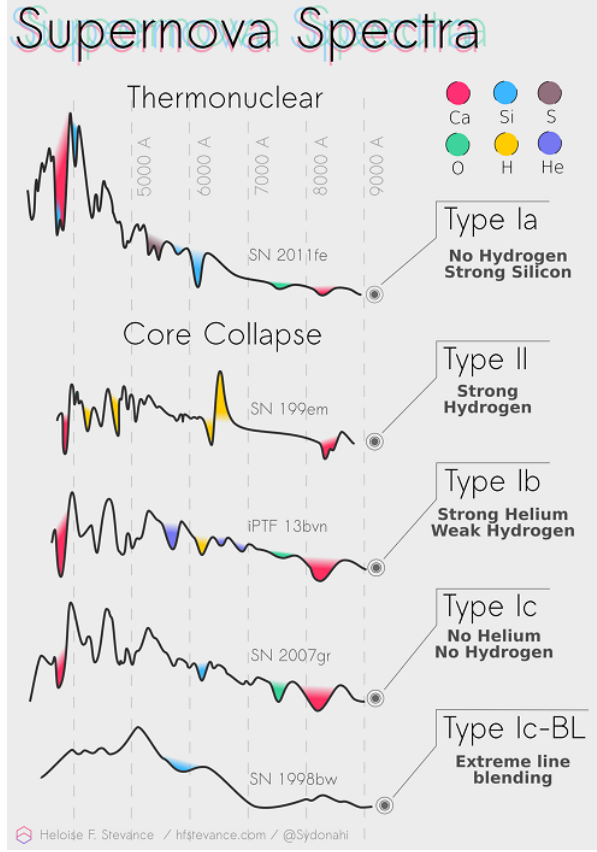


Figure 3.3: Spectral plots of supernovae for each type [40].

electrons. From Eq. (2.237), the Fermi momentum of degenerate electrons is given by

$$p_F = (3\pi^2 n_e)^{1/3} = \left(\frac{3\pi^2}{m_u} \rho Y_e \right)^{1/3}. \quad (3.1)$$

From this, when degenerate electrons behave relativistically, the electron Fermi energy is given by

$$E_F = \sqrt{p_F^2 + m_e^2} \simeq p_F = 11.1 (\rho_{10} Y_e)^{1/3} \text{ MeV}, \quad (3.2)$$

where $\rho_{10} \equiv \rho / (10^{10} \text{ g} \cdot \text{cm}^{-3})$. In natural units, the pressure of fully degenerate relativistic electrons given by Eq. (2.245) is

$$P_e = \frac{1}{3} \left(\frac{3}{8\pi m_u} \rho Y_e \right)^{4/3} \propto (\rho Y_e)^{4/3} \quad (3.3)$$

As the electron Fermi energy E_F increases, the electron capture, an endothermic process, can occur inside the stellar core. Electron capture reduces Y_e , thereby decreasing the electron degeneracy pressure, and at the same time cools the core through the emitted ν_e . In other words, it accelerates the collapse.

3.2.2. Photodisintegration

When the core temperature reaches $T_c \sim 1 \text{ MeV}$, nuclear fission begins, and fusion and fission reach equilibrium, leading to a state of nuclear statistical equilibrium (NSE). When NSE holds, the chemical potential of a nucleus with charge number Z and mass number A is obtained from the condition that the Helmholtz free energy remains stationary under changes in particle numbers,

$$0 = dF = -SdT - PdV + \sum_i \mu_i dN_i = \sum_i \mu_i dN_i, \quad (3.4)$$

and is given as follows.

$$\mu(Z, A) = Z\mu_p + (A - Z)\mu_n, \quad (3.5)$$

where μ_p and μ_n are the chemical potentials of free protons and free neutrons, respectively. For non-relativistic, non-degenerate ideal gas nuclei, the chemical potential is given as

$$\mu(Z, A) = m(Z, A) + k_B T \ln \left[\frac{n(Z, A)}{G(Z, A)} \left(\frac{2\pi}{m(Z, A)k_B T} \right)^{3/2} \right]. \quad (3.6)$$

Here, $G(Z, A)$ denotes the nuclear partition function,

$$G(Z, A) = \sum_i (2J_i + 1) e^{-E_i/k_B T}, \quad (3.7)$$

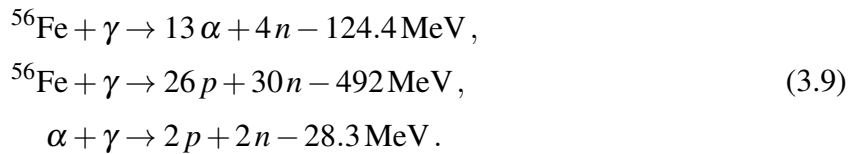
where J_i is the total angular momentum. $(2\pi/(m(Z, A)k_B T))^{1/2}$ in the Eq. (3.6) corresponds to the thermal de Broglie wavelength. Therefore, the non-relativistic form of the chemical potential defined as in Eq. (2.233) originates from the classical fugacity.

In regions where NSE holds, the nuclear reaction timescale is shorter than the hydrodynamical timescale, so local thermal equilibrium (LTE) can be assumed for the nuclear composition. Therefore, in a local region where NSE holds, the Helmholtz free energy

$$F = U - TS, \quad (3.8)$$

is minimized. That is, the internal composition changes in the direction that minimizes F . From Eq. (3.8), one can see that, in general, the contribution from U dominates at low temperature, whereas the contribution from S dominates at high temperature.

Representative photodisintegration processes occurring during core collapse are as follows.



Since these processes are strongly endothermic, they increase the internal energy. However, they also produce many light nuclei and therefore increase the entropy. Thus, in high-temperature environments, these processes reduce F . Consequently, photodisintegration is favored in this

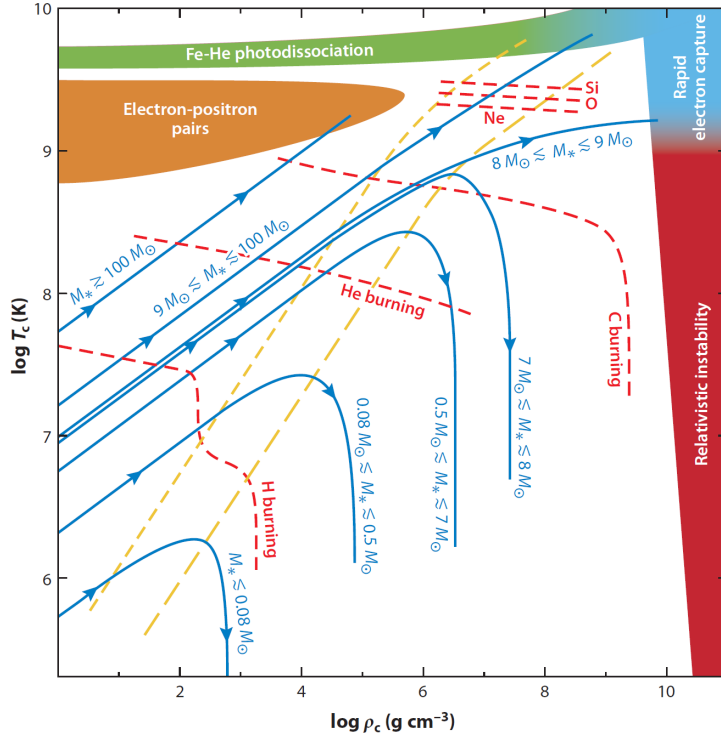


Figure 3.4: Schematic diagram of the stellar evolution tracks in the plane of core density and temperature as a function of the initial stellar mass, together with the mechanisms that drive collapse [41]. The red dashed lines indicate the points at which the corresponding stages of nuclear fusion begin in the core, while the yellow dashed lines correspond to the beginning of electron degeneracy (short-dashed) and strong electron degeneracy (long-dashed).

environment. Since photodisintegration is strongly endothermic, it lowers the core temperature and accelerates the collapse.

3.2.3. Subtypes of Core-Collapse Supernova

CCSN can be divided into electron-capture supernova (ECSN) and iron-core supernova, depending on the progenitor and the specific mechanism driving core collapse. In the narrow sense, CCSN refers to the latter, and we will use this term in this sense. As discussed in Section 1.6, ECSN and CCSN (iron-core SN) differ in their ZAMS masses and inner structures at the beginning of the core collapse.

Progenitors of ECSN are the stars with masses in the range $8-10M_{\odot}$, forming an O-Ne-Mg core during the super-AGB phase. The electrons in the core become degenerate before Ne burning can occur under hydrostatic equilibrium. The collapse is initiated and accelerated by electron capture, while the core temperature is not sufficiently high for photodisintegration to make a significant contribution. Therefore, the collapse is dominated by electron capture.

Progenitors of CCSN are the stars with masses above $10M_{\odot}$, ultimately forming an Fe core. Since iron nuclei cannot generate energy through fusion, the core contracts and the core temperature rises. Once collapse is initiated by electron capture, photodisintegration, which proceeds efficiently due to the high core temperature, accelerates the collapse.

A schematic illustration of the evolution of the core density and temperature during stellar evolution as a function of the initial stellar mass, together with the mechanisms that drive collapse, is shown in Fig. 3.4.

3.3. Core Collapse and Explosion Phases

Supernova explosion can be broadly divided into the collapse phase and the explosion phase. The detailed stages may differ slightly depending on the mechanisms contributing to collapse and explosion, but here we focus on iron-core supernova and neutrino-driven explosions. Each stage from the onset of core collapse to the supernova explosion is illustrated in Fig. 3.5.

3.3.1. Initial Phase of the Collapse

At the onset of collapse, the stellar iron core is in a high-temperature, high-pressure state, with $T_c \sim \mathcal{O}(10^9)$ K ~ 0.1 MeV and $\rho_c \sim \mathcal{O}(10^9)$ g/cm³. The onset of collapse and the infall dynamics are sensitive to Y_e and the entropy. Once electron capture reduces Y_e in the core and makes the core dynamically unstable, collapse begins. Since photodisintegration of nuclei requires about 2 MeV per nucleon, photodisintegration becomes efficient around $T_c \sim 10^{10}$ K ~ 1 MeV and further accelerates the collapse together with electron capture. The ν_e produced by electron capture transports energy from the core to the outer regions, thereby cooling the core. In the early stage of collapse, β decay is suppressed by Pauli blocking. However, as electron capture reduces Y_e , β decay gradually begins to occur and contributes to core cooling.

Fig. 3.6 show results from stellar evolution simulation using MESA code based on the test suite 20M_pre_ms_to_core_collapse, presenting profiles around the onset of the core collapse for a star with initial mass of $20 M_\odot$. In this simulation, the composition $X = 0.7237$, $Y = 0.2703$, and $Z = 0.0060$ was adopted. At this stage, the core temperature is approximately 0.75 MeV and the density is approximately 3.90×10^9 g/cm³, both of which gradually increase as the collapse proceeds.

3.3.2. Neutrino Trapping

When $\rho_c \sim 10^{12}$ g/cm³ is reached, neutrinos become trapped in the core through coherent scattering on heavy nuclei. In the inner core where neutrinos are trapped, the collapse proceeds faster than the neutrino diffusion time, so Y_L is approximately conserved. As a result, a sea of degenerate ν_e is formed. In the outer core, where neutrinos are not trapped, ν_e continues to be emitted through electron capture.

In this stage, the outer core collapses with supersonic velocity, while the inner core collapses with subsonic velocity, because the increasing density enhances the incompressibility of nuclear matter and slows down the collapse.

The fact that the outer core falls inward with supersonic velocity has both physical and practical implications. Physically, it means that the increase in pressure in the inner region

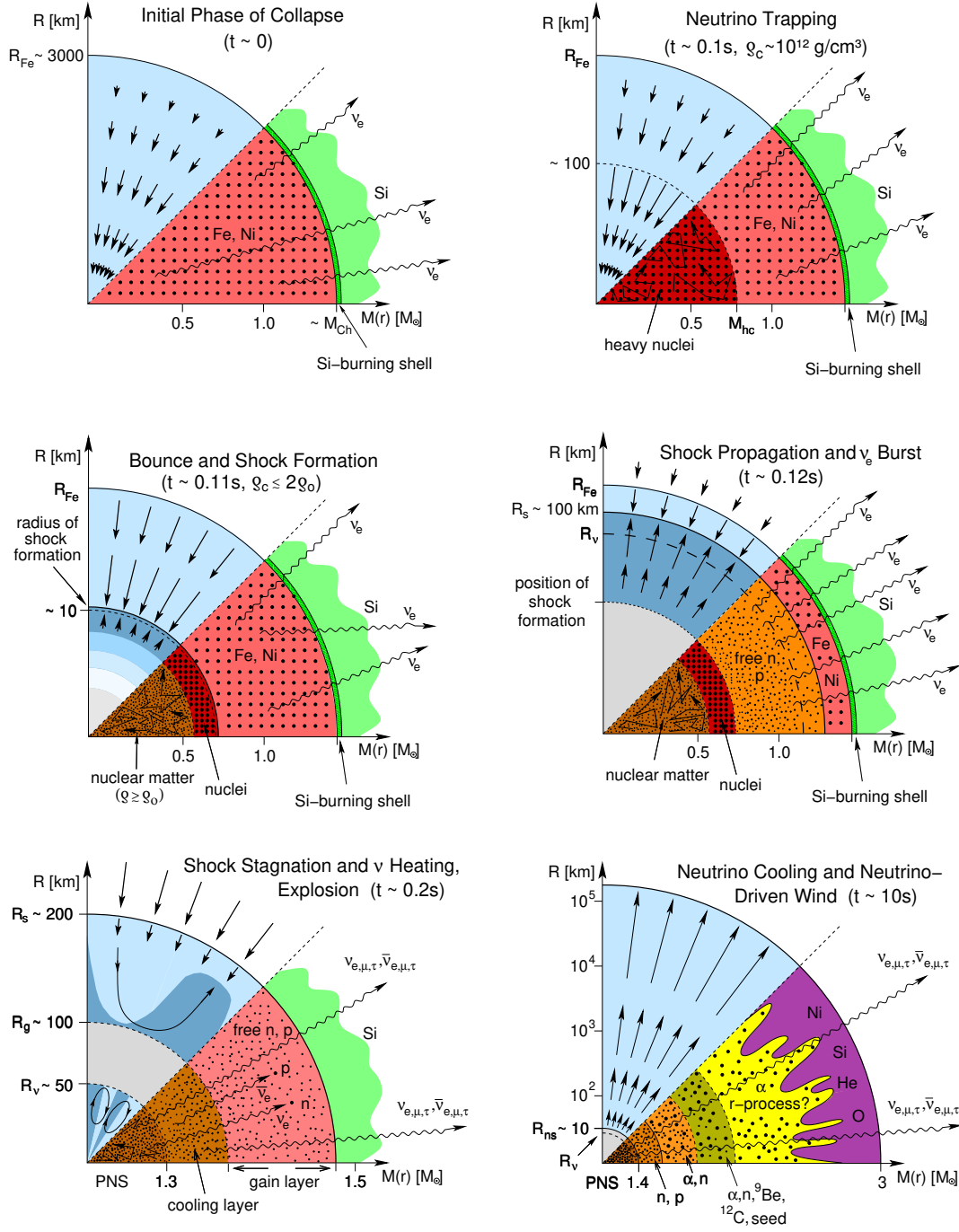


Figure 3.5: Scheme of the evolutionary stages from the onset of the core collapse to the supernova explosion [42]. M_{hc} deontes the mass of the subsonically collapsing, while R_{Fe} , R_s , R_g , R_{ns} , and R_v being the iron core radius, shock radius, gain radius, neutron star radius, and neutrinosphere, respectively.

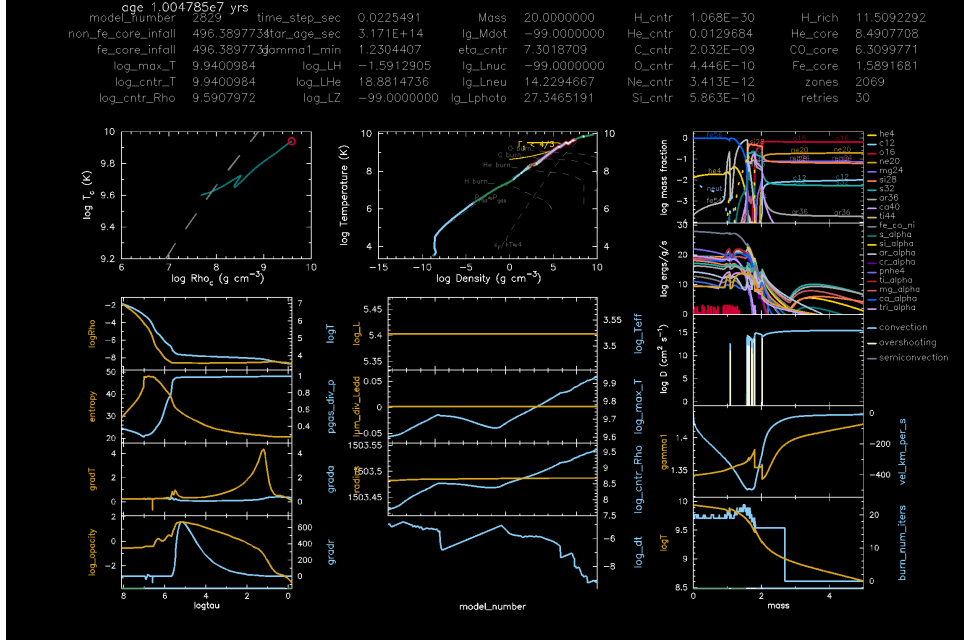


Figure 3.6: Stellar profiles around the onset of the core collapse for $20M_{\odot}$ star based on MESA test suite 20M_pre_ms_to_core_collapse [43]. The iron core at the onset of collapse has a temperature of approximately 0.75 MeV and a density of approximately $3.90 \times 10^9 \text{ g/cm}^3$.

cannot halt the infalling material in advance. Since information propagates through the medium at the sound speed, in a supersonically infalling region the infall velocity of the material exceeds the speed at which pressure information can propagate outward.

Practically, as mentioned in Section 2.9.2, the contribution of advective terms in the equations describing fluid motion during stellar evolution, can usually be neglected because the bulk motion of the medium is slower than the sound speed. However, during stellar collapse, when matter falls inward with supersonically, the contribution of the advective terms can no longer be neglected. Therefore, numerical simulations of SN explosions should be implemented fundamentally as hydrodynamic simulations.

3.3.3. Core Bounce and Shock Formation

When the density reaches $\rho_c \sim 10^{14} \text{ g/cm}^3$, the effective adiabatic exponent rises abruptly due to the repulsive short-range interactions of the strong force between nucleons. This is referred to as the stiffening of the EOS of nuclear matter. The increase in the adiabatic exponent Γ_1 given by Eq. (2.64) enhances the pressure and incompressibility of nuclear matter and, from Eq. (2.284), makes the inner core dynamically stable. As a result, the collapse of the homogeneously collapsing inner core halts.

Supersonically infalling material in the outer core falls inward with a density of about $\rho \sim 10^{12} \text{ g/cm}^3$ and a velocity of roughly 10% of the speed of light, eventually striking the PNS surface. However, the dynamically stable inner core withstands this tremendous impact. As a result of this impact, core bounce occurs and a shock wave is formed.

The resulting core bounce launches an outward-propagating pressure wave, which steepens into a shock wave as it encounters the infalling supersonic outer core. Through the formation and propagation of the shock wave, the collapse phase is turned into the explosion phase.

3.3.4. Shock Propagation and Electron Neutrino Burst

The outward-propagating shock wave dissociates heavy nuclei. This increases the abundance of free protons, making electron capture more efficient. The electron neutrinos produced by electron capture cannot escape freely in dense regions because of the large optical depth. However, as the shock moves into lower-density regions, the optical depth decreases rapidly, allowing them to be emitted freely and producing a short, intense ν_e burst. The energy released in the ν_e burst is about $\mathcal{O}(10^{51})$ erg, and the instantaneous luminosity is approximately $L_{\nu_e} \sim 10^{53}$ erg/s. Through this process, the shock wave loses energy and stalls.

Figs. 3.7 to 3.10 show the results of a 1D supernova explosion simulation of an $18M_{\odot}$ star using the GR1D code [44, 45]. The s18 progenitor profile was used [46], and the SFHo EOS was adopted [47]. Convection was treated using STIR based on MLT, with a mixing length parameter of $\alpha = 1.48$ [48].

Fig. 3.7 shows the electron neutrino luminosity in the lab frame as a function of the time after post-bounce t_{pb} . A short and intense ν_e burst can be seen around 4×10^{-3} s. Figs. 3.8 to 3.10 show the radial velocity, density, and temperature profiles as functions of radius at $t_{\text{pb}} = 10.02$ ms, respectively. In Fig. 3.8, two sharp changes in the radial velocity are shown. These correspond, in order, to the boundary of the inner core and the stalled shock wave. The density and temperature also change abruptly at these two boundaries, as shown in Figs. 3.9 and 3.10, respectively.

3.3.5. Shock Stagnation and Neutrino Heating

At the center, the proto-neutron star (PNS) is formed. From this stage onward, the density and temperature inside the PNS become sufficiently high that neutrinos of all flavors begin to be emitted in significant amounts. Inside the PNS, all neutrino species can be produced through four-Fermi interactions described by the following effective Lagrangian.

$$\mathcal{L}_{\text{eff}} \supset \frac{G_F}{\sqrt{2}} \left(\bar{\nu} \gamma^\mu (1 - \gamma^5) \nu \right) \left(\bar{f} \gamma_\mu (C_V - C_A \gamma^5) f \right). \quad (3.10)$$

Representative production processes are as follows.

$$\begin{aligned} e^- + e^+ &\rightarrow \nu_x + \bar{\nu}_x, \\ N + N &\rightarrow N + N + \nu_x + \bar{\nu}_x, \\ \nu_e + \bar{\nu}_e &\rightarrow \nu_x + \bar{\nu}_x, \end{aligned} \quad (3.11)$$

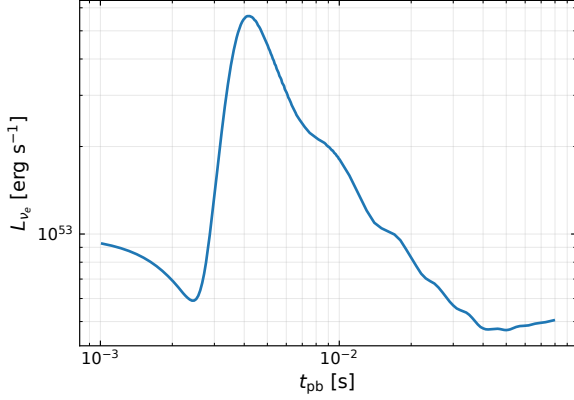


Figure 3.7: $t_{\text{pb}}-L_{\nu_e}$ graph during the shock propagation phase. A short and intense ν_e burst can be seen around $t_{\text{pb}} \sim 4 \times 10^{-3}$ s. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

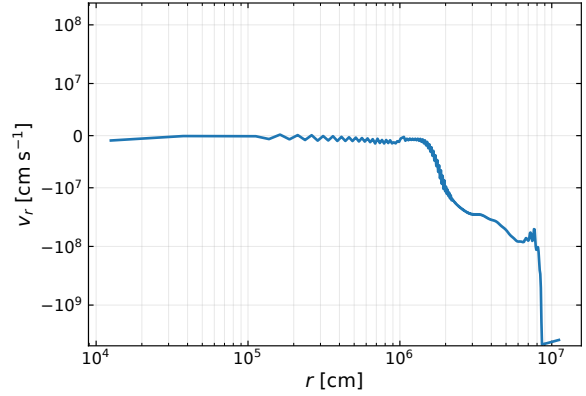


Figure 3.8: $r-v_r$ profile at $t_{\text{pb}} = 10.02$ ms. The radial velocity shows two sharp changes, which correspond to the inner-core boundary and the stalled shock wave, respectively. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

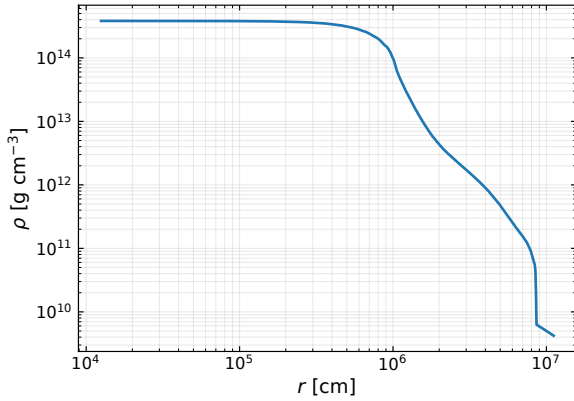


Figure 3.9: $r-\rho$ profile at $t_{\text{pb}} = 10.02$ ms. The two sharp density changes occur, in order, at the inner-core boundary and the stalled shock wave. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

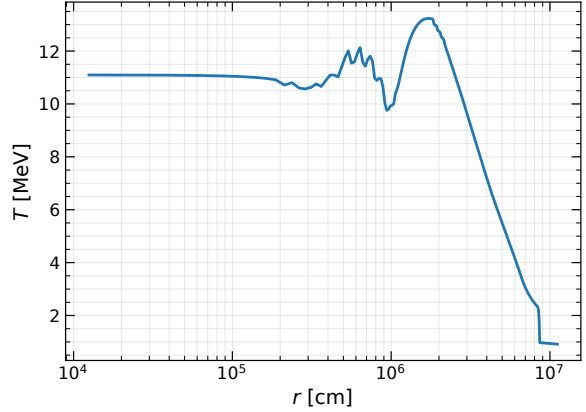


Figure 3.10: $r-T$ profile at $t_{\text{pb}} = 10.02$ ms. The temperature is nearly constant inside the inner core, reaches its maximum near the boundary, then gradually decreases outward before dropping sharply at the shock wave. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

where ν_x denotes ν_μ and ν_τ . Electron neutrinos can also be produced through electron capture in addition to the processes listed above. Since the corresponding charged leptons are almost absent in the PNS environment, ν_x and $\bar{\nu}_x$ are produced mainly through pair processes and neutral-current scattering. By contrast, ν_e and $\bar{\nu}_e$ can be produced not only through pair processes and neutral-current scattering, but also through charged-current interactions such as electron capture. Due to this property, after the ν_e burst phase, the average neutrino energies approximately satisfy $\langle E_{\nu_e} \rangle \lesssim \langle E_{\bar{\nu}_e} \rangle \lesssim \langle E_{\nu_x, \bar{\nu}_x} \rangle$. The difference between E_{ν_e} and $E_{\bar{\nu}_e}$ arises from the neutron-rich environment inside the star. In addition, the neutrinosphere radii of the different neutrino species are ordered as $R_{\nu_x, \bar{\nu}_x} \lesssim R_{\bar{\nu}_e} \lesssim R_{\nu_e}$.

Outside the stalled shock wave, matter continues to fall inward with supersonic velocity, while matter inside the shock wave falls inward with subsonic velocity. This material continuously accretes onto the core. As discussed in Section 1.6.2, at this stage there are three possible scenarios depending on whether the shock wave is revitalized: a supernova explosion with neutron star formation, a supernova explosion with black hole formation, and failed supernova explosion with black hole formation. Several mechanisms have been proposed for shock reheating, but here we describe the process in terms of the most widely accepted delayed neutrino-heating mechanism, in which the shock is reheated by neutrino reabsorption.

The shock reheating effect due to neutrino reabsorption can be assessed using the neutrino heating and cooling rates per nucleon. The neutrino heating rate per nucleon is

$$Q_v^+ \approx 110 \cdot \left(\frac{L_{\nu_e, 52} \langle \epsilon_{\bar{\nu}_e, 15}^2 \rangle}{r_7^2 \langle \mu \rangle_{\nu_e}} Y_n + \frac{L_{\bar{\nu}_e, 52} \langle \epsilon_{\nu_e, 15}^2 \rangle}{r_7^2 \langle \mu \rangle_{\bar{\nu}_e}} Y_p \right) \left[\frac{\text{MeV}}{\text{s} \cdot \text{N}} \right], \quad (3.12)$$

and, assuming $Y_p + Y_n = 1$ and the electron degeneracy parameter $\eta_{e-} \equiv \mu_{e-}/k_B T = -\eta_{e+} \approx 0$, the neutrino cooling rate per nucleon is

$$Q_v^- \approx 145 \cdot \left(\frac{k_B T}{2 \text{ MeV}} \right)^6 \left[\frac{\text{MeV}}{\text{s} \cdot \text{N}} \right]. \quad (3.13)$$

Here, L_{52} , ϵ_{15} and r_7 indicate normalization by 10^{52} erg/s , 15 MeV and 10^7 cm , respectively. The luminosity, average squared energy, and mean cosine angle between the radial direction and the neutrino propagation direction are given respectively as follows.

$$L_{\nu_i} = 4\pi r^2 c \frac{2\pi}{(hc)^3} \int_0^\infty d\epsilon_{\nu_i} \int_{-1}^1 d\mu_{\nu_i} \mu_{\nu_i} \epsilon_{\nu_i}^3 f_{\nu_i}(\epsilon_{\nu_i}, \mu_{\nu_i}), \quad (3.14)$$

$$\langle \epsilon_{\nu_i}^2 \rangle = \int_0^\infty d\epsilon_{\nu_i} \int_{-1}^1 d\mu_{\nu_i} \epsilon_{\nu_i}^5 f_{\nu_i} \left(\int_0^\infty d\epsilon_{\nu_i} \int_{-1}^1 d\mu_{\nu_i} \epsilon_{\nu_i}^3 f_{\nu_i} \right)^{-1}, \quad (3.15)$$

$$\langle \mu_{\nu_i} \rangle = \int_0^\infty d\epsilon_{\nu_i} \int_{-1}^1 d\mu_{\nu_i} \mu_{\nu_i} \epsilon_{\nu_i}^3 f_{\nu_i} \left(\int_0^\infty d\epsilon_{\nu_i} \int_{-1}^1 d\mu_{\nu_i} \epsilon_{\nu_i}^3 f_{\nu_i} \right)^{-1}, \quad (3.16)$$

$\langle \mu_{\nu_i} \rangle$ is called the flux factor, and can be understood as the neutrino energy flux, $L_{\nu_i}/4\pi r^2$, di-

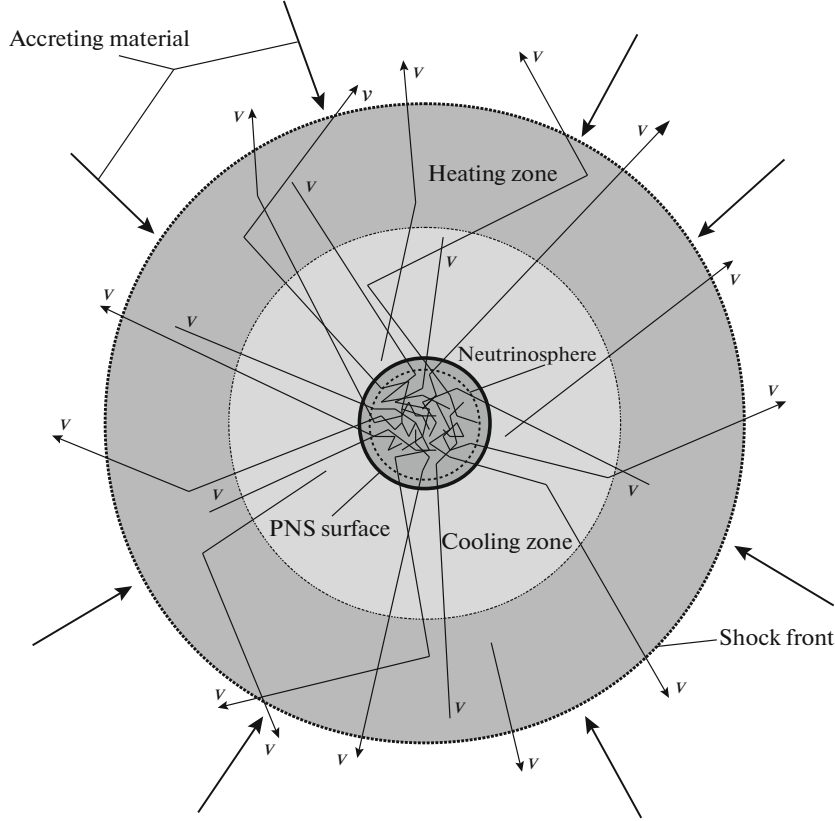


Figure 3.11: Schematic diagram of the stellar interior after shock stalling [49]. Neutrinos are emitted from the central PNS and the cooling layer (cooling zone) while neutrinos absorbed in the gain layer (heating zone) behind the shock front reheat the shock. If shock revitalization succeeds, the star explodes successfully; if it fails, the shock falls back onto the PNS, the explosion fails, and a black hole forms.

vided by the neutrino energy density. In general, this value is about 0.25 near the neutrinosphere and approaches 1 in the free-streaming region.

From Eqs. (3.12) and (3.13), one can define the region where heating occurs due to neutrino reabsorption and the region where cooling occurs due to neutrino emission. The radius at which $Q_{\nu}^{+} = Q_{\nu}^{-}$ is defined as the gain radius, R_g . The region where $Q_{\nu}^{+} > Q_{\nu}^{-}$ is called the gain layer (heating layer), while the region where $Q_{\nu}^{+} < Q_{\nu}^{-}$ is called the cooling layer. This is illustrated clearly in Fig. 3.11.

It takes about $\mathcal{O}(100)$ ms from the onset of collapse until whether the shock is successfully revitalized is determined.

3.3.6. Neutrino Cooling and Neutrino-Driven Wind

If the shock is successfully revitalized, it begins to propagate outward again, leading to a successful explosion and the formation of a NS at the center. Even if the shock is successfully revitalized, however, if a sufficiently large amount of stellar material falls back and accretes onto the PNS, a BH forms at the center instead of a NS. If the shock is not revitalized, it fails to

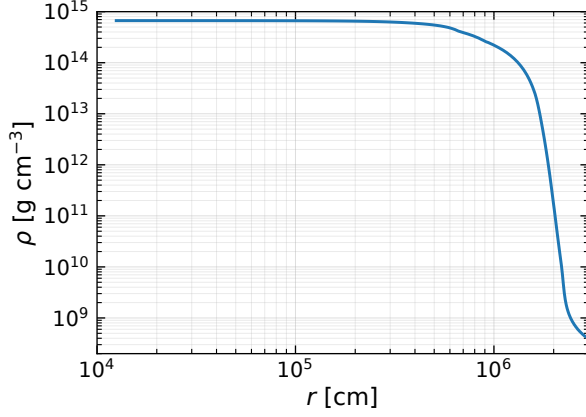


Figure 3.12: r - ρ profile at $t_{\text{pb}} = 1.00$ s. A dense PNS forms at the center of the supernova. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

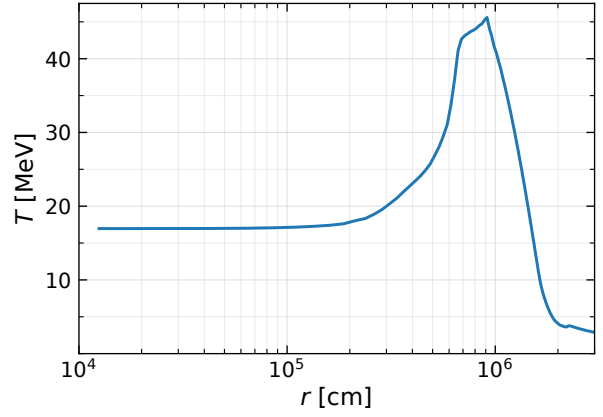


Figure 3.13: r - T profile at $t_{\text{pb}} = 1.00$ s. The interior of the PNS has a temperature of $T \sim \mathcal{O}(10)$ MeV, with the maximum temperature located near the boundary of the PNS inner core. The result is from a 1D simulation using the GR1D code, with an $18M_{\odot}$ progenitor profile, the SFHo EOS, and $\alpha = 1.48$.

eject the stellar envelope and falls back, resulting in a failed supernova explosion and leaving behind a BH at the center.

After a successful supernova explosion, the central PNS cools by emitting neutrinos of all flavors. The neutrino luminosity gradually decreases, producing a neutrino-driven wind that lasts for about ~ 10 s. Approximately 99% of the supernova explosion energy is emitted in the form of neutrinos, while only about 1% is emitted in a form observable as light. Most of the energy carried away by neutrinos is emitted during the first ~ 10 s.

Figs. 3.12 and 3.13 show the results of a 1D supernova explosion simulation of an $18M_{\odot}$ star at $t_{\text{pb}} = 1.00$ s using the GR1D code [44, 45]. The s18 progenitor profile was used [46], and the SFHo EOS was adopted [47]. Convection was treated using STIR based on MLT, with a mixing length parameter of $\alpha = 1.48$ [48]. The hot and dense interior of the PNS provides an extreme environment suitable for producing not only neutrinos of all flavors, but also BSM particles.

3.4. Shock Revitalization Mechanisms

After the prompt bounce shock stalls, the central problem of a CCSN becomes the revival of the stalled accretion shock. The revival mechanism must transfer a sufficient amount of energy and momentum to the post-shock material so that the shock overcomes the ram pressure of the infalling outer core and turns into an outward-propagating explosion shock.

The proposed mechanisms can be classified by the energy reservoir and the channel through which this energy is coupled to the stalled shock. In ordinary CCSNe, the leading framework is the neutrino-heating mechanism, aided by multidimensional hydrodynamic effects such as con-

vection, standing accretion shock instability (SASI), turbulence, and progenitor perturbations. Other mechanisms, such as magnetorotational explosions, hadron-quark phase transition explosions, acoustic explosions, jittering jets, and collapsar-like BH accretion engines, are usually associated with specific progenitors, special microphysics, or alternative explosion channels [41, 50, 51].

3.4.1. Neutrino-Heating Mechanism

The neutrino-heating mechanism (or delayed neutrino-heating mechanism) is the most widely accepted mechanism for ordinary CCSNe. In this mechanism, a small fraction of the neutrino energy emitted by the PNS is reabsorbed in the gain layer through charged-current reactions,

$$\begin{aligned} \nu_e + n &\rightarrow p + e^-, \\ \bar{\nu}_e + p &\rightarrow n + e^+. \end{aligned} \quad (3.17)$$

These charged-current absorption reactions are the dominant heating channels in the gain layer in the standard delayed neutrino-heating picture [52]. The local heating and cooling rates were given in Eqs. (3.12) and (3.13). These rates define the gain radius R_g by the condition $Q_\nu^+ = Q_\nu^-$. The region $R_g < r < R_s$ is the gain layer, where neutrino absorption can increase the entropy and pressure of the post-shock material.

Shock revival is determined by the competition between neutrino heating in the gain layer and the loss of matter by advection through the gain layer. A useful diagnostic is the ratio between the advection timescale and the heating timescale. The advection timescale is estimated as

$$\tau_{\text{adv}} \sim \frac{M_g}{\dot{M}}, \quad (3.18)$$

where M_g is the mass contained in the gain layer and $\dot{M}$ is the mass accretion rate through the shock. In a schematic Newtonian spherical notation, the gain mass can be written as

$$M_g = \int_{R_g}^{R_s} 4\pi r^2 \rho \, dr. \quad (3.19)$$

Here, ρ is the mass density, R_g is the gain radius, and R_s is the shock radius. The heating timescale is estimated as

$$\tau_{\text{heat}} \sim \frac{|E_g|}{\dot{Q}_\nu}, \quad (3.20)$$

where E_g is the total binding energy of matter in the gain layer and $\dot{Q}_\nu$ is the volume integrated net neutrino heating rate,

$$\dot{Q}_\nu = \int_{R_g}^{R_s} 4\pi r^2 \rho \, \dot{q}_\nu \, dr, \quad (3.21)$$

where $\dot{q}_\nu$ is the net neutrino heating rate per unit mass. If the rates in Eqs. (3.12) and (3.13) are used as rates per nucleon, then $\dot{q}_\nu$ is obtained from $Q_\nu^+ - Q_\nu^-$ after the appropriate conversion from per nucleon units to per mass units. In detailed general relativistic simulations, metric and

redshift factors are included in the corresponding volume integrals.

Shock revival becomes likely when

$$\frac{\tau_{\text{adv}}}{\tau_{\text{heat}}} \gtrsim 1. \quad (3.22)$$

This condition means that matter stays in the gain layer long enough to absorb sufficient neutrino energy before being advected into the cooling layer and accreted onto the PNS [53].

Another useful formulation is the critical neutrino luminosity condition, originally formulated in terms of the absence of steady-state accretion-shock solutions above a critical luminosity [54]. A commonly used semi-analytic scaling relates the critical condition to the electron-flavor neutrino luminosity and mean squared neutrino energy. For a given mass accretion rate $\dot{M}$ and PNS mass M_{PNS} , there is a critical luminosity above which a steady stalled shock solution cannot be maintained [55],

$$(L_{\nu} \langle E_{\nu}^2 \rangle)_{\text{crit}} \propto (\dot{M} M_{\text{PNS}})^{3/5} |\bar{e}_{\text{tot,g}}|^{3/5} R_{\text{g}}^{-2/5} \left(1 + \frac{4}{3} \langle \text{Ma}^2 \rangle \right)^{-3/5}. \quad (3.23)$$

Here, $L_{\nu} \equiv L_{\nu_e} + L_{\bar{\nu}_e}$ is the electron-flavor neutrino luminosity, $\langle E_{\nu}^2 \rangle$ denotes the weighted average of the mean squared energies of electron neutrinos and antineutrinos,

$$\langle E_{\nu}^2 \rangle = \frac{L_{\nu_e} \langle E_{\nu_e}^2 \rangle + L_{\bar{\nu}_e} \langle E_{\bar{\nu}_e}^2 \rangle}{L_{\nu}}, \quad (3.24)$$

$\bar{e}_{\text{tot,g}} = E_{\text{tot,g}}/M_{\text{g}}$ is the average specific binding energy in the gain layer, and $\langle \text{Ma}^2 \rangle$ is the average squared turbulent Mach number in the gain layer. The last factor represents the reduction of the critical condition by turbulent pressure and non-radial motion.

The basic sequence of the neutrino-heating mechanism is as follows:

1. The prompt bounce shock stalls and becomes an accretion shock.
2. Neutrinos emitted from the PNS and cooling layer heat the material in the gain layer.
3. The post-shock pressure increases, and the shock radius gradually expands.
4. The expansion of the shock increases the mass and dwell time of matter in the gain layer.
5. Neutrino heating becomes more efficient and the shock expansion becomes runaway.

The mechanism is therefore delayed because the prompt shock itself is not energetic enough to explode the star. Instead, the explosion is powered by the continuous deposition of neutrino energy after shock stagnation.

The energy reservoir of this mechanism is the gravitational binding energy released during PNS formation, which is of order

$$E_{\text{PNS,bind}} \sim 10^{53} \text{ erg}. \quad (3.25)$$

A canonical supernova explosion energy is approximately

$$E_{\text{exp}} \sim 10^{51} \text{ erg}. \quad (3.26)$$

This energy scale has its own unit, called Bethe (B), or FOE (means ten to the power of Fifty-One Erg). Thus, an efficiency of $\mathcal{O}(1)\%$ is sufficient in principle. Here, the efficiency does not mean that $\mathcal{O}(1)\%$ of the total neutrino energy is absorbed everywhere in the star. Rather, it means that a small fraction of the PNS neutrino output must be deposited in the gain layer in a dynamically useful way before the accreted matter leaves the heating region. The difficulty is that this energy must be deposited in the gain layer before the matter is advected onto the PNS and before the shock is overwhelmed by the ram pressure of the infalling material.

3.4.2. Multidimensional Hydrodynamic Effects

Although the neutrino-heating mechanism can be described in spherical symmetry, multidimensional hydrodynamic effects play an essential role in realistic shock revival. These effects do not usually provide an independent energy source. Instead, they help the delayed neutrino-heating mechanism by increasing the efficiency of neutrino heating and by reducing the critical neutrino luminosity [50]. The most important multidimensional effects are neutrino-driven convection, SASI, turbulent pressure, and pre-collapse progenitor perturbations.

Neutrino-driven convection. Neutrino-driven convection develops when neutrino heating produces an unstable entropy gradient in the gain layer. Hot and high-entropy matter rises toward the shock, while cooler material is transported toward smaller radii, where it can be heated again. This circulation increases the dwell time of matter in the gain layer and helps push the shock outward. The growth of neutrino-driven convection is affected by the competition between buoyant growth and advection through the gain layer. This competition is often characterized by the Foglizzo parameter χ [56],

$$\chi = \int_{R_g}^{R_s} \frac{\omega_{\text{BV}}}{v_r} dr, \quad (3.27)$$

where ω_{BV} is the buoyancy frequency given in Eq. (2.145). If advection is too rapid, buoyant perturbations may be advected out of the gain layer before they grow appreciably.

Standing accretion shock instability (SASI). SASI is a global instability of the stalled accretion shock. It produces large-scale sloshing or spiral motions of the shock. In the advective-acoustic picture, entropy and vorticity perturbations generated at the shock are advected inward and generate acoustic feedback near the PNS, which then propagates outward and perturbs the shock again. SASI can increase the shock radius and the residence time of matter in the gain layer, but it is usually regarded as an aid to neutrino heating rather than an independent explosion engine [57].

Turbulent pressure. Turbulent motion in the gain layer also contributes an effective pressure behind the shock. More precisely, this effect is associated with the radial component of the Reynolds stress generated by non-spherical turbulent motion. In a schematic Newtonian notation, one may write

$$P_{\text{eff}} \simeq P_{\text{th}} + P_{\text{turb}}. \quad (3.28)$$

Here, P_{th} is the ordinary thermal pressure and P_{turb} denotes the effective turbulent pressure. The latter is not an independent thermodynamic pressure, but represents the dynamical support associated with turbulent kinetic energy. Roughly,

$$P_{\text{turb}} \sim \rho \langle \delta v_r^2 \rangle, \quad (3.29)$$

where ρ is the mass density and δv_r is the fluctuating radial velocity component in the gain layer. This turbulent pressure helps support the shock against the ram pressure of the infalling material. Therefore, multidimensional turbulence can increase the average shock radius and reduce the neutrino luminosity required for shock revival.

Pre-collapse perturbations. Pre-collapse perturbations generated by late-stage convective shell burning can also affect explodability. When velocity and density perturbations in the Si/O layers are advected through the shock, they seed non-radial motion and enhance turbulence in the gain layer [58]. Therefore, the explodability of a progenitor is sensitive not only to the spherically averaged density profile, but also to the multidimensional structure of the progenitor.

In one-dimensional simulations, these multidimensional effects are absent unless they are modeled effectively. For example, in STIR-type calculations, turbulent convection is incorporated through a time-dependent mixing-length treatment. The mixing-length parameter then controls the strength of the effective turbulent transport and turbulent pressure. Such a prescription is not a separate physical explosion mechanism, but an effective one-dimensional model of multidimensional turbulence.

3.4.3. Magnetorotational Mechanism

The magnetorotational mechanism is a separate explosion channel in which the rotational energy of the collapsing core or the PNS is extracted by magnetic fields. The available rotational energy is approximately

$$E_{\text{rot}} \sim \frac{1}{2} I \Omega^2. \quad (3.30)$$

Here, I is the moment of inertia of the PNS and Ω is its angular velocity. For a rapidly rotating millisecond PNS, this energy can reach $\sim 10^{52}$ erg, which is large enough to power energetic and strongly aspherical explosions [59].

During collapse and after bounce, magnetic fields can be amplified by compression, differential winding, the magnetorotational instability, and turbulent dynamo-like processes. If

the magnetic field becomes dynamically important, magnetic pressure and magnetic stresses can launch bipolar outflows or jets. The resulting explosion is typically more aspherical than a neutrino-driven explosion. Recent multidimensional magnetohydrodynamic (MHD) simulations support this channel for rapidly rotating and strongly magnetized progenitors, although it is not expected to be generic for ordinary Type II supernovae [60].

This mechanism is most relevant for rapidly rotating progenitors, hypernovae, broad-lined Type Ic supernovae, long gamma-ray burst supernovae, and possibly magnetar formation. It is not expected to be the generic mechanism for ordinary Type II supernovae because most stellar cores are likely not rotating rapidly enough at collapse.

3.4.4. Acoustic Mechanism

The acoustic mechanism proposes that accretion excites oscillation modes of the PNS, especially low-order core gravity modes (g -modes). These oscillations emit acoustic waves into the post-shock region. The sound waves steepen into shocks and deposit energy behind the stalled shock. This mechanism was proposed as an accretion-powered acoustic explosion mechanism in which the PNS acts as a transducer converting accretion energy into sound [61, 62].

The acoustic mechanism can be summarized as the following energy transfer sequence:

1. Accretion onto the PNS supplies gravitational energy.
2. Aspherical accretion flows excite oscillation modes of the PNS, especially low-order core g -modes.
3. The oscillating PNS emits acoustic waves into the post-shock region.
4. These sound waves steepen into weak shocks as they propagate outward.
5. The resulting shocks deposit energy and momentum behind the stalled shock, helping to push it outward.

In this sense, the PNS acts as a transducer that converts accretion energy into acoustic power. This mechanism is conceptually distinct from neutrino heating because the main carrier of energy from the inner core to the post-shock region is sound rather than neutrino absorption.

However, the acoustic mechanism is not currently regarded as the standard mechanism for ordinary CCSNe. Its main uncertainties are whether sufficiently large PNS oscillation amplitudes can be generated self-consistently and whether the resulting acoustic power is large enough to dominate over, or at least significantly supplement, neutrino heating. It is therefore usually treated as a historically important alternative rather than the leading explanation.

3.4.5. Hadron-Quark Phase Transition Mechanism

If a first-order phase transition from hadronic matter to quark matter occurs inside the PNS during the post-bounce accretion phase, the high-density EOS can soften. This softening may trigger a secondary collapse of the PNS core. If the newly formed dense phase becomes stiff again, the collapse can be halted by a second bounce, launching a second shock [63].

The sequence can be summarized as

1. The ordinary bounce shock stalls.
2. Accretion increases the PNS mass and central density.
3. A first-order phase transition softens the high-density EOS.
4. The PNS core undergoes a secondary collapse.
5. The core rebounds on the hybrid-star branch.
6. A second shock is launched.
7. The second shock catches up with the stalled shock and may trigger explosion.

This mechanism is highly dependent on the high-density EOS. It requires a sufficiently strong first-order phase transition at densities and temperatures reached during the post-bounce evolution. If realized, it may produce distinctive neutrino and gravitational-wave signatures, such as a second neutrino burst associated with the second shock.

3.4.6. Jet-based and Collapsar-like Mechanisms

Jet-based mechanisms form another class of proposed explosion engines. In the jittering-jets scenario, the explosion is powered by intermittent jets launched by transient accretion disks or accretion belts around the newly formed compact object. The angular momentum of the accreted material is assumed to be stochastic rather than fixed by a globally rapidly rotating progenitor, so the jet direction changes with time. The jets deposit kinetic energy inside the collapsing core and inflate hot bubbles that help eject the stellar mantle.

The relevant energy reservoir is accretion power,

$$E_{\text{acc}} \sim \frac{GM_{\text{NS}}M_{\text{acc}}}{R_{\text{NS}}}. \quad (3.31)$$

Here, M_{NS} and R_{NS} are the mass and radius of the central neutron star, and M_{acc} is the mass accreted through the transient disk or belt. The mechanism is often described as a feedback cycle: accretion powers jets, jets eject the surrounding material, and successful ejection shuts off the accretion [64]. This mechanism is an alternative proposal and is not the standard interpretation of ordinary CCSN explosions.

In very massive and rapidly rotating progenitors, the stalled shock may fail to produce an ordinary PNS-centered explosion. If the compact remnant collapses to a BH and the infalling material has enough angular momentum, an accretion disk can form around the BH. The disk and BH can then power relativistic jets or disk winds through ν pair annihilation, magnetically driven disk winds, or Blandford-Znajek extraction of BH spin energy. This collapsar-like mechanism is most relevant for long gamma-ray bursts, hypernovae, and broad-lined Type Ic supernovae, and should be distinguished from ordinary PNS-centered stalled-shock revival [65].

3.4.7. Nuclear Recombination and Explosive Burning

Nuclear recombination and explosive burning can contribute to the final explosion energy, but they are usually not primary shock revival mechanisms. During the expansion of neutrino-heated ejecta, free nucleons can recombine into α particles and heavy nuclei, releasing nuclear binding energy. This contribution can be important for the diagnostic explosion energy, but it becomes effective mainly after matter has already been lifted out of the deep gravitational potential [66].

Similarly, explosive O and Si burning can add energy after the revived shock propagates through the stellar mantle. These processes affect the final explosion energy and nucleosynthesis, but they are better understood as post-revival energy contributions rather than as the initial trigger of shock revival.

3.5. Observational Signatures of Core-Collapse Supernovae

Observations of CCSNe provide several complementary probes of the collapse, explosion, and remnant formation. Since photons are processed by the stellar envelope and the ejecta before escaping, electromagnetic observations mainly probe the outer layers, the ejecta composition, the circumstellar environment, and the radioactive energy deposition. By contrast, neutrinos and gravitational waves can carry information from much deeper regions. Neutrinos directly probe the thermodynamic evolution and cooling of the PNS, while gravitational waves probe non-spherical dynamics in the inner core and the PNS. These are summarized in Table 3.1.

Observation channel	Current status	Observables	Physical information
Electromagnetic	routinely observed	light curve, spectra, polarization	ejecta, envelope, circumstellar medium (CSM), nucleosynthesis
Neutrino	directly observed only from SN 1987A	event time, energy, flavor channel	PNS cooling, deleptonization, core energetics
Gravitational wave	no confirmed detection	strain, frequency, duration	inner-core dynamics, rotation, SASI, PNS oscillations
Remnant	observed for nearby remnants	compact object, morphology, abundances	explosion asymmetry, fallback, nucleosynthesis

Table 3.1: Observation channels of CCSNe and the corresponding physical information. Electromagnetic observations mainly probe the outer ejecta and late-time remnant, whereas neutrinos and gravitational waves can probe the inner core more directly.

This distinction is important in exploring new physics. Electromagnetic observations constrain the successful explosion, the ejecta energy, the progenitor structure, and the amount of

radioactive material. Neutrino observations constrain the cooling time scale and the total energy emitted from the PNS. Gravitational wave observations, if detected from a future Galactic CCSN, would provide additional information on the multidimensional dynamics and compact remnant formation. Therefore, CCSN observations provide the physical basis for the BSM constraints.

3.5.1. Electromagnetic Observations

Electromagnetic observations are the most common observational probes of CCSNe. They include light curves, spectra, polarization, and late-time imaging of the remnant. These observables are not direct probes of the central PNS. Instead, they mainly describe how the explosion energy deposited in the ejecta is converted into radiation as the shock propagates through the stellar envelope and as radioactive isotopes decay.

The early electromagnetic signal begins with shock breakout. When the explosion shock reaches the stellar surface, radiation trapped behind the shock is released. The duration and luminosity of this signal are sensitive to the progenitor radius, the envelope density profile, and the circumstellar material. After shock breakout, the expanding ejecta cools, and the light curve enters a phase controlled by the envelope structure and radiative diffusion.

For Type II SNe, the optical light curve often shows a plateau. The plateau is caused by hydrogen recombination in the expanding envelope. During this phase, the photosphere recedes in mass coordinate as the ejecta expands and cools. The duration and luminosity of the plateau depend on the progenitor radius, the hydrogen envelope mass, the explosion energy, and the ejecta opacity.

At late times, the luminosity is often powered by radioactive decay. The most important chain is



The half-life of ^{56}Ni is about 6.1 days, and ^{56}Co is about 77.3 days. The late-time radioactive tail is therefore used to estimate the synthesized ^{56}Ni mass. In SN 1987A, the light curve was not a standard Type II-P or Type II-L, but instead showed a long-rising behavior associated with a compact blue supergiant progenitor. Nevertheless, its late-time tail is a representative example of radioactive powering by the decay of ^{56}Co , clearly shown in Fig. 3.2.

Radioactive decays of isotopes are important not only for explaining the late-time light curves of CCSNe, but also for understanding low-energy SNe (LESNe). Among SNe, some events have optically inferred explosion energies lower than the canonical CCSN explosion energy; these are referred to as LESNe. The explosion energies of LESNe are about 10% of those of canonical CCSNe, and their expansion velocities are lower by a factor of about 2-3. A notable feature is that the mass of ^{56}Ni contained in their ejecta is also only about 10% of that in typical CCSNe. As shown in Figs. 3.14 and 3.15, electromagnetic observations of SNe suggest a strong correlation among the mass of radioactive nickel in the ejecta, the SN explosion energy, and the expansion velocity.

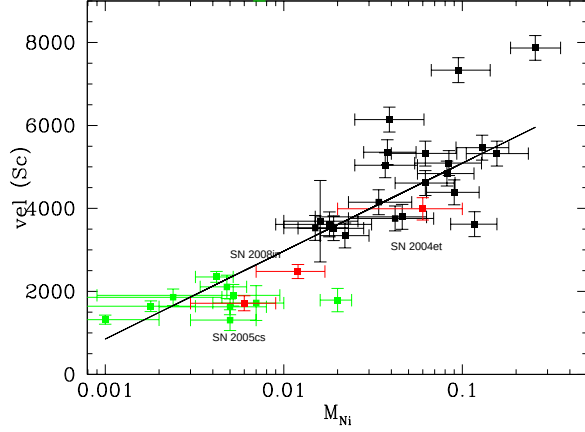


Figure 3.14: Ni masses-ejecta velocities plot [67]. The mass is given in units of $M_{\odot}$, and the velocity is given in units of km/s. Sc indicates that the velocity was measured using the Sc II 6246 Å line. Green squares denote Type II LESNe, black squares denote canonical Type II CCSNe, and red squares denote SN 2004et, SN 2005cs, and SN 2008in.

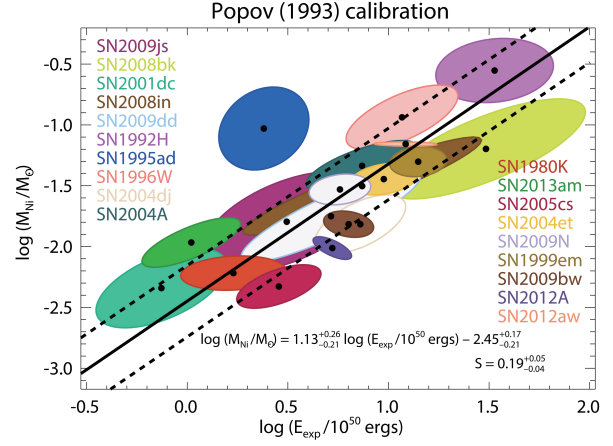


Figure 3.15: Explosion energy-Ni masses plot [68]. The best-fit relation is shown by the solid line, while the Gaussian intrinsic scatter (Σ) is indicated by the dashed lines. Scaling relation of Popov was used [69].

Spectra provide complementary information. During the photospheric phase, absorption and emission features identify the chemical composition and ionization state of the ejecta. During the nebular phase, the ejecta becomes optically thin, and emission lines from O, Ca, Fe, and other elements probe the inner ejecta and nucleosynthesis products. Narrow emission lines, when present, indicate interaction between the ejecta and dense circumstellar material.

Another useful electromagnetic signal is a null observation. One of the representative examples is the no prompt MeV and sub-MeV scale gamma-ray burst associated with SN 1987A. If they exist, they should have been detected by the Solar Maximum Mission (SMM) and the Pioneer Venus Orbiter (PVO). However, as can be seen from Figs. 3.16 and 3.17, such signals were not observed. Such non-detections are not ordinary CCSN observables in the same sense as optical light curves, but they are important in BSM physics because they can constrain radiative decays or delayed energy deposition outside the PNS.

Electromagnetic observations are therefore essential for reconstructing the progenitor, explosion energy, ejecta mass, circumstellar environment, and nucleosynthetic yields. However, they are indirect probes of the central engine. For this reason, electromagnetic observations alone do not directly determine the PNS cooling history. In BSM studies, they are most useful as consistency checks: a new energy loss channel or energy injection channel must not distort the successful explosion, the observed light curve, or the inferred nucleosynthesis.

3.5.2. Neutrino Signal from SN 1987A

Neutrinos are the most direct observed messengers of the CCSN core. During collapse and after bounce, neutrinos are produced by electron capture, nucleon-nucleon bremsstrahlung, pair

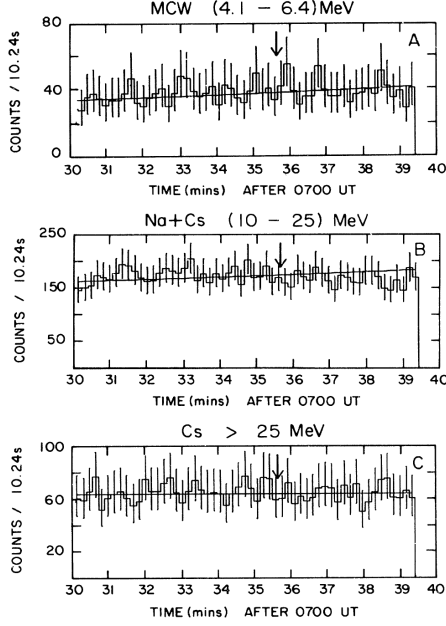


Figure 3.16: Gamma-ray spectrometer data from the Solar Maximum Mission [70]. The three panels correspond to the energy intervals (A) 4.1-6.4 MeV, (B) 10-25 MeV, and (C) >25 MeV. The arrows indicate the first neutrino arrival time at IMB, 7:35:41.37 UTC.

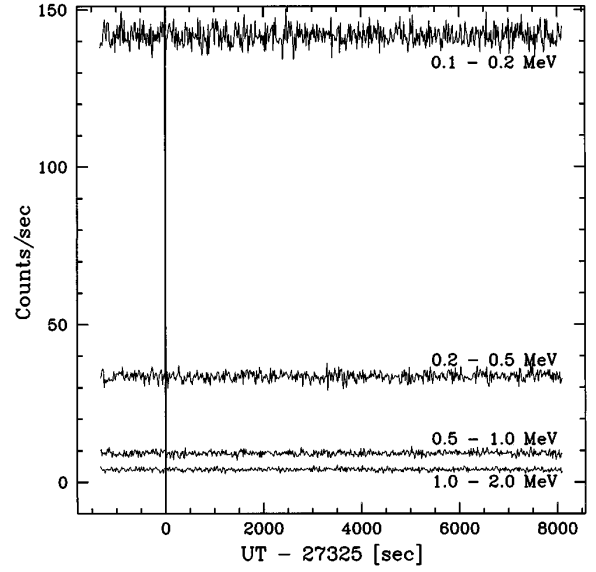


Figure 3.17: Pioneer Venus Orbiter Gamma Burst Detector data for the time immediately before and after the arrival of the neutrinos from SN 1987A [71]. The arrival time of the neutrinos at PVO is 7:35:25 UTC, indicated by the vertical black solid line.

processes, and scattering processes in the hot and dense PNS. Since most of the gravitational binding energy released in the formation of the compact remnant is emitted in neutrinos, the neutrino signal carries direct information about the PNS cooling and deleptonization.

The neutrino signal can be divided into three stages. The first is the neutronization burst, dominated by ν_e emission when the shock crosses the neutrinosphere. The second is the accretion phase, during which matter continues to accrete through the stalled shock and the neutrino luminosity is powered partly by accretion. The third is the cooling phase, during which the PNS cools and deleptonizes by emitting neutrinos of all flavors. The total duration of the cooling signal is typically of order several to ten seconds. Each phase is illustrated in Fig. 3.18.

The neutrino fluence at Earth can be written as

$$\mathcal{F}_{\nu_i}(E_\nu) = \int dt F_{\nu_i}(E_\nu, t) = \frac{1}{4\pi D^2} \int dt \frac{d^2 N_{\nu_i}}{dE_\nu dt}, \quad (3.33)$$

where D is the distance to the supernova and $d^2 N_{\nu_i}/dE_\nu dt$ is the source emission spectrum. For a detector channel a , the expected number of events is

$$N_a = N_{\text{tar}} \int dt \int dE_\nu F_{\nu_i}(E_\nu, t) \sigma_a(E_\nu) \epsilon_a(E_\nu), \quad (3.34)$$

where N_{tar} is the number of targets, and σ_a is the relevant interaction cross section. ϵ_a is the dimensionless detection efficiency and represents the probability that an event produced in

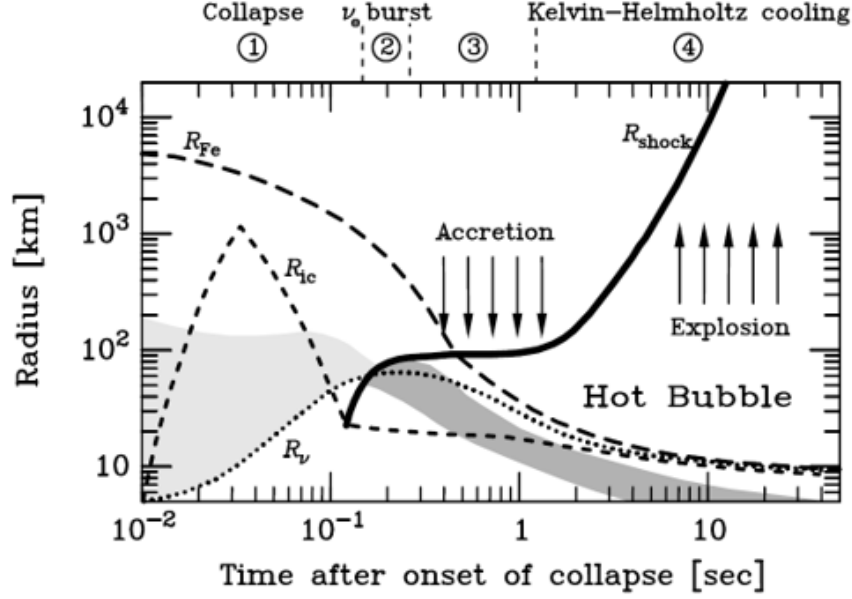


Figure 3.18: Schematic diagram of the stages of the core-collapse supernova explosion [72]. ② marks the neutronization burst phase, ③ the accretion phase, and ④ the cooling phase of the neutrino emission signal. R_{Fe} , R_{ic} , R_{ν} , and R_{shock} denote the iron-core radius, inner-core radius, neutrinosphere radius, and bounce-shock radius, respectively.

channel a is triggered, reconstructed, and selected. In a more detailed detector analysis, this factor is replaced by an energy-dependent detector response or smearing function. In water Cherenkov and liquid scintillator detectors, the dominant channel for $\bar{\nu}_e$ is inverse beta decay,

$$\bar{\nu}_e + p \rightarrow n + e^+. \quad (3.35)$$

The only confirmed neutrino observation from an extragalactic CCSN is SN 1987A. Neutrino events were observed by IMB, Kamiokande-II, and Baksan as shown in Fig. 3.19. Although the statistics were small, this observation confirmed the basic picture that a CCSN releases most of its gravitational binding energy in neutrinos with typical energies of order tens of MeV.

This historical neutrino signal is also the basis of many CCSN constraints on new physics. If a weakly coupled BSM particle is produced efficiently in the hot and dense PNS and escapes freely, it can carry away energy in addition to neutrinos. This would shorten the neutrino cooling time or reduce the observed neutrino luminosity. Conversely, if the new particle is trapped, decays, or deposits energy outside the PNS, it may modify the explosion, the neutrino signal, or the electromagnetic observables. Therefore, the observed duration and energetics of the SN 1987A neutrino signal provide a benchmark for anomalous cooling and energy transport constraints.

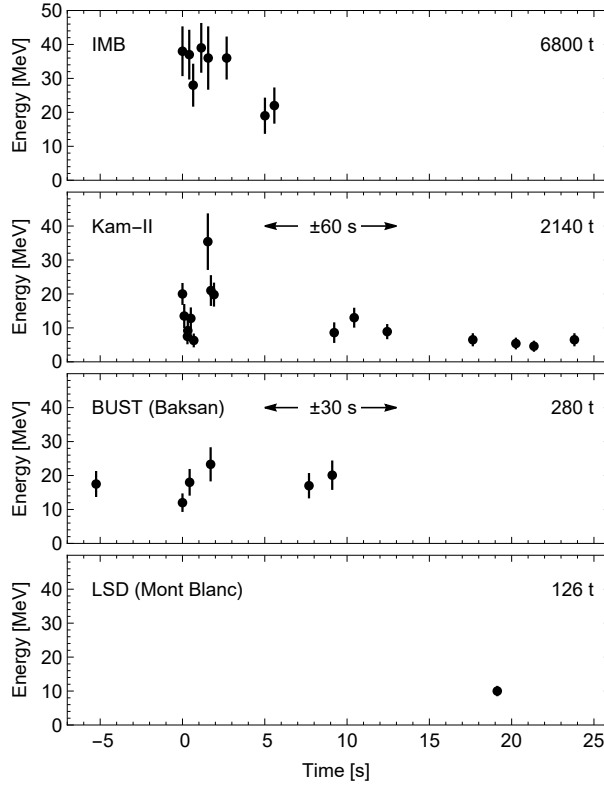


Figure 3.19: SN 1987A neutrino data from IMB, Kamiokande-II, BUST and LSD [73]. The observed positron energies are shown relative to the time at which the first neutrino event was observed at IMB (7:35:41.374 UTC). The first detections at Kamiokande-II and BUST are aligned with the time of the first IMB event, because the two detectors had clock uncertainties at the time.

3.5.3. Gravitational Wave Signatures

Gravitational waves provide a potential probe of the non-spherical dynamics of the CCSN core. Unlike electromagnetic radiation, gravitational waves are not significantly affected by the stellar envelope. They can therefore carry information from the inner core, the gain layer, and the PNS. However, no gravitational wave signal from a CCSN has yet been confirmed.

To illustrate the present observational status of this channel, it is useful to compare the detection efficiency of current gravitational wave searches for different CCSN models. The detection efficiency is the fraction of simulated signals that are recovered by a search pipeline after they are injected into real detector data for a given source distance, sky position, and waveform model. Thus, the distance at which the efficiency reaches 50% gives a representative detection distance for that model. As an example, Fig. 3.20 shows the detection efficiency for simulated CCSN gravitational wave signals associated with the optically observed supernova SN 2019ejj. Current searches are mainly sensitive to ordinary neutrino-driven CCSN gravitational wave signals only within the Milky Way, whereas more energetic or extreme emission models can be probed at larger distances.

The gravitational wave observable is the dimensionless strain $h(t)$, which represents the fractional deformation of distance caused by the wave, schematically $h \sim \Delta l/l$. More precisely,

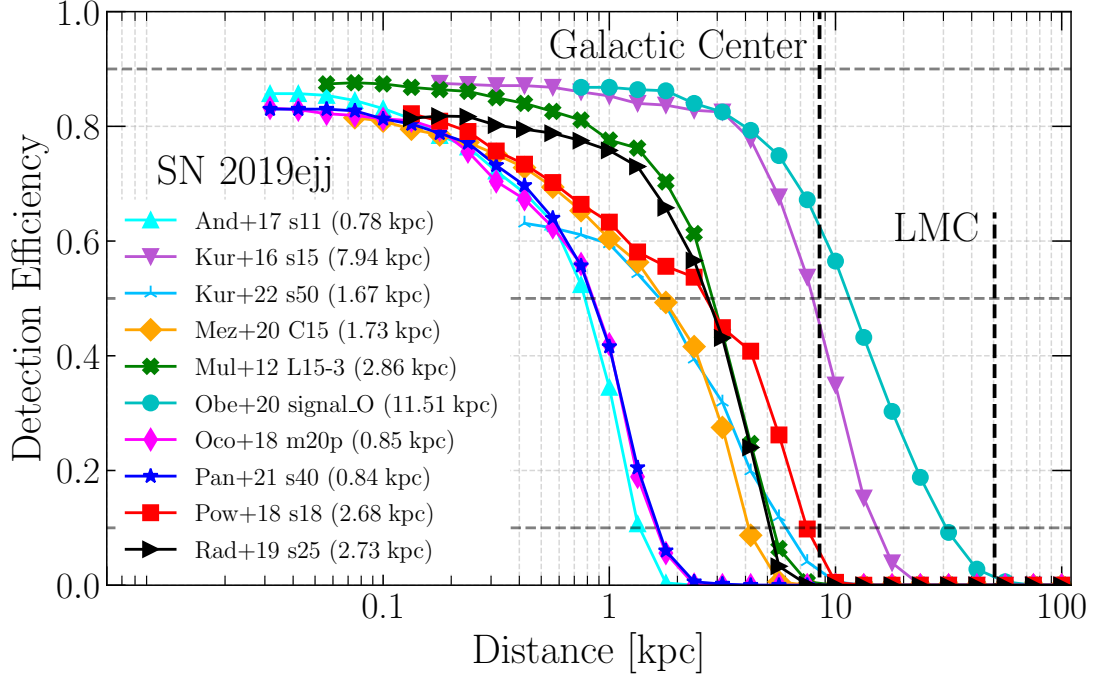


Figure 3.20: Detection efficiency as a function of the assumed source distance for simulated CCSN gravitational wave signals associated with the optically observed Type II supernova SN 2019ejj, located at a distance of 15.7 Mpc, using data from the third observing run of Advanced LIGO and Advanced Virgo [74]. The plot shows the efficiencies for 12 CCSN waveform models derived from multidimensional CCSN simulations, with the Galactic Center and Large Magellanic Cloud (LMC) distances indicated for reference. The horizontal axis denotes the assumed source distance, and the numbers in parentheses indicate the distances at 50% detection efficiencies. The result illustrates that ordinary neutrino-driven CCSN gravitational wave signals are mainly Galactic targets for current detectors.

a detector measures a linear combination of the two tensor polarizations,

$$h_{\text{det}}(t) = F_+ h_+(t) + F_\times h_\times(t), \quad (3.36)$$

where F_+ and $F_\times$ are the detector antenna pattern functions. In the quadrupole approximation, the transverse-traceless part of the strain is

$$h_{ij}^{\text{TT}} \sim \frac{2G}{c^4 D} \frac{d^2 Q_{ij}^{\text{TT}}}{dt^2}, \quad (3.37)$$

where D is the distance to the source and Q_{ij} is the mass quadrupole moment. In a Newtonian notation,

$$Q_{ij} = \int d^3x \rho \left(x_i x_j - \frac{1}{3} \delta_{ij} \right). \quad (3.38)$$

Therefore, purely spherical collapse does not emit gravitational waves. A detectable signal requires non-spherical motion, such as rotation, convection, SASI, anisotropic accretion, PNS oscillations, or asymmetric neutrino emission.

Different phases of a CCSN can generate different gravitational wave features. Rapid ro-

tation can produce a bounce signal. Prompt convection and neutrino-driven convection generate stochastic emission after bounce. SASI can produce lower-frequency modulations associated with large-scale shock motion. PNS oscillations and surface modes can generate higher-frequency emission that is sensitive to the PNS mass, radius, and EOS. If the PNS collapses to a BH, the gravitational wave and neutrino signals may terminate abruptly.

A future Galactic CCSN detected in gravitational waves would therefore provide a direct probe of multidimensional dynamics. Combined with neutrino observations, it could help identify the time of core bounce, the presence of strong rotation, the development of SASI or convection, and the compact remnant formation history. In BSM studies, gravitational waves are less direct than neutrino cooling bounds, but they can provide an important consistency check on whether the explosion dynamics and PNS evolution are modified.

3.5.4. Remnants and Late-Time Observations

Late-time observations of supernova remnants provide information about the outcome of the explosion. These observations include X-ray, optical, infrared, radio, and gamma-ray measurements of the ejecta, circumstellar interaction, compact remnant, and radioactive isotopes. They probe much later times than the explosion phase, but they preserve information about explosion asymmetry, nucleosynthesis, and fallback.

A representative example is the young Galactic supernova remnant Cassiopeia A, shown in Fig. 3.21. Multiwavelength observations of Cas A demonstrate how different tracers reveal different components of the remnant. The near-infrared JWST image and the radio image trace the large-scale remnant morphology and synchrotron-emitting structures, while the X-ray emission traces shock-heated material. The NuSTAR ^{44}Ti map is particularly important because ^{44}Ti is synthesized in the innermost ejecta and therefore provides a late-time tracer of asymmetric nucleosynthesis. Thus, spatially resolved remnant observations preserve information about multidimensional explosion dynamics long after the prompt neutrino and gravitational-wave signals have disappeared.

The compact remnant also provides important information. A NS, pulsar, magnetar, or BH remnant constrains the final baryonic and gravitational mass of the central object, the fallback history, and the explosion energy. For example, a successful explosion with a NS remnant, a successful explosion with BH formation after fallback, and a failed explosion with prompt BH formation correspond to different observable outcomes.

For BSM studies, remnant observations are usually indirect. They do not directly measure the cooling luminosity of a newly formed PNS. However, they provide important consistency conditions. A new particle model that strongly modifies the energy budget, the electron fraction, or the explosion dynamics may also affect nucleosynthesis, remnant morphology, or compact-remnant formation. Therefore, late-time observables can be used together with neutrino and electromagnetic data to test whether a proposed BSM scenario remains compatible with the observed CCSN population.

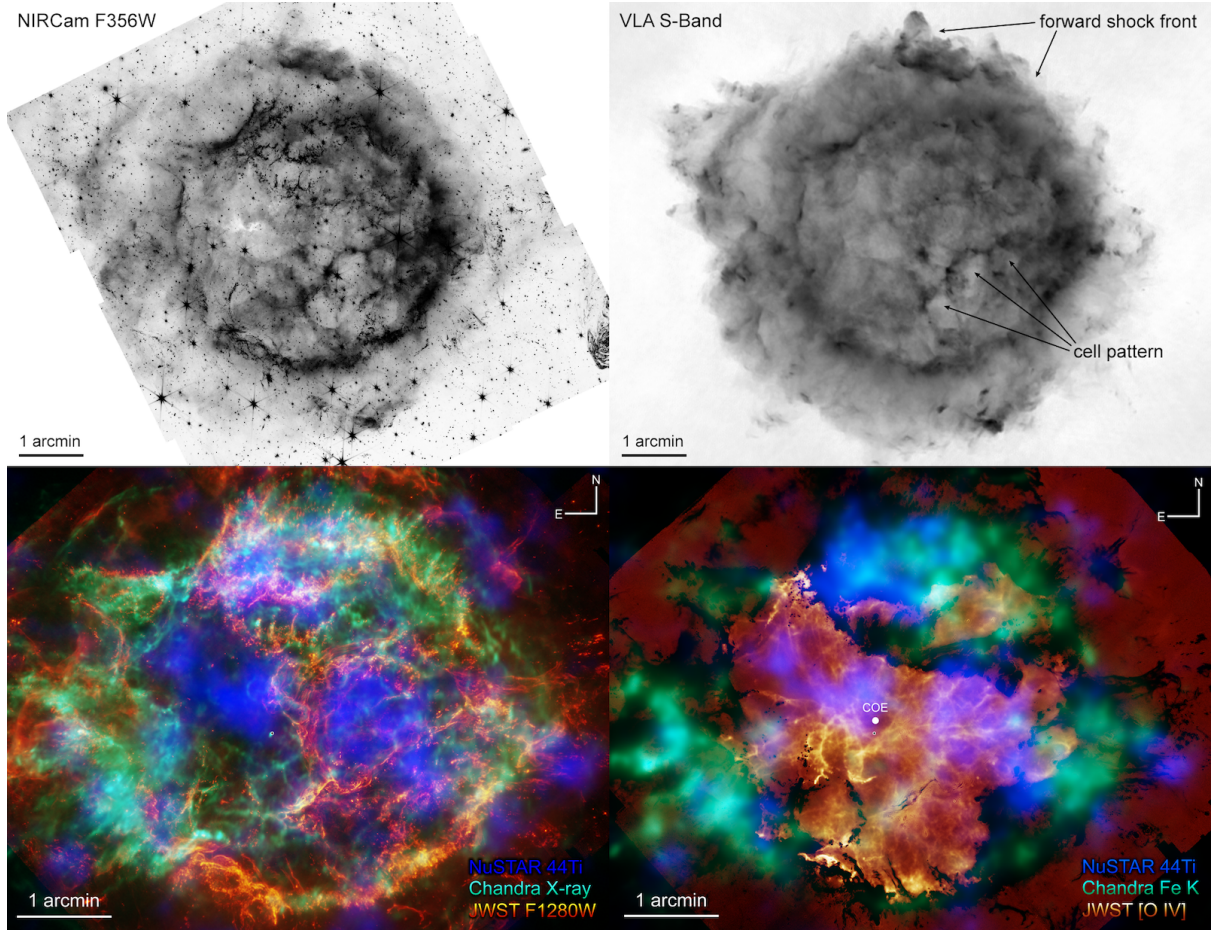


Figure 3.21: Multiwavelength views of the young Galactic supernova remnant Cassiopeia A [75]. The upper panels compare the JWST/NIRCam F356W image with a VLA S-band radio image. The lower left panel combines the NuSTAR ^{44}Ti map, the Chandra 1 Msec X-ray image, and the JWST/MIRI F1280W image, while the lower right panel combines the JWST unshocked ejecta $25.89\,\mu\text{m}$ line map, the NuSTAR ^{44}Ti map, and the Chandra Fe-K map. The radio and X-ray emission trace nonthermal synchrotron radiation and shock-heated material, whereas the infrared emission probes dust and ejecta/CSM structures. The radioactive isotope ^{44}Ti traces material synthesized in the innermost ejecta and provides a late-time diagnostic of asymmetric nucleosynthesis and explosion dynamics. The black circle marks the location of the central neutron star, while the white circle marks the location of the center of expansion (COE).

3.5.5. Multimessenger Picture

The different observational channels probe different regions and time scales of a CCSN. Electromagnetic signals mainly probe the shock-heated envelope, expanding ejecta, CSM interaction, and radioactive decay. Neutrinos probe the PNS cooling, deleptonization, and total gravitational binding energy release. Gravitational waves probe non-spherical inner-core and PNS dynamics. Remnant observations probe the final ejecta morphology, nucleosynthesis products, and compact-remnant outcome.

Thus, CCSN observations are naturally multimessenger. A future Galactic CCSN observed

simultaneously in electromagnetic radiation, neutrinos, and gravitational waves would provide a direct probe of the sequence

collapse $\rightarrow$ bounce $\rightarrow$ shock stagnation $\rightarrow$ shock revival $\rightarrow$ PNS cooling $\rightarrow$ remnant formation.

In the context of new physics, each channel constrains a different aspect of the model. Neutrinos constrain anomalous cooling and energy transport in the PNS. Electromagnetic signals constrain the explosion energy, ejecta, and late-time energy deposition. Gravitational waves constrain multidimensional dynamics and compact remnant evolution. Remnant observations constrain nucleosynthesis, fallback, and the final compact object.

3.6. Proposed Strategies to Explore New Physics

CCSNe provide several strategies to explore physics beyond the Standard Model. The central object formed after core bounce, the PNS, is hot, dense, and lepton-rich. Therefore, weakly coupled new particles can be produced efficiently in the PNS, in the cooling layer, or in the gain layer. In addition, if these particles decay or deposit energy outside the PNS, they may affect the ejecta, the circumstellar medium, or diffuse astrophysical backgrounds. In this context, we classify CCSN-based new physics searches not by the specific interaction channel, but by the physical or observational mechanism used to impose the constraint. The main strategies to investigate new physics with CCSNe are follows:

1. PNS cooling.
2. Neutrino signal distortions.
3. Shock cooling and explosion failure.
4. Energy deposition in low-energy SNe.
5. Circumstellar medium precursor signals.
6. Gamma-ray and X-ray signals.
7. Positron injection and the 511 keV line.
8. Diffuse photon and neutrino backgrounds.
9. Dense matter phase transition signatures.

Based on these strategies, axions, ALPs, dark photons, dark Z bosons, sterile neutrinos, majorons, and other feebly interacting particles can all be constrained by similar arguments if they lead to anomalous cooling, energy deposition, or observable photon or neutrino signals ¹.

¹For example, CCSN-based bounds on axions and dark photons can be found in the constraint plots provided on webpages <https://cajohare.github.io/AxionLimits/> and <https://cajohare.github.io/AxionLimits/docs/dp>, respectively.

3.6.1. Proto-Neutron Star Cooling

The most widely used CCSN constraint on new physics is the anomalous cooling argument based on the SN 1987A neutrino signal. This PNS cooling criterion was first formulated as a general constraint on new particle emission from SN 1987A by Raffelt and Seckel [76], therefore it is also known as the Raffelt criterion.

In the neutrino-driven mechanism, most of the gravitational binding energy released during compact remnant formation is emitted as neutrinos over a time scale of several to ten seconds. If a new particle X is produced efficiently in the PNS and escapes freely, it provides an additional cooling channel. This cooling effect reduces the energy available for neutrino production. Therefore, if this exotic energy-loss rate is too large, the neutrino signal would be shortened or significantly modified.

The commonly used PNS cooling criterion was established from the SN 1987A neutrino observations summarized in Fig. 3.19. In order for the observed neutrino signal to be explained, the luminosity L_X of particle X at $t_{\text{pb}} \sim 1$ s must satisfy the following condition.

$$L_X \lesssim \mathcal{O}(10^{52}) \text{ erg/s}. \quad (3.39)$$

Assuming spherical symmetry, L_X can be written as

$$L_X = 4\pi \int_0^{R_v} dr r^2 \int_{m_X/\alpha}^{\infty} d\omega \alpha^2 \frac{dQ_X}{d\omega} e^{-\tau_X}, \quad (3.40)$$

where Q_X is the emissivity of X , i.e., the energy emitted per unit time and per unit volume, and corresponds to the production rate of particle X . τ_X is the optical depth of X . If a particle is produced at radius r and travels a distance l with an angle θ_X with respect to the outward radial direction, it can be written as follows.

$$\tau_X = \int_0^{l_{\text{max}}} dl \frac{\Gamma_X \left(\sqrt{r^2 + l^2 + 2rl \cos \theta_X} \right)}{v_X}. \quad (3.41)$$

Here, l_{max} is the maximum travel distance of the particle. For example, if the particle is produced at $r < R_v$ and travels to R_v ,

$$l_{\text{max}} = \sqrt{R_v^2 - r^2 \sin^2 \theta_X} - r \cos \theta_X. \quad (3.42)$$

Since it is difficult to perform a numerical evaluation due to the θ_X dependence of τ_X , one may remove the angular dependence by taking an angular average as

$$\langle e^{-\tau_X} \rangle_{\theta_X} = \frac{\int d\theta_X \omega(\theta_X) e^{-\tau_X}}{\int d\theta_X \omega(\theta_X)} = \frac{1}{2} \int d\theta_X e^{-\tau_X}. \quad (3.43)$$

The last equality holds when ω has no θ_X dependence. From the definition of optical depth, the

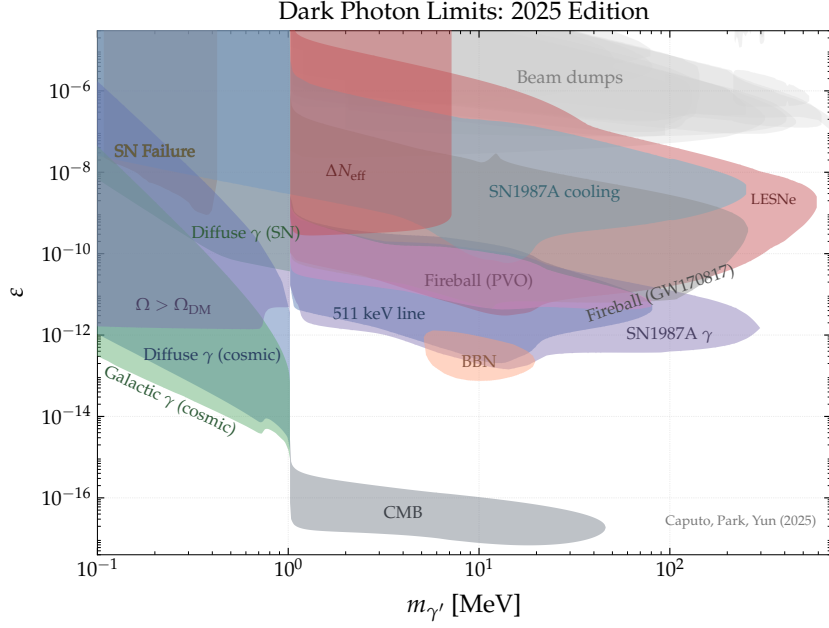


Figure 3.22: Astrophysical, cosmological, and collider constraints on the dark photon-photon kinetic mixing as a function of the dark photon mass [77]. The constraints from core-collapse supernovae include SN 1987A cooling (dark blue), failed supernovae (mustard), low-energy supernovae (light red), SN 1987A gamma-ray (lilac), fireball formation from SN 1987A (magenta), the 511 keV line (violet blue), and diffuse supernova gamma-ray background.

flux I_0 at the production point of X and the observed flux I after passing through the medium satisfy the following relation.

$$e^{-\tau_X} = \frac{I}{I_0}. \quad (3.44)$$

Therefore, this factor corresponds to the escape probability accounting for absorption of X . From a practical point of view, since Eq. (3.40) includes X particles produced at R_v , the observation point of X should not be set exactly at R_v , but rather at a position slightly outside R_v . α is the gravitational lapse function. In the ADM formalism, for a spherically symmetric metric with vanishing shift vector, it is given by

$$\alpha = \sqrt{-g_{tt}}. \quad (3.45)$$

So, the factor $1/\alpha$ in the lower bound of the ω integration in Eq. (3.40) corresponds to gravitational trapping, while the factor α^2 inside the integration accounts for energy redshift and time dilation.

Therefore, Eq. (3.40) gives the luminosity of X particles produced inside R_v that escape outside R_v without being absorbed by the supernova medium. It is the basic physical quantity used to evaluate the anomalous PNS cooling effect caused by X . Since Q_X and τ_X depend on the interaction channels of X , for a given BSM model one can compute these quantities analytically, evaluate Eq. (3.40) numerically using SN simulation profiles, and compare the result with the criterion in Eq. (3.39) to impose the PNS cooling constraint.

One example of imposing PNS cooling bounds on a BSM particle is shown in Fig. 3.22. Fig. 3.22 summarizes astrophysical, cosmological, and collider bounds on the dark photon mass $m_{\gamma'}$ and kinetic mixing parameter ε of a MeV scale dark photon, which is the gauge boson of $U(1)_D$ that has kinetic mixing with $U(1)_{EM}$.

The dark blue shaded region labeled SN1987A cooling corresponds to the PNS cooling constraint, and dark photons in this region are excluded by the PNS cooling criterion. Along the ε axis, the upper region not excluded by the PNS cooling bound, corresponding to large ε and therefore strong coupling, is called the trapping regime. The lower region, corresponding to small ε and therefore weak coupling, is called the free-streaming regime.

In the trapping regime, a large number of X particles are produced inside the SN due to the strong coupling, but they are also efficiently absorbed by the medium, so L_X can satisfy the criterion given in Eq. (3.39). In this case, the emission of X resembles neutrino emission from the neutrinosphere, and has the character of surface emission.

By contrast, in the free-streaming regime, the weak coupling suppresses the production of X . In this case, the produced X particles escape freely from the SN, and the emission has the character of volume emission.

3.6.2. Neutrino Signal Distortions

A more refined strategy is to use the detailed time, energy, and flavor structure of the neutrino signal. This is different from the PNS cooling argument. The cooling argument constrains the total exotic energy loss, whereas signal distortion constraints ask whether the observable neutrino event distribution is compatible with the predicted source emission, flavor conversion, and detector response.

One important example is neutrino flavor conversion. After neutrinos decouple from the PNS, their flavor content can be modified by Mikheyev-Smirnov-Wolfenstein (MSW) conversion in the stellar envelope, shock-wave effects, turbulence, collective neutrino oscillations, fast flavor conversion, and Earth matter effects. A future Galactic CCSN could therefore provide information on the neutrino mass ordering, flavor mixing, and possible nonstandard neutrino interactions.

In the standard treatment of supernova neutrino flavor conversion, the fluxes arriving at Earth can be written in terms of survival probabilities. For example, if the original non-electron flavor spectra are approximately equal, the electron-flavor fluxes can be written as [78]

$$F_{\nu_e} = p F_{\nu_e}^0 + (1 - p) F_{\nu_x}^0, \quad (3.46)$$

and

$$F_{\bar{\nu}_e} = \bar{p} F_{\bar{\nu}_e}^0 + (1 - \bar{p}) F_{\bar{\nu}_x}^0, \quad (3.47)$$

where $F_{\nu_\alpha}^0$ denotes the original flux emitted from the supernova, F_{ν_α} denotes the flux after flavor conversion, and p and $\bar{p}$ are the survival probabilities for ν_e and $\bar{\nu}_e$, respectively. Since p and

$\bar{p}$ depend on the mass ordering, matter profile, shock propagation, and Earth matter effects, the observed time and energy spectra can carry information on neutrino flavor physics.

The neutronization burst is especially useful because it is dominated by ν_e emission and occurs shortly after bounce. Liquid-argon detectors such as DUNE are particularly sensitive to ν_e charged-current interactions, whereas water Cherenkov and liquid-scintillator detectors mainly probe $\bar{\nu}_e$ through inverse beta decay. Therefore, a combination of detectors can test the flavor-dependent structure of the CCSN neutrino burst.

The same idea can also be applied to the existing SN 1987A data, although in a statistically different regime. Unlike a future Galactic CCSN, SN 1987A provides only a small number of events, so the information is extracted through an event-level likelihood rather than through high-statistics spectral reconstruction.

For SN 1987A, the observed data are sparse and must be treated statistically. A standard approach is to write a likelihood for the detected event times and energies. For an isotropic signal rate, the likelihood can be written in the form [79]

$$\mathcal{L}(P) = \exp\left(-f \int_T dt \int d\epsilon \bar{\eta}(\epsilon) R(\epsilon, t)\right) \prod_{i=1}^{N_d} e^{R_{\text{eff}}(t_i)\tau} \left(B_i + \int d\epsilon L_i(\epsilon) R(\epsilon, t_i)\right). \quad (3.48)$$

Here, $R(\epsilon, t)$ is the signal positron-production rate in the detector, $\bar{\eta}(\epsilon)$ is the averaged detection efficiency, $L_i(\epsilon)$ is the event energy likelihood for the i -th event, B_i is the background contribution, τ is the detector dead time, and f is the live-time fraction. This likelihood illustrates that a realistic analysis of the neutrino signal should include not only the emitted spectrum, but also detector efficiency, event-by-event energy uncertainty, background, and dead time.

This event-level viewpoint motivates a direct comparison between simulated detector-level neutrino signals and the observed SN 1987A events. Representative examples are shown in Figs. 3.23 and 3.24. In Fig. 3.23, modern CCSN simulations are compared with the first-second neutrino data from IMB and Kamiokande-II in the nominal case without neutrino oscillations. The predictions from different modern simulations are broadly consistent with one another, despite differences in dimensionality, numerical methods, and input physics. However, their agreement with the SN 1987A data is poor. In the nominal comparison, the predicted event counts are generally too high, and the predicted average detected positron energies do not simultaneously match both detectors. This demonstrates that the early neutrino signal can be tested through detector-level event counts and energy distributions, not only through the total duration of the burst.

Since flavor conversion is one of the most important sources of neutrino signal distortion, it is natural to ask whether ordinary oscillation effects can reduce this tension. This is illustrated in Fig. 3.24, where representative oscillation scenarios are applied to the ALCAR three-dimensional $20M_\odot$ model. The oscillation comparison shows how the predicted detector-level signal moves when the flavor conversion scenario is changed. In this example, oscillations tend to decrease the predicted counts and increase the average detected positron energies, reflecting the fact that non-electron antineutrinos have lower luminosities but higher average energies

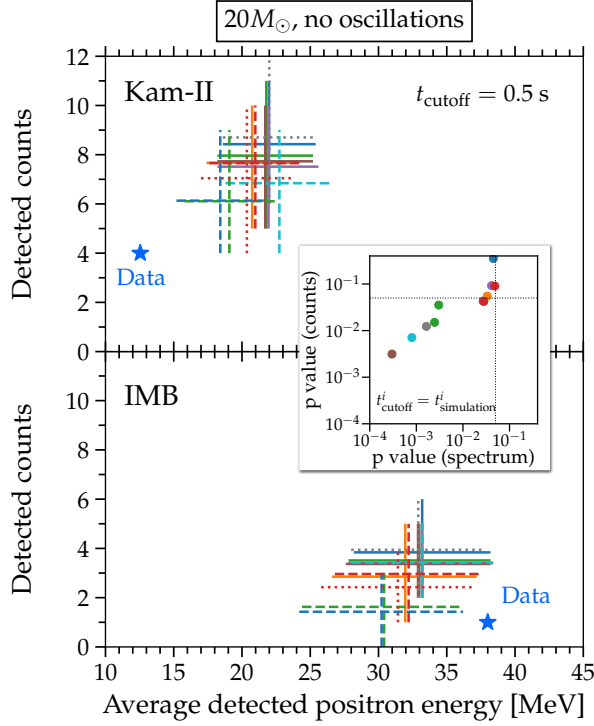


Figure 3.23: Predicted detected counts and average detected positron energies of modern CCSN simulations for a $20M_{\odot}$ progenitor without neutrino oscillations, compared with the first-second SN 1987A data from IMB and Kamiokande-II [80]. Solid, dashed, and dotted lines show the 1D, 2D, and 3D simulation results, respectively. The insets show the corresponding goodness-of-fit p -values.

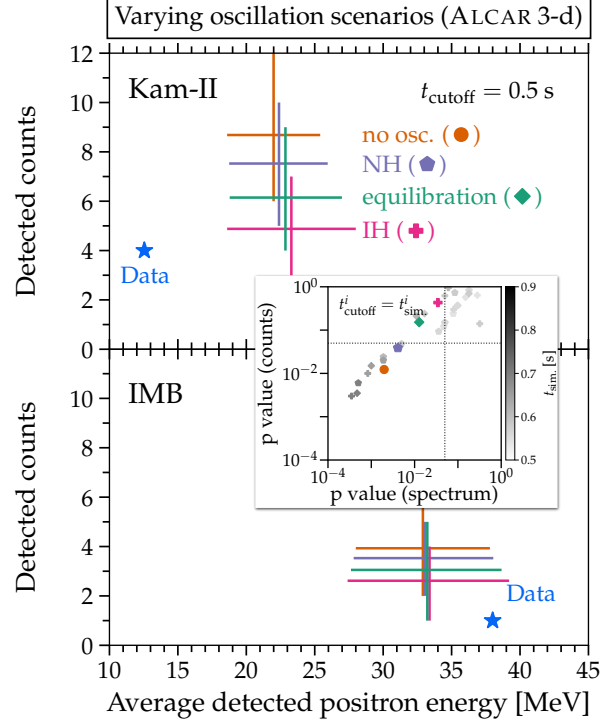


Figure 3.24: Predicted detected counts and average detected positron energies for the ALCAR three-dimensional $20M_{\odot}$ model under representative neutrino oscillation scenarios, compared with the first-second SN 1987A data from IMB and Kamiokande-II [80]. The inset shows the corresponding goodness-of-fit p -values.

than $\bar{\nu}_e$. However, the agreement with the SN 1987A data remains poor. Thus, within this analysis, standard neutrino oscillations do not appear to be sufficient to remove the discrepancy between modern CCSN simulations and the first-second SN 1987A neutrino data.

The lesson is that the key observable is not only the total duration of the neutrino burst, but also the event-by-event time and energy distribution. A future Galactic CCSN with high-statistics neutrino data would therefore make it possible to test the consistency of the source model, flavor conversion, and possible new physics much more sharply than SN 1987A.

3.6.3. Shock Cooling and Explosion Failure

Another strategy is to require that new particles should not prevent shock revival. This argument is different from the PNS cooling bound. This shock failure criterion was introduced by Caputo, Janka, Raffelt, and Yun in the context of dark photons resonantly produced in the gain region [81].

In the delayed neutrino-heating mechanism, the shock stalls after core bounce and is later

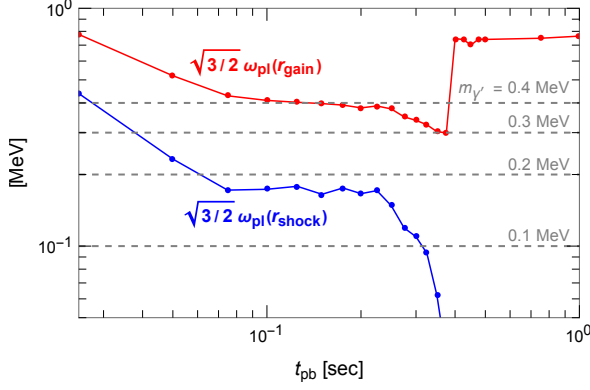


Figure 3.25: Time evolution of the photon plasma frequency at the gain radius and shock radius in s18.88-LS220 1D simulation results [81]. Horizontal dashed lines indicate representative dark photon masses.

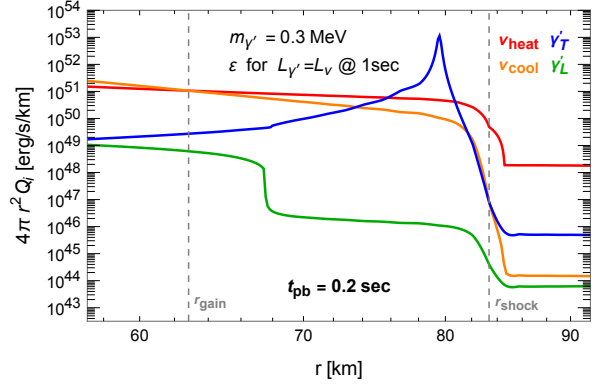


Figure 3.26: Radial profiles of the differential dark photon luminosity and neutrino heating/cooling rates in the gain region before shock revival [81]. Vertical dashed lines indicate the gain radius and shock radius.

revived by neutrino energy deposition in the gain layer,

$$R_g < r < R_s. \quad (3.49)$$

If X is produced efficiently in this region and escapes, it can remove energy from the same layer where neutrinos are trying to reheat the shock. In this case, the explosion can fail even if the exotic luminosity from the PNS core is too small to violate the traditional SN 1987A cooling bound.

Using the volume integrated net neutrino heating rate $\dot{Q}_v$ defined in Eq. (3.21), the impact of an additional cooling channel can be characterized by the ratio

$$\xi_{X/v} = \frac{\int_{t_{st}}^{t_{sh}} dt \dot{Q}_X(t)}{\int_{t_{st}}^{t_{sh}} dt \dot{Q}_v(t)}, \quad (3.50)$$

where $\dot{Q}_X$ is the cooling rate of X integrated over the gain layer. The time interval t_{st}, t_{sh} is chosen to cover the shock-reheating phase, from shock stagnation to the onset of revitalized shock runaway. A successful explosion requires $\xi_{X/v}$ not to be too large, because otherwise the new cooling channel would cancel a significant fraction of the neutrino energy deposition.

A representative realization of this idea is the dark photon shock cooling bound [81]. In this case, dark photons can be resonantly produced when the in-medium photon plasma frequency matches the dark photon mass. The exotic cooling rate in the gain layer can be written as

$$\dot{Q}_{\gamma'}(t) = \int_{R_g}^{R_s} \frac{dL_{\gamma'}}{dr} dr, \quad \frac{dL_{\gamma'}}{dr} = 4\pi r^2 Q_{\gamma'}, \quad (3.51)$$

where $Q_{\gamma'}$ is the local dark photon cooling rate per unit volume.

This mechanism is illustrated in Figs. 3.25 and 3.26. If the dark photon mass lies between

the plasma frequencies at the gain radius and the shock radius, transverse dark photons can be resonantly produced inside the gain layer. As shown in Fig. 3.25, dark photons with masses of $\mathcal{O}(0.1)$ MeV can be resonantly produced in the gain layer. The resulting cooling can then be compared directly with the neutrino heating and cooling rates in the same region, based on the Eq. (3.50). In the example shown in Fig. 3.26, the resonant dark photon contribution appears as a localized cooling peak in the gain layer.

The resulting constraint is imposed by requiring that the integrated exotic cooling should not remove too large a fraction of the neutrino energy deposition during the shock-revival phase. For ordinary iron-core CCSNe, a representative critical value is taken to be

$$\xi_{\gamma/\nu} \lesssim 0.2. \quad (3.52)$$

The precise threshold is model dependent and ultimately should be calibrated by self-consistent multidimensional simulations, but the resulting bound depends only weakly on this choice because the kinetic mixing scales ε approximately as

$$\varepsilon \propto \xi_{\gamma/\nu}^{1/2}. \quad (3.53)$$

The corresponding excluded region is shown in mustard in Fig. 3.22, labeled as SN Failure.

Thus, shock cooling provides a consistency condition that precedes the usual PNS cooling argument. If the new particle already prevents the explosion, then a later-time SN 1987A cooling bound is no longer the relevant first constraint. This makes shock cooling a conceptually distinct strategy: it tests whether the explosion mechanism itself remains viable in the presence of a new energy-loss channel.

3.6.4. Energy Deposition in Low-Energy Supernovae

If a new particle escapes the PNS but decays or deposits energy inside the stellar envelope or ejecta, it may enhance the explosion energy or modify the light curve. The broader calorimetric idea that visible particle decays inside the progenitor star should not over-energize the explosion goes back to early work on radiative neutrino decays, while the specific use of LESNe as especially sensitive calorimeters was introduced by Caputo, Janka, Raffelt, and Vitagliano [82].

The relevant regime is therefore roughly

$$R_{\text{PNS}} \lesssim \lambda_X \lesssim R_*, \quad (3.54)$$

where R_* is the progenitor photosphere radius, and λ_X is the decay or absorption mean free path of X . If X decays into photons or e^-e^+ pairs inside the star, the decay products quickly thermalize and transfer their energy to the stellar material.

LESNe are especially useful because their explosion energies are much smaller than the canonical CCSN. They are observationally associated with low luminosities, low ejecta veloc-

ities, and small ejected ^{56}Ni masses. Their explosion energies can be of order

$$E_{\text{exp, LESNe}} \sim \mathcal{O}(0.1) B, \quad (3.55)$$

or even smaller. Therefore, the energy deposition by X larger than this scale would over-energize the LESNe explosion energy. The resulting calorimetric constraint can be written as

$$E_{X, \text{dep}} \lesssim E_{\text{LESNe}} \simeq 0.1 B, \quad (3.56)$$

where $E_{X, \text{dep}}$ is the energy deposited by X inside the progenitor star and E_{LESNe} is the characteristic explosion energy of LESNe.

For a radiatively decaying particle X , the deposited energy in the mantle can be evaluated by integrating the emitted luminosity with the probability that the particle decays between the PNS surface and the stellar surface. In the notation of Ref. [82], this deposited energy is

$$E_{\text{mantle}} = \int dt \int_0^{R_{\text{PNS}}} dr \int_{m_X/\alpha}^{\infty} d\omega \frac{dL_X}{d\omega dr} \left[e^{-(R_{\text{PNS}}-r)/\lambda_X} - e^{-(R_*-r)/\lambda_X} \right]. \quad (3.57)$$

Here, $dL_X/d\omega dr$ is the differential luminosity of the emitted particle, and the exponential factor accounts for decays occurring between R_{PNS} and R_* . The corresponding excluded region for dark photon is shown in light red in Fig. 3.22, labeled as LESNe.

Physically, the bound follows from the fact that an energy injection of order $0.1 B$ is already comparable to the total explosion energy of the faintest LESNe. If such an additional energy source were present, the event would no longer look like a LESN. This makes LESNe powerful calorimeters for radiatively decaying or visibly interacting particles.

3.6.5. Circumstellar Medium Precursor Signals

Dense circumstellar material can provide another probe of new particles. This CSM precursor strategy was proposed by Cheng, Kong, Lin, Wu, and Yun in the context of feebly interacting particles produced in the PNS and decaying in the dense CSM before shock breakout [83].

The relevant spatial regime is therefore

$$R_* \lesssim \lambda_X \lesssim R_{\text{CSM}}, \quad (3.58)$$

where R_{CSM} is the characteristic radius of the dense CSM. In this regime, X does not simply over-energize the ejecta. Instead, it deposits visible energy outside the stellar surface but before the shock reaches the CSM, producing a precursor emission component.

A representative realization is given by MeV scale dark photons. For a dark photon with momentum k , the decay length is

$$l_{\gamma'} = \frac{k}{m_{\gamma'}} \Gamma_{\gamma'}^{-1}, \quad (3.59)$$

where $\Gamma_{\gamma'}$ is the decay rate in the rest frame. If the dark photons propagate approximately

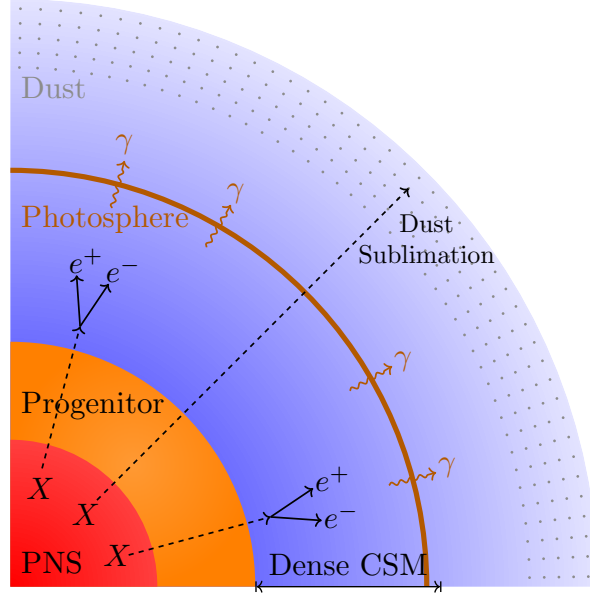


Figure 3.27: Schematic illustration of the energy deposition by X in the dense CSM before shock breakout [83].

radially through the CSM, the differential visible energy injection per unit radius can be written as

$$\frac{dQ}{dr} = \int dk \frac{1}{l_{\gamma'}} \frac{d\mathcal{N}_{\gamma'}}{dk} e^{-r/l_d} \sum_i \int dE_i E_i \frac{dN_i}{dE_i}. \quad (3.60)$$

Here, $d\mathcal{N}_{\gamma'}/dk$ is the dark photon spectrum integrated over the production volume and emission time, and the sum runs over visible final states. For $m_{\gamma'} > 2m_e$, the dominant visible channel is typically $\gamma' \rightarrow e^-e^+$.

If the CSM is dense enough, the produced electrons and positrons efficiently transfer their kinetic energy to the medium. The deposited energy raises the gas temperature and modifies the opacity, leading to a new photosphere inside the CSM. The resulting emission can appear before the ordinary shock breakout, because the FIP induced energy deposition occurs ahead of the outgoing shock.

This mechanism is illustrated in Fig. 3.27. X produced in the PNS can escape the star and decay in the dense CSM before the SN shock reaches the outer layers. The visible decay products then heat and ionize the CSM, producing an additional photosphere and an early precursor signal. Since the precursor is generated before the usual shock breakout, early-time photometric observations can constrain this scenario.

The first application of this strategy used the nearby Type II supernova SN 2023ixf. Early observations of SN 2023ixf indicate the presence of dense and confined CSM, making it a useful benchmark system. The absence of excessive precursor luminosity before shock breakout can be translated into an upper bound on the visible energy deposited by new particles in the CSM,

$$Q_{X,\text{CSM}} \lesssim Q_{\text{obs}}, \quad (3.61)$$

where Q_{obs} is inferred from early-time observations. In the dark photon application, this condition excludes parameter regions in which the decay length is comparable to the CSM scale and the visible energy deposition is large enough to produce an observable precursor.

Compared with the LESNe bound, the CSM precursor bound probes particles that deposit energy at larger radii and earlier observational times. Thus, CSM precursor signals provide a new strategy: they test whether escaping new particles produce an additional early electromagnetic component before the ordinary shock breakout.

3.6.6. Gamma-Ray and X-Ray Signals

New particles produced in CCSNe can also produce high-energy photon signals. This idea was already applied to SN 1987A in early searches for gamma rays from radiative particle decays, using the absence of coincident signals in instruments such as the SMM and the PVO [70, 71].

The most direct case is a particle X that decays into visible final states after escaping the optically thick region. If X decays into photons, electrons, or positrons, the resulting electromagnetic signal can appear as a prompt or delayed gamma-ray burst. The signal is most relevant when the decay length is long enough for X to escape the PNS and the photosphere, but short enough that a sizable fraction of the particles decay before reaching Earth. The observational constraint is then imposed by requiring that the predicted photon fluence should not exceed the instrumental upper limit,

$$\mathcal{F}_\gamma^{\text{th}} \lesssim \mathcal{F}_\gamma^{\text{obs}}. \quad (3.62)$$

Here, $\mathcal{F}_\gamma^{\text{th}}$ denotes the predicted photon fluence from the new-particle signal, integrated over the observational energy range ΔE and time window Δt , while $\mathcal{F}_\gamma^{\text{obs}}$ is the corresponding instrumental upper limit.

For SN 1987A, the non-observation of a prompt MeV gamma-ray burst by the SMM provides a classic example of this strategy. The same logic can also apply to softer X-ray or gamma-ray signatures if the decay products form an electromagnetic cascade or fireball outside the star [84]. In this case, the original decay energy may be reprocessed into lower-energy photons before detection, as in analyses based on PVO data.

A more model-specific realization is photon conversion in an external magnetic field. This conversion strategy was introduced in the SN 1987A context by Brockway, Carlson, and Raffelt, and independently by Grifols, Masso, and Toldra [85, 86]. For example, ultralight ALPs produced in the SN core can convert into gamma rays while propagating through the Galactic magnetic field. In this case, the photon fluence at Earth is related to the ALP fluence by

$$\frac{d\mathcal{F}_\gamma}{d\omega} = P_{a\gamma}(\omega) \frac{d\mathcal{F}_a}{d\omega}, \quad (3.63)$$

where $P_{a\gamma}$ is the ALP to photon conversion probability.

In the constraint plot shown in Fig. 3.22, the SMM non-observation appears as the lilac region labeled SN 1987A gamma-ray, while the PVO fireball constraint appears as the magenta

region labeled Fireball (PVO). Here the diagnostic is not the hydrodynamic response of the star or the duration of the neutrino signal, but the absence of a coincident or delayed photon fluence. This makes the method most sensitive to particles that escape the core and reappear as electromagnetic radiation at larger radii or during propagation.

3.6.7. Positron Injection and the 511 keV Line

If a new particle produced in a CCSN decays into final states containing positrons, the accumulated Galactic population of supernovae can be constrained by the observed 511 keV line. The basic idea was already considered in the context of decaying heavy neutrinos by Dar, Goodman, and Nussinov [87], and has recently been developed as a constraint on feebly interacting particles produced in supernovae [88].

The 511 keV line is produced when nonrelativistic positrons annihilate with electrons in the Galactic medium. Observations by INTEGRAL/SPI show a bright annihilation signal from the inner Galaxy [89], but the origin of the required positrons remains uncertain. This makes the line a useful probe of any source that positrons into the Galaxy.

In the CCSN scenario, the relevant particles are produced in the PNS, escape the stellar envelope, and decay into final states containing positrons before leaving the Galaxy. The produced positrons lose energy in the interstellar medium and eventually annihilate with electrons, producing 511 keV photons. Therefore, the key quantity is the number of positrons injected into the Galaxy per supernova, denoted by N_{pos} .

Given a SFR of the Galactic CCSN R_{SN} , the positron injection rate from this mechanism is approximately

$$\dot{N}_{\text{pos}}^X \simeq R_{\text{SN}} N_{\text{pos}}. \quad (3.64)$$

Here, $\dot{N}_{\text{pos}}^X$ is the Galactic positron injection rate induced by X , and N_{pos} is the average number of positrons injected into the Galactic medium per CCSN. The observed 511 keV emission then imposes

$$R_{\text{SN}} N_{\text{pos}} \lesssim \dot{N}_{\text{pos}}^{\text{obs}}, \quad (3.65)$$

where $\dot{N}_{\text{pos}}^{\text{obs}}$ is the positron injection rate required to account for the observed Galactic annihilation signal.

A more detailed treatment should include the spatial distribution of Galactic supernovae, the probability that the new particle decays outside the stellar envelope but inside the Galaxy, positron propagation, and the angular morphology of the observed 511 keV emission. In applications to MeV scale sterile neutrinos and dark photons [88], using $R_{\text{SN}} = 2.30/\text{century}$ and comparing the predicted annihilation signal with INTEGRAL/SPI data gives the representative upper limit

$$N_{\text{pos}} \lesssim 1.4 \times 10^{52}. \quad (3.66)$$

Thus, particles that produce more than about 10^{52} positrons per Galactic supernova are excluded, with the precise value depending on the assumed SN distribution, positron propagation,

and observational region used in the comparison.

The corresponding excluded region is shown in violet blue in Fig. 3.22, labeled as 511 keV line. This bound is most relevant for particles with masses above the electron-positron threshold and decay channels containing positrons in the final state.

3.6.8. Diffuse Backgrounds

If a signal is produced in every CCSN and survives propagation over cosmological distances, the relevant observable is not a single nearby event but the cumulative background from all past CCSNe. This idea is familiar from the diffuse supernova neutrino background (DSNB), whose existence was proposed in early studies of relic supernova neutrinos [90]. The same logic can also be applied to new particles. Even if the signal from each individual SN is too weak to detect, the integrated contribution from the cosmic CCSN population can be constrained by diffuse photon or neutrino backgrounds.

For neutrinos, the diffuse flux at Earth is obtained by integrating the emitted neutrino spectrum over the cosmic CCSN history. A standard expression is

$$\frac{d\Phi_{\nu\beta}}{dE_\nu} = c \int dz R_{\text{SN}}(z) \frac{dN_{\nu\alpha}}{dE'_\nu} P_{\alpha\beta}(E'_\nu, z) (1+z) \left| \frac{dt}{dz} \right|, \quad (3.67)$$

where $E'_\nu = (1+z)E_\nu$ is the emitted neutrino energy, $R_{\text{SN}}(z)$ is the SFR of the cosmic CCSN per comoving volume and per source frame time, $dN_{\nu\alpha}/dE'_\nu$ is the time-integrated neutrino spectrum emitted per CCSN, and $P_{\alpha\beta}$ is the flavor transition probability from the emitted flavor α to the detected flavor β . Equivalently, since

$$(1+z) \left| \frac{dt}{dz} \right| = \frac{1}{H(z)}, \quad (3.68)$$

this expression can be written as a convolution of the cosmic CCSN rate with the source neutrino spectrum over $dz/H(z)$.

The DSNB is sensitive to the cosmic CCSN rate, the average neutrino spectrum of each collapse, the fraction of failed SNe, and neutrino flavor conversion. It can also be used to test new physics in neutrino sector. Examples include neutrino decay, pseudo-Dirac neutrinos, sterile neutrino mixing, and secret neutrino interactions [91]. In this case, the strategy is to compare the predicted DSNB spectrum with the observed limits or future measurements. Since the DSNB is a population signal, its interpretation is limited not only by detector backgrounds but also by astrophysical uncertainties in the cosmic CCSN population.

For photon signals, the analogous strategy is to require that the diffuse X-ray or gamma-ray flux induced by new particles from all past CCSNe does not exceed the observed diffuse photon background. This has been applied, for example, to ALPs produced in CCSNe, where the accumulated ALP flux can lead to photons through decay or conversion in magnetic fields

[92]. The observational condition can be written as

$$\frac{d\Phi_{\gamma}^{\text{th}}}{dE_{\gamma}} \lesssim \frac{d\Phi_{\gamma}^{\text{obs}}}{dE_{\gamma}}, \quad (3.69)$$

where $d\Phi_{\gamma}^{\text{th}}/dE_{\gamma}$ is the predicted diffuse photon flux from the new particle, and $d\Phi_{\gamma}^{\text{obs}}/dE_{\gamma}$ is the observed diffuse photon background in the same energy range and sky region.

The diffuse background strategy is therefore complementary to event-based constraints such as SN 1987A gamma-ray searches. Event-based constraints use the absence of a coincident signal from one nearby SN, whereas diffuse background constraints use the accumulated contribution of many unresolved CCSNe. The price is that the result depends on population level inputs, such as the cosmic CCSN rate, redshift evolution, source spectra, and propagation effects.

3.6.9. Dense Matter Phase Transition Signatures

The previous strategies mainly constrain weakly coupled particles produced in CCSNe. Another class of new physics searches uses CCSNe to probe the high-density EOS and possible phase transitions in dense-matter. If the PNS reaches densities at which hadronic matter undergoes a transition to a new high-density phase, such as deconfined quark matter, the CCSN dynamics itself can carry the imprint of dense matter physics. This possibility has been studied as a signature of a QCD or hadron-quark phase transition in CCSNe [93–95].

The basic mechanism is tied to the EOS. During the postbounce accretion phase, the PNS mass and central density continue to increase. If a first-order phase transition is reached, the EOS can soften in the mixed-phase region, reducing pressure support and triggering a secondary contraction of the PNS core. If the newly formed high-density phase becomes sufficiently stiff, this contraction can be halted by a second bounce, launching an additional shock wave from deep inside the PNS.

Such a secondary collapse and bounce can leave characteristic neutrino signatures. When the second shock propagates outward and crosses the neutrinospheres, it can produce a short and sharp neutrino burst, separated in time from the ordinary neutronization burst. Depending on the phase-transition model and the flavor channel, this burst can appear prominently in the $\bar{\nu}_e$ signal or modify the luminosities and mean energies of several neutrino species. The delay time of this burst relative to core bounce is sensitive to the onset density of the phase transition, the accretion history, and the PNS mass.

A dense-matter phase transition can also generate gravitational wave signatures. The rapid secondary contraction and bounce of the PNS core can produce a short gravitational wave burst, while subsequent oscillations of the compact remnant can modify the high-frequency gravitational wave spectrum [96]. Therefore, a coincident neutrino and gravitational wave signal from a future Galactic CCSN would provide a direct probe of the high-density EOS.

This strategy differs from the previous constraints on weakly coupled particles. It does not rely on excessive exotic cooling, visible energy deposition, or decay products outside the

star. Instead, the observable is the dynamical response of the PNS itself: a secondary collapse, a second shock, and correlated neutrino and gravitational wave features. A future Galactic CCSN observation could therefore test whether the dense-matter EOS admits a strong phase transition during the postbounce evolution.

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